

A

Appendices

A.1 Functions

Informally, a function f can be viewed as a rule that associates with every point x in some domain D an image value $f(x)$. More formally, a function can be defined as a collection of ordered pairs with the property that if (x, y_1) and (x, y_2) are both in the collection, then y_1 and y_2 must be equal. The *domain* of the function can then be defined as $D = \{x : (x, y) \in f\}$, and the *range* of the function is $R = \{y : (x, y) \in f\}$. For $x \in D$, $f(x)$ denotes the unique value $y \in R$ with $(x, y) \in f$, and we say f maps x to this value $f(x)$.

A function f with domain D and range R is said to be *into* a set S if $R \subset S$, and *onto* S if $R = S$. The notation $f : D \rightarrow S$ means that f has domain D and maps this domain into S .

A function is *one-to-one* if every value y in the range R is the image of a *unique* value x in the domain D . In this case the collection formed reversing all of the ordered pairs, $\{(y, x) : (x, y) \in f\}$, is also a function, with domain R and range D , called the *inverse function*, denoted $f^{\leftarrow}$. Then $f^{\leftarrow}(y)$ for $y \in R$ is the unique value x with $f(x) = y$. Functions also have an inverse that maps sets to sets, denoted f^{-1} , given by

$$f^{-1}(S) = \{x \in D : f(x) \in S\}.$$

This inverse always exists, even if f is not one-to-one.

Example A.1. If $D = \{HH, HT, TH, TT\}$, perhaps viewed as all possible outcomes of tossing a coin twice, then

$$f = \{(HH, 2), (HT, 1), (TH, 1), (TT, 0)\}$$

is the function mapping the outcome to the number of heads. The range of this function is $R = \{0, 1, 2\}$. The function is not one-to-one because the value $1 \in R$ is the image of HT and TH , $f(HT) = f(TH)$. So the inverse function $f^{\leftarrow}$ does not exist. The other inverse, f^{-1} , does exist. For instance, $f^{-1}(1, 2) = \{HT, TH, HH\}$ and $f^{-1}(4) = \emptyset$.

The definitions given above are quite general, covering situations where D and R are very complicated sets. For instance, a measure is a function with domain D a σ -field of subsets of some sample space. The domain for a function could even be itself a collection of functions. For instance, if $C[0, 1]$ is the set of all continuous functions on $[0, 1]$, then

$$h \rightsquigarrow f(h) = \int_0^1 h(x) dx$$

describes a function f with domain $D = C[0, 1]$ and range $R = \mathbb{R}$, $f : C[0, 1] \rightarrow \mathbb{R}$. A function is called *real-valued* if $R \subset \mathbb{R}$ and *real-valued vector-valued* if $R \subset \mathbb{R}^n$ for some n .

A.2 Topology and Continuity in $\mathbb{R}^n$

For $x \in \mathbb{R}^n$ and $\epsilon > 0$, let $B_\epsilon(x) = \{y : \|y - x\| < \epsilon\}$, the open ball around x with radius ϵ . A set $O \subset \mathbb{R}^n$ is called *open* if it has the property that for every point x in O there is some open ball $B_\epsilon(x)$ that is contained in O ($B_\epsilon(x) \subset O$). The collection $\mathcal{T}$ of all open sets is called a *topology*. A set C is *closed* if its complement is open.

Topologies can be used to characterize convergence and continuity in general settings. Note that the topology $\mathcal{T}$ for $\mathbb{R}^n$ is closed under finite intersections and arbitrary unions. Also, $\mathbb{R}^n$ itself and the empty set $\emptyset$ are both open sets in $\mathcal{T}$. In general, a topology is any collection of sets with these properties. One example arises when we are only concerned with points in a subset S of $\mathbb{R}^n$. In this case, we use the *relative topology* in which all sets of the form $O \cap S$ with $O \in \mathcal{T}$ are open. For instance, in the relative topology with $S = [0, 2]$, $[0, 1)$ is an open set, even though this set is not open in $\mathbb{R}$. Sets in the relative topology are called *open relative to S* .

A set N is called a *neighborhood* of x if N contains an open set O with $x \in O$. A sequence of vectors x_n , $n \geq 1$, in $\mathbb{R}^n$ *converges* to $x \in \mathbb{R}^n$ if for any neighborhood N of x , x_n lies in N for all sufficiently large n . This definition is equivalent to the usual definition in calculus courses. But because it is based only on the topology $\mathcal{T}$, this definition of convergence can be used to define convergence on any space with a topology of open sets, even if there is no notion of distance between elements in the space.

A point x lies in the boundary ∂S of a set $S \subset \mathbb{R}^n$ if for any $\epsilon > 0$, $B_\epsilon(x)$ contains at least one point in S and at least one point in S^c . The *closure* $\overline{S}$ of a set S is the union of S with its boundary ∂S , $\overline{S} = S \cup \partial S$. This closure $\overline{S}$ is the smallest closed set that contains S . The *interior* S° of S is $S - \partial S$. The interior is the largest open set contained in S .

A function $f : D \rightarrow \mathbb{R}$ is *continuous* at a point $x \in D$ if $f(x_n) \rightarrow f(x)$ whenever x_n , $n \geq 1$, is a sequence in D converging to x . The function f is called continuous if it is continuous at every point x in D . Continuity can

also be characterized using open sets. A function $f : D \rightarrow \mathbb{R}$ is continuous if $f^{-1}(O)$ is open relative to D for any $O \in \mathcal{T}$.

A collection $\{O_\alpha : \alpha \in A\}$ of open sets is called an *open cover* of K if K is a subset of the union,

$$K \subset \bigcup_{\alpha \in A} O_\alpha.$$

A set K in $\mathbb{R}^n$ (or any other topological space) is *compact* if any open cover $\{O_\alpha : \alpha \in A\} \subset \mathcal{T}$ has a finite subcover, $\{O_{\alpha_1}, \dots, O_{\alpha_m}\}$ with $K \subset \bigcup_{i=1}^m O_{\alpha_i}$. The following result provides a useful characterization of compact sets in $\mathbb{R}^n$.

Theorem A.2 (Heine–Borel). *A set $K \subset \mathbb{R}^n$ is compact if and only if it is closed and bounded, $\sup_{x \in K} \|x\| < \infty$.*

If x_n , $n \geq 1$, is a sequence of points in some space, and if $n_1 < n_2 < \dots$ are positive integers, then x_{n_m} , $m \geq 1$, is called a *subsequence*. A set K is called *sequentially compact* if any subsequence x_n , $n \geq 1$, has a convergent subsequence, $x_{n_m} \rightarrow x$ with $x \in K$. In general, compactness implies sequential compactness, but in $\mathbb{R}^n$ (or any metric space), compactness and sequential compactness are the same. Let $C(K)$ denote the collection of continuous real-valued functions on K . The next result shows that functions in $C(K)$ achieve their supremum if K is compact.

Proposition A.3. *Suppose K is compact and $f \in C(K)$, and let $M = \sup_{x \in K} f(x)$. Then $f(x) = M$ for some $x \in K$.*

Proof. There must be values in the range of f arbitrarily close to M . Thus for any $n \geq 1$ there exists x_n with $f(x_n) > M - 1/n$. Since the points x_n , $n \geq 1$, lie in K , by compactness there is a subsequence $x_{n_m} \rightarrow x \in K$. Since f is continuous, $f(x_{n_m}) \rightarrow f(x)$. But because $M - 1/n_m < f(x_{n_m}) \leq M$, $f(x_{n_m}) \rightarrow M$. Thus $f(x) = M$. $\square$

If $f : D \rightarrow \mathbb{R}$ is continuous, then for any $x \in D$ and any $\epsilon > 0$, there exists $\delta > 0$ such that $|f(x) - f(y)| < \epsilon$ whenever $\|x - y\| < \delta$. In general, the value for δ will need to depend on both ϵ and x . If the choice can be made independently of x , then f is called *uniformly continuous*. Equivalently, f is uniformly continuous if

$$\sup_{\|x-y\| < \epsilon} |f(x) - f(y)| \rightarrow 0$$

as $\epsilon \downarrow 0$.

Proposition A.4. *If K is compact and $f \in C(K)$, then f is uniformly continuous.*

Proof. Fix $\epsilon > 0$ and for every $x \in D$ choose δ_x so that $|f(x) - f(y)| < \epsilon/2$ whenever $y \in B_{\delta_x}(x)$. Since $x \in B_{\delta_x/2}(x)$, these balls $B_{\delta_x/2}(x)$, $x \in D$, form

an open cover of K . Because K is compact, there must be a finite subcover $B_{\delta_{x_i}/2}(x_i)$, $i = 1, \dots, m$. Define $\delta = \min_{i=1, \dots, m} \delta_{x_i}/2$. Suppose $\|x - y\| < \delta$. Since x is in one of the sets in the finite open cover, $\|x - x_i\| < \delta_{x_i}/2$ for some i . By the triangle inequality,

$$\|y - x_i\| \leq \|y - x\| + \|x - x_i\| < \frac{1}{2}\delta + \frac{1}{2}\delta_{x_i} \leq \delta_{x_i}.$$

From the definition of δ_x ,

$$|f(x) - f(x_i)| < \frac{\epsilon}{2} \quad \text{and} \quad |f(y) - f(x_i)| < \frac{\epsilon}{2},$$

and so, by the triangle inequality,

$$|f(x) - f(y)| \leq |f(x) - f(x_i)| + |f(x_i) - f(y)| < \epsilon. \quad \square$$

Functions in $C(K)$, K compact, achieve their supremum and are uniformly continuous. A final useful property of $C(K)$ is that monotone sequences converge uniformly.

Theorem A.5 (Dini). *If $f_1 \geq f_2 \geq \dots$ are positive functions in $C(K)$, K compact, and if $f_n(x) \rightarrow 0$ as $n \rightarrow \infty$ for every $x \in K$, then $\sup_{x \in D} f_n(x) \rightarrow 0$ as $n \rightarrow \infty$.*

Proof. Fix $\epsilon > 0$ and define

$$O_n = f_n^{-1}[(-\infty, \epsilon)] = \{x \in K : f_n(x) < \epsilon\}.$$

Since f is continuous these sets O_n , $n \geq 1$, are open relative to K , and since the functions are decreasing, $O_1 \subset O_2 \subset \dots$. Because $f_n(x) \rightarrow 0$, the point x will be in O_n once n is large enough, and so these sets cover K . By compactness, there is a finite subcover, $O_{n_1}, \dots, O_{n_m}$. If $N = \max n_i$, then the union of these sets is O_N , and thus $O_N = K$. So $f_N(x) < \epsilon$ for all $x \in K$, and since the functions are decreasing, $f_n(x) < \epsilon$ for all $x \in K$ and all $n \geq N$. So

$$\limsup_{n \rightarrow \infty} \sup_{x \in K} f_n(x) \leq \epsilon,$$

and since $\epsilon > 0$ is arbitrary, $\sup_{x \in K} f_n(x) \rightarrow 0$. $\square$

A.3 Vector Spaces and the Geometry of $\mathbb{R}^n$

Definition A.6. *A set V is a vector space over the real numbers¹ $\mathbb{R}$ if elements of V can be added, with the sum again an element of V , and multiplied by a constant, with the product an element of V , so that:*

¹ Vector spaces can be defined analogously over complex numbers or any other field.

1. (Commutative and associative laws for addition) If $u \in V$, $v \in V$, and $w \in V$,

$$u + v = v + u \quad \text{and} \quad (u + v) + w = u + (v + w).$$

2. There is a zero element in V , denoted $\mathbb{O}$, such that $\mathbb{O} + u = u$, for all $u \in V$.

3. Given any $u \in V$ there exists an element $-u \in V$ such that $u + (-u) = \mathbb{O}$.

4. (Associative law for multiplication) If $v \in V$, $a \in \mathbb{R}$, and $b \in \mathbb{R}$, then $(ab)v = a(bv)$.

5. (Distributive laws) If $u \in V$, $v \in V$, $a \in \mathbb{R}$, and $b \in \mathbb{R}$, then

$$(a + b)v = av + bv \quad \text{and} \quad a(u + v) = au + av.$$

6. If $v \in V$, $1v = v$.

A subset $W \subset V$ is called a *subspace* if W is closed under addition (if $u \in W$ and $v \in W$, then $u + v \in W$) and multiplication (if $c \in \mathbb{R}$ and $v \in W$, then $cv \in W$). If these hold, then W must also be a vector space.

Most of the vector spaces in this book are $\mathbb{R}^n$ or subspaces of $\mathbb{R}^n$. Other examples of interest include the set of all $n \times p$ matrices with real entries. Collections of functions can also form vector spaces. For instance, the collection of all functions f with form $f(x) = a \sin x + b \cos x$ is a vector space. And the set $C[0, 1]$ of all continuous real-valued functions on $[0, 1]$ is a vector space.

A vector u is a linear combination of vectors $v_1, \dots, v_n$ if $u = c_1v_1 + \dots + c_nv_n$ for some constants $c_i \in \mathbb{R}$, $i = 1, \dots, n$. The set of all linear combinations of vectors from a set S is called the *linear span* of S , and the linear span of any set is then a vector space. For instance, $\mathbb{R}^2$ is the linear span of $S = \{(1, 0), (0, 1)\}$ because an arbitrary vector $(x, y) \in \mathbb{R}^2$ can be expressed as $x(1, 0) + y(0, 1)$. The *dimension* of a vector space V is the smallest number of vectors needed to span V . If V is not the span of any finite set, then V is called infinite-dimensional. By convention, the dimension of the trivial vector space $\{\mathbb{O}\}$ is zero.

Vectors $v_1, \dots, v_p$ are *linearly independent* if the linear combination $c_1v_1 + \dots + c_pv_p = \mathbb{O}$ only if $c_1 = \dots = c_p = 0$. In this case, the linear span W of $S = \{v_1, \dots, v_p\}$ has dimension p . If these vectors are points in $\mathbb{R}^n$, and if $X = (v_1, \dots, v_p)$, an $n \times p$ matrix with columns $v_1, \dots, v_p$, then $X\beta = \beta_1v_1 + \dots + \beta_pv_p$ for $\beta \in \mathbb{R}^p$. Since this is an arbitrary linear combination of the vectors in S ,

$$W = \text{span}(S) = \{X\beta : \beta \in \mathbb{R}^p\}.$$

The rank^2 of X is the dimension of W . Because X has p columns, its rank is at most p . If the columns of X are linearly independent, then the rank of X is p and X is called *full rank*.³

The *dot* or *inner* product of two vectors u and v in $\mathbb{R}^n$ is $u \cdot v = u_1v_1 + \cdots + u_nv_n$, and these vectors are called *orthogonal*, denoted $u \perp v$, if $u \cdot v = 0$. If W is a subspace of $\mathbb{R}^n$, the set of vectors orthogonal to all points in W is called the orthogonal complement of W , denoted $W^\perp$. The sum of the dimensions of W and $W^\perp$ is n .

The (Euclidean) *length* $\|u\|$ of a vector $u \in \mathbb{R}^n$ is

$$\|u\| = \sqrt{u \cdot u} = \sqrt{u_1^2 + \cdots + u_n^2},$$

and u is called a *unit vector* if $\|u\| = 1$. If a subspace W of $\mathbb{R}^n$ has dimension p , then there exist p unit vectors $e_1, \dots, e_p$ that are mutually orthogonal, $e_i \cdot e_j = 0$, $i \neq j$, and span W . This collection $\{e_1, \dots, e_p\}$ is called an *orthonormal basis* for W , and any vector $v \in W$ can be expressed uniquely as $v = c_1e_1 + \cdots + c_pe_p$. Then if $e_{p+1}, \dots, e_n$ is an orthonormal basis for $W^\perp$, $e_1, \dots, e_n$ forms an orthonormal basis for $\mathbb{R}^n$. Writing $w \in \mathbb{R}^n$ uniquely as $c_1e_1 + \cdots + c_ne_n$, we see that $w = u + v$ with $u = c_1e_1 + \cdots + c_pe_p \in W$ and $v = c_{p+1}e_{p+1} + \cdots + c_ne_n \in W^\perp$. These vectors u and v are called the (orthogonal) *projections* of w onto W and $W^\perp$. The projection u can be characterized as the vector in W closest to w . We can show this using the Pythagorean formula that if $u \perp v$,

$$\|u + v\|^2 = \|u\|^2 + \|v\|^2.$$

If x is an arbitrary point in W , then

$$\|w - x\|^2 = \|(u - x) + (w - u)\|^2 = \|u - x\|^2 + \|w - u\|^2 \leq \|w - u\|^2,$$

because $w - x \in W$ and $w - u = v \in W^\perp$ are orthogonal.

A.4 Manifolds and Tangent Spaces

Manifolds are sets in $\mathbb{R}^r$ that look locally like $\mathbb{R}^q$ for some q called the *dimension of the manifold*. For instance, the unit circle in $\mathbb{R}^2$ is a manifold with dimension one. A precise definition involves reparameterizing the manifold locally by a differentiable function. If U is an open subset of $\mathbb{R}^q$, let $C_1(U)$

² This might also be called the column rank of X since it is the dimension of the linear span of the columns of X . But the row rank of X can be defined similarly, and these two ranks must agree.

³ Since the row and column ranks agree, the rank of X is also at most n . So it is also natural to call X full rank if $n < p$ and the rank of X is n . This happens if the rows of X are linearly independent.

be the collection of all functions $h : U \rightarrow \mathbb{R}^r$ with continuous first partial derivatives. For $h \in \mathcal{C}_1(U)$, let Dh be the matrix of partial derivatives,

$$[Dh(x)]_{i,j} = \frac{\partial h_i(x)}{\partial x_j}.$$

When U is not open, a function $h : U \rightarrow \mathbb{R}^r$ is in $\mathcal{C}_1(U)$ if there exists an open set $V \supset U$ and a function $H \in \mathcal{C}_1(V)$ that coincides with h on U ; that is, $h(x) = H(x)$ for $x \in U$. A set $\Omega_0 \subset \mathbb{R}^r$ is a *manifold* of dimension $q < r$ if for every $\theta_0 \in \Omega_0$ there exists an open neighborhood N_0 of θ_0 , an open set $M_0 \subset \mathbb{R}^q$ and a one-to-one function $h : M_0 \rightarrow N_0 \cap \Omega_0$ with $h \in \mathcal{C}_1(M_0)$ and $h^{-1} \in \mathcal{C}_1(N_0 \cap \Omega_0)$. Using the inverse function theorem, the condition that h^{-1} is differentiable can be replaced with the condition that $Dh(x)$ has full rank for all $x \in M_0$. The definition of a manifold is sometimes given in terms of constraints: $\Omega_0 \subset \mathbb{R}^r$ is a manifold of dimension $q < r$ if for every $\theta_0 \in \Omega_0$ there exists an open neighborhood N_0 of θ_0 and a function $g : N_0 \rightarrow \mathbb{R}^{r-q}$ in $\mathcal{C}_1(N_0)$ such that $Dg(x)$ has full rank for all $x \in N_0$ and

$$\Omega_0 \cap N_0 = \{x \in N_0 : g(x) = 0\}.$$

Because g maps onto $\mathbb{R}^{r-q}$, the last assertion means that in some neighborhood of θ_0 , the points in Ω_0 are those that satisfy $r - q$ nonlinear constraints. The assumption on the rank of Dg is needed so that the constraints are not redundant.

Tangent spaces of a manifold Ω_0 are defined so that if points θ_0 and θ_1 in Ω_0 are close to each other, their difference should lie approximately in the tangent space at θ_0 . To be specific, let θ_0 be an arbitrary point of the manifold Ω_0 and let h be the local reparameterization given in our first definition of a manifold. Assume that $h(y) = \theta_0$. First-order Taylor expansion of h_i about y gives

$$h_i(x) \approx h_i(y) + \sum_{j=1}^q \frac{\partial}{\partial y_j} h_i(y)(x_j - y_j),$$

or, in matrix notation,

$$h(x) \approx \theta_0 + Dh(y)(x - y).$$

The *tangent space* at θ_0 is defined as

$$V_{\theta_0} = \{Dh(y)x : x \in \mathbb{R}^q\}.$$

Equivalently, V_{θ_0} is the linear span of the columns of $Dh(y)$. It is worth noting that V_{θ_0} is a vector space that passes through the origin, but typically does not pass through θ_0 . Since $h(x)$ lies in Ω_0 for x near y , by the Taylor expansion above

$$h(x) - \theta_0 \approx Dh(y)(x - y) \in V_{\theta_0},$$

so if $\theta_1 \in \Omega_0$ is close to θ_0 , then $\theta_1 - \theta_0$ is “almost” in V_{θ_0} . The tangent space V_{θ_0} can also be identified from a local constraint function g . By first-order Taylor expansion, if $\theta \in \Omega_0$ is close to θ_0 , then

$$0 = g(\theta) - g(\theta_0) \approx Dg(\theta_0)(\theta - \theta_0).$$

Hence $\theta - \theta_0$ (which should lie almost in V_{θ_0}) is approximately orthogonal to each of the rows of $Dg(\theta_0)$. A careful argument along these lines shows that $V_{\theta_0}^\perp$ is the vector space spanned by the rows of $Dg(\theta_0)$.

Example A.7. Let Ω_0 be the unit circle in $\mathbb{R}^2$. Then Ω_0 is a manifold with dimension $q = 1$. Let

$$\theta_0 = \begin{pmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix} \in \Theta_0.$$

The tangent space at θ_0 is just the line $V_{\theta_0} = \{x \in \mathbb{R}^2 : x_1 + x_2 = 0\}$. The graph to the left in [Figure A.1](#) shows the circle Ω_0 with the tangent line, and the graph to the right shows the tangent space V_{θ_0} .

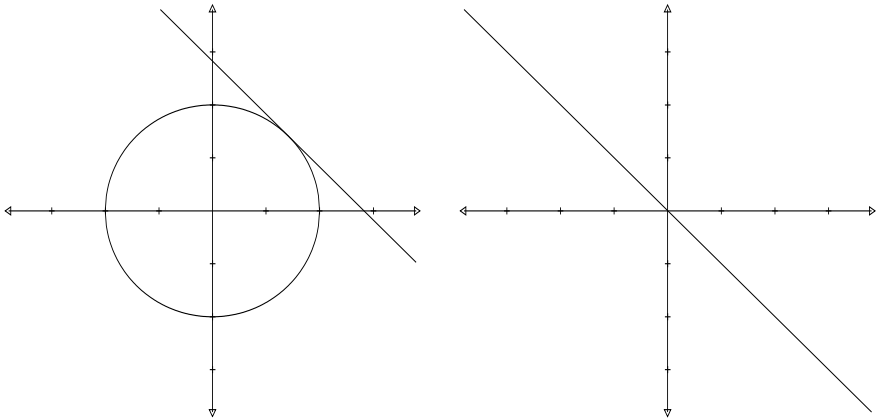


Fig. A.1. Tangent lines and spaces.

A.5 Taylor Expansion for Functions of Several Variables

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ have a continuous derivative f' . Taylor's theorem with Lagrange's form for the remainder asserts that

$$f(x) = f(x_0) + (x - x_0)f'(x^*), \quad (\text{A.1})$$

where x^* is an intermediate point between x and x_0 . If f'' is continuous, then

$$f(x) = f(x_0) + (x - x_0)f'(x_0) + \frac{1}{2}(x - x_0)^2 f''(x^{**}), \quad (\text{A.2})$$

where x^{**} is an intermediate value between x and x_0 . The goal of this section is to derive analogous results when $f : \mathbb{R}^n \rightarrow \mathbb{R}$. Define the function $\nabla_i f$ by

$$\nabla_i f(x) = \frac{\partial}{\partial x_i} f(x),$$

and let

$$\nabla f(x) = \begin{pmatrix} \nabla_1 f(x) \\ \vdots \\ \nabla_n f(x) \end{pmatrix}.$$

The first step is deriving a chain rule formula for computing $\partial f(hx)/\partial h$. We assume that ∇f is continuous. Using the definition of derivatives,

$$\begin{aligned} \frac{\partial}{\partial h} f(hx) &= \lim_{\epsilon \rightarrow 0} \frac{f((h + \epsilon)x_1, \dots, (h + \epsilon)x_n) - f(hx_1, \dots, hx_n)}{\epsilon} \\ &= \lim_{\epsilon \rightarrow 0} \frac{f((h + \epsilon)x_1, \dots, (h + \epsilon)x_n) - f(hx_1, (h + \epsilon)x_2, \dots, (h + \epsilon)x_n)}{\epsilon} \\ &\quad + \lim_{\epsilon \rightarrow 0} \frac{f(hx_1, (h + \epsilon)x_2, \dots, (h + \epsilon)x_n) - f(hx_1, \dots, hx_n)}{\epsilon}, \end{aligned}$$

provided both limits exist. By (A.1), the argument of the first limit equals

$$\begin{aligned} &\frac{((h + \epsilon)x_1 - hx_1) \nabla_1 f(z^*, (h + \epsilon)x_2, \dots, (h + \epsilon)x_n)}{\epsilon} \\ &= x_1 \nabla_1 f(z^*, (h + \epsilon)x_2, \dots, (h + \epsilon)x_n), \end{aligned}$$

where z^* lies between $(h + \epsilon)x_1$ and hx_1 . Since ∇f is continuous, this term approaches $x_1 \nabla_1 f(hx)$ as $\epsilon \rightarrow 0$. The other limit is like the original derivative except that ϵ does not appear in the first argument. Repeating the argument we can remove ϵ from the second argument, but an additional term arises: $x_2 \nabla_2 f(hx)$. Iteration n times gives

$$\frac{\partial}{\partial h} f(hx) = \sum_{i=1}^n x_i \nabla_i f(hx) = x' \nabla f(hx).$$

Also,

$$\begin{aligned} \frac{\partial^2}{\partial h^2} f(hx) &= \sum_{i=1}^n x_i \frac{\partial}{\partial h} \nabla_i f(hx) \\ &= \sum_{i=1}^n x_i \sum_{j=1}^n x_j \nabla_j \nabla_i f(hx) \\ &= x' \nabla^2 f(hx) x, \end{aligned}$$

where $\nabla^2 f$ is the Hessian matrix of partial derivatives of f :

$$\nabla^2 f = \begin{pmatrix} \nabla_1 \nabla_1 f & \nabla_1 \nabla_2 f & \cdots \\ \nabla_2 \nabla_1 f & \nabla_2 \nabla_2 f & \cdots \\ \vdots & \vdots & \ddots \end{pmatrix}.$$

Viewing $f(hx)$ as a function of h with x a fixed constant, we can use (A.1) and (A.2) to express the value of the function at $h = 1$ by an expansion about $h = 0$. This gives

$$f(x) = f(0) + x' \nabla f(x^*)$$

and

$$f(x) = f(0) + x' \nabla f(0) + \frac{1}{2} x' \nabla^2 f(x^{**}) x,$$

where $x^* = h^* x$, $x^{**} = h^{**} x$, and h^* and h^{**} are intermediate points between 0 and 1; so x^* and x^{**} lie on the chord between 0 and x . As $x \rightarrow 0$, $\nabla f(x^*) \rightarrow \nabla f(0)$ and $\nabla^2 f(x^{**}) \rightarrow \nabla^2 f(0)$, which justifies the approximations

$$f(x) \approx f(0) + x' \nabla f(0)$$

and

$$f(x) \approx f(0) + x' \nabla f(0) + \frac{1}{2} x' \nabla^2 f(0) x.$$

The corresponding Taylor approximations expanding about a point $x_0 \neq 0$ are

$$f(x) \approx f(x_0) + (x - x_0)' \nabla f(x_0)$$

and

$$f(x) \approx f(x_0) + (x - x_0)' \nabla f(x_0) + \frac{1}{2} (x - x_0)' \nabla^2 f(x_0) (x - x_0).$$

Both of these approximations hold with equality if the argument of the highest derivative is changed to an intermediate point on the chord between x and x_0 .

A.6 Inverting a Partitioned Matrix

Let A be a nonsingular matrix partitioned into blocks,

$$A = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix},$$

and let $B = A^{-1}$, also partitioned into blocks,

$$B = \begin{pmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{pmatrix},$$

Then

$$AB = \begin{pmatrix} A_{11}B_{11} + A_{12}B_{21} & A_{11}B_{12} + A_{12}B_{22} \\ A_{21}B_{11} + A_{22}B_{21} & A_{21}B_{12} + A_{22}B_{22} \end{pmatrix} = \begin{pmatrix} I & 0 \\ 0 & I \end{pmatrix}.$$

This leads to the following two equations:

$$A_{11}B_{12} + A_{12}B_{22} = 0 \quad (\text{A.3})$$

and

$$A_{21}B_{12} + A_{22}B_{22} = I. \quad (\text{A.4})$$

Using (A.3),

$$B_{12} = -A_{11}^{-1}A_{12}B_{22}.$$

Using this to eliminate B_{12} in (A.4),

$$-A_{21}A_{11}^{-1}A_{12}B_{22} + A_{22}B_{22} = I$$

and hence

$$B_{22} = (A_{22} - A_{21}A_{11}^{-1}A_{12})^{-1}.$$

Using this in the equation for B_{12} ,

$$B_{12} = -A_{11}^{-1}A_{12}(A_{22} - A_{21}A_{11}^{-1}A_{12})^{-1}.$$

Similar calculations show that

$$B_{11} = (A_{11} - A_{12}A_{22}^{-1}A_{21})^{-1}$$

and

$$B_{21} = -A_{22}^{-1}A_{21}B_{11} = -A_{22}^{-1}A_{21}(A_{11} - A_{12}A_{22}^{-1}A_{21})^{-1}.$$

So A^{-1} equals

$$\begin{pmatrix} (A_{11} - A_{12}A_{22}^{-1}A_{21})^{-1} & -A_{11}^{-1}A_{12}(A_{22} - A_{21}A_{11}^{-1}A_{12})^{-1} \\ -A_{22}^{-1}A_{21}(A_{11} - A_{12}A_{22}^{-1}A_{21})^{-1} & (A_{22} - A_{21}A_{11}^{-1}A_{12})^{-1} \end{pmatrix}.$$

A.7 Central Limit Theory

This appendix derives the central limit theorem and a few of the extensions used in the main body of this text. The approach is based on an inversion formula for characteristic functions and smoothing.

A.7.1 Characteristic Functions

If U and V are random variables, then the function $X = U + iV$, mapping the sample space to the complex numbers $\mathcal{C}$, is called a *complex random variable*. The mean of X is defined as

$$EX \stackrel{\text{def}}{=} EU + iEV.$$

As with ordinary random variables,

$$|EX| \leq E|X|, \quad (\text{A.5})$$

which follows from Jensen's inequality. The integral of a complex function $u + iv$ against a measure μ is defined similarly as

$$\int (u + iv) d\mu = \int u d\mu + i \int v d\mu.$$

The *characteristic function* of a random variable $X \sim F$ is

$$\mathfrak{f}(t) = Ee^{itX} = \int e^{itx} dF(x), \quad t \in \mathbb{R}.$$

Formally, the characteristic function is just the moment generating function for X evaluated at the imaginary argument it , so it is natural that derivatives of $\mathfrak{f}$ are related to moments for X . By dominated convergence, if $E|Y|^k < \infty$,

$$\frac{d^k}{dt^k} \mathfrak{f}(t) = E \frac{\partial^k}{\partial t^k} e^{itY} = E(iY)^k e^{itY}.$$

In particular, the k th derivative at zero is $i^k EX^k$, and Taylor expansion gives

$$\mathfrak{f}(t) = 1 + itEY - \frac{1}{2}t^2 EY^2 + \cdots + \frac{1}{k!}(it)^k EY^k + o(t^k) \quad (\text{A.6})$$

as $t \rightarrow 0$.

Suppose $X \sim F$ and $Y \sim G$ are independent with characteristic functions $\mathfrak{f}$ and $\mathfrak{g}$. Then

$$e^{-ity} \mathfrak{f}(y) = \int e^{iy(x-t)} dF(x).$$

Integrating this against G ,

$$\int e^{-ity} \mathfrak{f}(y) dG(y) = \int \mathfrak{g}(x-t) dF(x),$$

an important identity called *Parseval's relation*.

Example A.8. If $Z \sim N(0, 1)$, then

$$\begin{aligned} Ee^{itZ} &= \int \frac{\exp(itz - z^2/2)}{\sqrt{2\pi}} dz \\ &= e^{-t^2/2} \int \frac{\exp[-\frac{1}{2}(z - it)^2]}{\sqrt{2\pi}} dz \\ &= e^{-t^2/2}. \end{aligned}$$

From this, if $X \sim N(\mu, \sigma^2)$, then

$$Ee^{itX} = Ee^{it(\mu + \sigma Z)} = e^{it\mu} Ee^{it\sigma Z} = e^{it\mu - t^2\sigma^2/2}.$$

Suppose $Y = Z/a$ in Parseval's relation. Then

$$\mathbf{g}(t) = Ee^{itZ/a} = \exp[-\frac{1}{2}(t/a)^2].$$

The density of Y is $a\phi(ay)$, and so

$$\int e^{-ity} \mathbf{f}(y) a\phi(ay) dy = \int \exp\left[-\frac{1}{2}\left(\frac{x-t}{a}\right)^2\right] dF(x),$$

or

$$\frac{1}{2\pi} \int \mathbf{f}(y) e^{-ity - a^2 y^2/2} dy = \int \frac{1}{a} \phi\left(\frac{x-t}{a}\right) dF(x). \quad (\text{A.7})$$

The right-hand side of this equation is the density of $X + aZ$, and by this formula, if $\mathbf{f}$ is known the density for $X + aZ$ can be computed for any $a > 0$. Because $X + aZ \Rightarrow X$ as $a \downarrow 0$, we have the following result.

Theorem A.9. *Distinct probability distributions on $\mathbb{R}$ have distinct characteristic functions.*

The next result is a bit more constructive. It gives an inversion formula for the density when the characteristic function is integrable.

Theorem A.10. *Suppose $\int |\mathbf{f}(t)| dt < \infty$. Then F is absolutely continuous with a bounded density given by*

$$f(x) = \frac{1}{2\pi} \int e^{-itx} \mathbf{f}(t) dt.$$

Proof. Let f_a be the density of $X + aZ$ in (A.7). Then $f_a(x) \rightarrow f(x)$ as $a \downarrow 0$ for every $x \in \mathbb{R}$ by dominated convergence. Also,

$$f_a(x) \leq \frac{1}{2\pi} \int |\mathbf{f}(t)| dt, \quad \forall x \in \mathbb{R}, \quad a > 0.$$

Because $X + aZ \Rightarrow X$, by the portmanteau theorem (Theorem 9.25) and dominated convergence, for any $b < c$,

$$\begin{aligned}
P[X \in (b, c)] &\leq \liminf_{a \downarrow 0} P[X + aZ \in (b, c)] \\
&= \liminf_{a \downarrow 0} \int_b^c f_a(x) dx = \int_b^c f(x) dx,
\end{aligned}$$

and

$$P[X \in [b, c]] \geq \limsup_{a \downarrow 0} \int_b^c f_a(x) dx = \int_b^c f(x) dx.$$

So $P[X \in (b, c)] = \int_b^c f(x) dx$, and X has density f . $\square$

Remark A.11. When F is absolutely continuous with density f , the characteristic function $\mathfrak{f}$ is also called the *Fourier transform* of f . Because $\mathfrak{f}(t) = \int e^{itx} f(x) dx$, Fourier transforms can be defined for measurable functions f that are not densities, provided f is integrable, $\int |f(x)| dx < \infty$. When $\mathfrak{f}$ is integrable, the inversion formula in this theorem remains valid and gives a constructive way to compute f from $\mathfrak{f}$.

Remark A.12. The inversion formula for the standard normal distribution gives

$$\phi(x) = \frac{1}{2\pi} \int e^{-itx - t^2/2} dt. \quad (\text{A.8})$$

Dominated convergence and repeated differentiation then give

$$\frac{1}{2\pi} \int (-it)^k e^{-itx - t^2/2} dt = \phi^{(k)}(x). \quad (\text{A.9})$$

A.7.2 Central Limit Theorem

Let $X, X_1, X_2, \dots$ be i.i.d. with common mean μ and finite variance σ^2 , and define

$$Z_n = \frac{\sqrt{n}(\overline{X}_n - \mu)}{\sigma}.$$

Let $\mathfrak{f}$ denote the characteristic function of $Y = (X - \mu)/\sigma$

$$\mathfrak{f}(t) = Ee^{itY},$$

and let $\mathfrak{f}_n$ denote the characteristic function of Z_n . Noting that

$$e^{itZ_n} = \prod_{i=1}^n \exp\left[\frac{it}{\sqrt{n}} \frac{X_i - \mu}{\sigma}\right],$$

a product of independent variables, we have

$$\mathfrak{f}_n(t) = \mathfrak{f}^n(t/\sqrt{n}).$$

Since $EY = 0$ and $EY^2 = 1$, by Taylor expansion as in (A.6),

$$\mathfrak{f}(t/\sqrt{n}) = 1 - \frac{t^2}{2n} + o(1/n)$$

as $n \rightarrow \infty$ with t fixed. It follows that

$$\log \mathfrak{f}_n(t) = n \log \mathfrak{f}(t/\sqrt{n}) \rightarrow -t^2/2,$$

and so

$$\mathfrak{f}_n(t) \rightarrow e^{-t^2/2},$$

the characteristic function for the standard normal distribution. Convergence of these characteristic functions certainly suggests that the corresponding distributions should converge, but a careful argument using our inversion formula is a bit delicate, mainly because $\mathfrak{f}_n$ need not be integrable. To circumvent this problem, we use a smoothing approach due to Berry. Let h be a density with support $[-1, 1]$ and bounded derivatives of all orders. One concrete possibility is

$$h(x) = c \exp[-1/(1-x^2)] 1_{(-1,1)}(x).$$

Let $\mathfrak{h}$ be the corresponding characteristic function, and let W be a random variable with density h independent of Z_n . Repeated integration by parts gives

$$\mathfrak{h}(t) = \int h(x) e^{itx} dx = \frac{i}{t} \int h'(x) e^{itx} dx = \cdots = \left(\frac{i}{t}\right)^j \int h^{(j)}(x) e^{itx} dx.$$

So $\mathfrak{h}(t) = O(|t|^{-j})$ as $t \rightarrow \pm\infty$, for any $j = 1, 2, \dots$

Instead of approximating the distribution of Z_n directly, we consider the distribution of

$$\tilde{Z}_n = Z_n + \epsilon_n W,$$

with ϵ_n , $n \geq 1$, a sequence of constants tending to zero. By the independence, $\tilde{Z}_n$ has characteristic function

$$\tilde{\mathfrak{f}}_n(t) = \mathfrak{f}_n(t) \mathfrak{h}(\epsilon_n t),$$

and since $\mathfrak{h}$ is integrable, $\tilde{Z}_n$ has a bounded density $\tilde{f}_n$ given by the inversion formula in Theorem A.10. Because $|W| \leq 1$, we have the bounds

$$P(\tilde{Z}_n \leq x - \epsilon_n) \leq P(Z_n \leq x) \leq P(\tilde{Z}_n \leq x + \epsilon_n),$$

or

$$\tilde{F}_n(x - \epsilon_n) \leq F_n(x) \leq \tilde{F}_n(x + \epsilon_n),$$

where F_n and $\tilde{F}_n$ denote the cumulative distribution functions for Z_n and $\tilde{Z}_n$. Since differences $|\tilde{F}_n(x \pm \epsilon_n) - F_n(x)|$ are at most $\epsilon_n \|\tilde{f}_n\|_\infty$, these bounds imply

$$\|F_n - \tilde{F}_n\|_\infty \leq \epsilon_n \|\tilde{f}_n\|_\infty. \quad (\text{A.10})$$

With this bound, the central limit theorem follows easily from the following proposition.

Proposition A.13. Taking $\epsilon_n = 1/n^{1/4}$, as $n \rightarrow \infty$,

$$\tilde{f}_n(x) \rightarrow \phi(x).$$

Proof. The desired result essentially follows by dominated convergence from the inversion formula

$$\tilde{f}_n(x) = \frac{1}{2\pi} \int e^{-itx} \mathfrak{f}_n(t) \mathfrak{h}(\epsilon_n t) dt. \quad (\text{A.11})$$

To be specific, since $\log \mathfrak{f}(t) \sim -t^2/2$ as $t \rightarrow 0$, then in some neighborhood of zero, $(-\delta, \delta)$ say, we have

$$\Re[\log \mathfrak{f}(t)] \leq -\frac{t^2}{4},$$

which implies

$$|\mathfrak{f}(t)| \leq e^{-t^2/4}, \quad |t| < \delta,$$

and

$$|\mathfrak{f}_n(t)| \leq e^{-t^2/4}, \quad |t| < \delta\sqrt{n}.$$

Then

$$\tilde{f}_n(x) = \frac{1}{2\pi} \int_{|t| < \delta\sqrt{n}} e^{-itx} \mathfrak{f}_n(t) \mathfrak{h}(\epsilon_n t) dt + \frac{1}{2\pi} \int_{|t| \geq \delta\sqrt{n}} e^{-itx} \mathfrak{f}_n(t) \mathfrak{h}(\epsilon_n t) dt.$$

If we take $\epsilon_n = 1/n^{1/4}$, it is easy to see that the second term here tends to zero, since $|\mathfrak{f}_n| \leq 1$ and $\mathfrak{h}(t) = O(|t|^{-j})$ as $t \rightarrow \infty$ for any j . The integrand in the first term is dominated by $e^{-t^2/4}/(2\pi)$, an integrable function. So by dominated convergence the first term tends to

$$\frac{1}{2\pi} \int e^{-t^2/2 - itx} dt.$$

This integral equals $\phi(x)$ by (A.8), proving the proposition. $\square$

By Proposition A.13, the densities for $\tilde{Z}_n$ converge pointwise to ϕ . So by Scheffé's theorem, given in Problem 2.19,

$$\tilde{Z}_n \Rightarrow N(0, 1).$$

Because $Z_n = \tilde{Z}_n - \epsilon_n W$ and $\epsilon_n W \xrightarrow{p} 0$, it follows that

$$Z_n \Rightarrow N(0, 1),$$

proving the central limit theorem (Theorem 8.12).

A.7.3 Extensions

The central limit theorem in the last subsection was proved using the inversion formula, smoothing, and Taylor expansion to approximate f_n . The Taylor approximation for f_n can be improved keeping an extra term in the expansion. By (A.6), if $E|X|^3 < \infty$,

$$f(t/\sqrt{n}) = 1 - \frac{t^2}{2n} - \frac{i\gamma t^3}{6\sigma^3 n\sqrt{n}} + o(n^{-3/2}),$$

as $n \rightarrow \infty$, where $\gamma = E(X - \mu)^3$. Because $\log(1 + \epsilon) = \epsilon + O(\epsilon^2)$,

$$\log f_n(t) = n \log f(t/\sqrt{n}) = -\frac{1}{2}t^2 - \frac{i\gamma t^3}{6\sigma^3 \sqrt{n}} + o(1/\sqrt{n}).$$

Finally, since $e^\epsilon = 1 + \epsilon + O(\epsilon^2)$ as $\epsilon \rightarrow 0$,

$$f_n(t) = e^{-t^2/2} \left(1 - \frac{i\gamma t^3}{6\sigma^3 \sqrt{n}} \right) + o(1/\sqrt{n}). \quad (\text{A.12})$$

By Fourier inversion, if we define

$$\begin{aligned} g_n(x) &= \frac{1}{2\pi} \int e^{-itx} e^{-t^2/2} \left(1 - \frac{i\gamma t^3}{6\sigma^3 \sqrt{n}} \right) dt \\ &= \phi(x) - \frac{\gamma \phi'''(x)}{6\sigma^3 \sqrt{n}}, \end{aligned}$$

with the integral in this expression evaluated using (A.9), then the Fourier transform of g_n is the approximation for f_n in (A.12). Integrating g_n , the function G_n , defined as

$$G_n(x) = \int_{-\infty}^x g_n(u) du = \Phi(x) - \frac{\gamma \phi''(x)}{6\sigma^3 \sqrt{n}} = \Phi(x) - \frac{\gamma(x^2 - 1)\phi(x)}{6\sigma^3 \sqrt{n}},$$

is the natural candidate to approximate F_n . There is one technical issue worth noting. Because g_n/ϕ is a cubic polynomial, $g_n(x)$ will be negative for some x (unless $\gamma = 0$), and so G_n may not be nondecreasing. We call a function, such as G_n , that tends to zero at $-\infty$ and one at $+\infty$ a *pseudo-cdf*, and if a pseudo-cdf is differentiable, we define its characteristic function as the Fourier transform of its derivative.

To use (A.12) to argue that G_n is a good approximation for F_n we need the following technical lemma. Although there are technical differences, the approach used to prove this result is similar to that used above to prove the central limit theorem. For details, see Section 16.3 of Feller (1971).

Lemma A.14. *Let F be the cumulative distribution function for a distribution with mean zero and characteristic function f , and let G be a differentiable pseudo-cdf with density $g = G'$ and characteristic function g satisfying $g'(0) = 0$. Then for any $T > 0$,*

$$\|F - G\|_\infty \leq \frac{1}{\pi} \int_{-T}^T \left| \frac{f(t) - g(t)}{t} \right| dt + \frac{24\|g\|_\infty}{\pi T}.$$

Proof of Theorem 19.3. Let us first show that the nonlattice assumption implies $|f(t)| < 1$ for $t \neq 0$. To see this, first note that by (A.5),

$$|f(t)| = |Ee^{itX}| \leq E|e^{itX}| = 1,$$

and we only need to rule out $|f(t)| = 1$, that is, that $f(t) = e^{i\omega}$ for some $\omega \in \mathbb{R}$. But in this case

$$1 = \Re(f(t)e^{-i\omega}) = \Re(Ee^{itX-i\omega}) = E \cos(tX - \omega),$$

and since the cosine function is at most one, this implies

$$P(\cos(tX - \omega) = 1) = P(X = \omega + 2\pi k/t, \exists k \in \mathbb{Z}) = 1,$$

and Q is a lattice distribution.

Next, fix $\epsilon > 0$ and take

$$c = 24 \sup_{n \geq 1} \frac{\|g_n\|_\infty}{\pi \epsilon}.$$

By Lemma A.14 with $T = c\sqrt{n}$,

$$\|F_n - G_n\|_\infty \leq \frac{1}{\pi} \int_{-c\sqrt{n}}^{c\sqrt{n}} \left| \frac{f_n(t) - g_n(t)}{t} \right| dt + \frac{\epsilon}{\sqrt{n}}. \quad (\text{A.13})$$

Because $\epsilon > 0$ is arbitrary, the theorem will follow if the integral in this formula is $o(1/\sqrt{n})$. For some $\delta > 0$, the contribution integrating over $|t| < \delta\sqrt{n}$ will be $o(1/\sqrt{n})$ by dominated convergence and the expansion (A.12). If instead, $|t| \in [\delta\sqrt{n}, c\sqrt{n}]$,

$$|f_n(t)| = |f^n(t/\sqrt{n})| \leq M^n$$

with

$$M = \sup_{|t| \in [\delta, c]} |f(t)|.$$

Since f is continuous and $|f(t)| < 1$ for $t \neq 0$, $M < 1$, and f_n is exponentially small for $\delta\sqrt{n} \leq |t| \leq c\sqrt{n}$. Because g_n is also exponentially small over this region, the contribution to the integral in (A.13) over $|t| \geq \delta\sqrt{n}$ is also $o(1/\sqrt{n})$, and the theorem follows. $\square$

The approximation in Theorem 19.3 is called an *Edgeworth expansion* for F_n . The same method can be used to obtain higher-order expansions. For regularity, higher-order moments of X must be finite to improve the Taylor approximation for f_n . In addition, the nonlattice assumption needs to be strengthened. This occurs because T in Lemma A.14 will need to grow at a

rate faster than $\sqrt{n}$, and exponential decay for f_n over this region can fail. A suitable replacement, due to Cramér, is that

$$\limsup_{|t| \rightarrow \infty} |f(t)| < 1. \quad (\text{A.14})$$

This assumption fails if Q is discrete, but holds when Q is absolutely continuous.⁴ Similar expansions are possible if Q is a lattice distribution. For these results and a derivation of the Berry–Esséen theorem (equation (8.2)) based on Taylor expansion and Lemma A.14, see Feller (1971). For Edgeworth expansions in higher dimensions, see Bhattacharya and Rao (1976).

⁴ The Riemann–Lebesgue lemma asserts that $f(t) \rightarrow 0$ as $|t| \rightarrow \infty$ when Q is absolutely continuous.

B

Solutions

B.1 Problems of Chapter 1

1. If $j < k$, then $B_j \subseteq B_{k-1}$, which implies $B_{k-1}^c \subseteq B_j^c$. Since $A_j \subset B_k$ and $A_k \subset B_{k-1}^c \subseteq B_k^c$, A_j and A_k are disjoint. By induction, $B_n = \bigcup_{j=1}^n A_j$. Also, $\bigcup_{j=1}^{\infty} A_j = B$, for if $x \in B$, $x \in B_n$ for some n , and then $x \in A_j$ for some $j \leq n$. Conversely, if $x \in \bigcup_{j=1}^{\infty} A_j$, $x \in A_n \subset B_n \subset B$ for some n . By countable additivity,

$$\mu(B) = \sum_{j=1}^{\infty} \mu(A_j) = \lim_{n \rightarrow \infty} \sum_{j=1}^n \mu(A_j) = \lim_{n \rightarrow \infty} \mu(B_n).$$

8. If B is the union of the B_i , then $1_B \leq \sum 1_{B_i}$, and by Fubini's theorem, viewing summation as integration against counting measure,

$$\mu(B) = \int 1_B d\mu \leq \int \sum_i 1_{B_i} d\mu = \sum_i \int 1_{B_i} d\mu = \sum_i \mu(B_i).$$

10. a) Let B be an arbitrary set in $\mathcal{B}$. Since $\mu(B) \geq 0$ and $\nu(B) \geq 0$, $\eta(B) = \mu(B) + \nu(B) \geq 0$. Thus $\mu : \mathcal{B} \rightarrow [0, \infty]$. Next, if $B_1, B_2, \dots$ are disjoint sets in $\mathcal{B}$, then

$$\begin{aligned} \eta\left(\bigcup_{i=1}^{\infty} B_i\right) &= \mu\left(\bigcup_{i=1}^{\infty} B_i\right) + \nu\left(\bigcup_{i=1}^{\infty} B_i\right) \\ &= \sum_{i=1}^{\infty} \mu(B_i) + \sum_{i=1}^{\infty} \nu(B_i) = \sum_{i=1}^{\infty} [\mu(B_i) + \nu(B_i)] = \sum_{i=1}^{\infty} \eta(B_i). \end{aligned}$$

Thus η is a measure.

b) To establish the integration identity, suppose f is a simple function: $f = \sum_{i=1}^n c_i 1_{A_i}$. Then

$$\begin{aligned}\int f \, d\eta &= \sum_{i=1}^n c_i \eta(A_i) = \sum_{i=1}^n c_i [\mu(A_i) + \nu(A_i)] \\ &= \sum_{i=1}^n c_i \mu(A_i) + \sum_{i=1}^n c_i \nu(A_i) = \int f \, d\mu + \int f \, d\nu.\end{aligned}$$

So the identity holds for simple functions. For the general case, let f_n be simple functions increasing to f . Then from our definition of the integral,

$$\begin{aligned}\int f \, d\eta &= \lim_{n \rightarrow \infty} \int f_n \, d\eta = \lim_{n \rightarrow \infty} \left\{ \int f_n \, d\mu + \int f_n \, d\nu \right\} \\ &= \lim_{n \rightarrow \infty} \int f_n \, d\mu + \lim_{n \rightarrow \infty} \int f_n \, d\nu = \int f \, d\mu + \int f \, d\nu.\end{aligned}$$

11. By finite additivity,

$$\mu((0, 1/2]) + \mu((1/2, \pi]) = \mu((0, \pi]).$$

Since $\mu((0, 1/2]) = 1/\sqrt{2}$ and $\mu((0, \pi]) = \sqrt{\pi}$,

$$\mu((1/2, \pi]) = \sqrt{\pi} - \frac{1}{\sqrt{2}}.$$

Similarly $\mu((1, 2]) = \sqrt{2} - 1$. Then

$$\int f \, d\mu = \mu((1/2, \pi]) + 2\mu((1, 2]) = \sqrt{\pi} + 2\sqrt{2} - 2 - \frac{1}{\sqrt{2}}.$$

12. The integral is $\int f \, d\mu = 4 + 21\pi$.

13. For $x \in (0, 1]$, let $f_n(x) = \lfloor 2^n x \rfloor / 2^n$, where $\lfloor y \rfloor$ is y rounded down to the nearest integer. If $x \notin (0, 1]$, let $f_n(x) = 0$. So, for instance, $f_1(x)$ is $1/2$ for $x \in [1/2, 1)$, $f_1(1) = 1$, and $f_1(x) = 0$, $x \notin [1/2, 1]$. (Draw a picture of f_2 .) Then

$$\int f_n \, d\mu = \frac{1 + 2 + \cdots + (2^n - 1)}{4^n} = \frac{2^n(2^n - 1)}{2 \times 4^n} \rightarrow \frac{1}{2}.$$

16. Define $B_n = \{X \leq a - 1/n\}$. Then $B_1 \subset B_2 \subset \cdots$. Also, $\{X < a\} = \bigcup_{n=1}^{\infty} B_n$, for if $X(e) < a$, $e \in \{X \leq a - 1/n\}$ for some n , and if $e \in \bigcup_{n=1}^{\infty} B_n$, $e \in \{X \leq a - 1/n\}$ for some n , which implies $e \in \{X < a\}$. Then by Problem 1.1,

$$P(X < a) = \lim_{n \rightarrow \infty} P(X \leq a - 1/n) = \lim_{n \rightarrow \infty} F(a - 1/n) = F(a-).$$

For the second part, $\{X < a\}$ and $\{X = a\}$ are disjoint with union $\{X \leq a\}$, and so $P(X < a) + P(X = a) = P(X \leq a)$; that is, $F_X(a-) + P(X = a) = F_X(a)$.

17. By countable additivity, the chance X is even is

$$P(X = 0) + P(X = 2) + \cdots = \theta + \theta(1 - \theta)^2 + \cdots = \frac{1}{2 - \theta}.$$

18. For the first assertion, an outcome e lies in $X^{-1}(A \cap B)$ if and only if $X(e) \in A \cap B$ if and only if $X(e) \in A$ and $X(e) \in B$ if and only if $e \in X^{-1}(A)$ and $e \in X^{-1}(B)$ if and only if $e \in X^{-1}(A) \cap X^{-1}(B)$. For the third assertion, an outcome e lies in $X^{-1}(\bigcup_{i=1}^{\infty} A_i)$ if and only if $X(e) \in \bigcup_{i=1}^{\infty} A_i$ if and only if $X(e) \in A_i$, for some i , if and only if $e \in X^{-1}(A_i)$, for some i , if and only if $e \in \bigcup_{i=1}^{\infty} X^{-1}(A_i)$. The second assertion follows in the same way.
19. Since $P_X(B) = P(X^{-1}(B)) \geq 0$, we only need to establish countable additivity. Let us first show that $X^{-1}(\bigcup_i B_i) = \bigcup_i X^{-1}(B_i)$. Suppose $e \in X^{-1}(\bigcup_i B_i)$. Then $X(e) \in \bigcup_i B_i$, which implies $X(e) \in B_j$ for some j . But then $e \in X^{-1}(B_j)$, and so $e \in \bigcup_i X^{-1}(B_i)$. Conversely, if $e \in \bigcup_i X^{-1}(B_i)$, then $e \in X^{-1}(B_j)$ for some j , which implies $X(e) \in B_j$, and so $X(e) \in \bigcup_i B_i$. Thus $e \in X^{-1}(\bigcup_i B_i)$. Next, suppose B_i and B_j are disjoint. Then $X^{-1}(B_i)$ and $X^{-1}(B_j)$ are disjoint, for if e lies in both of these sets, $X(e)$ lies in B_i and B_j . Finally, if $B_1, B_2, \dots$ are disjoint Borel sets with union $B = \bigcup_i B_i$, then $X^{-1}(B_1), X^{-1}(B_2), \dots$ are disjoint sets with union $X^{-1}(B)$, and so

$$\sum_i P_X(B_i) = \sum_i P(X^{-1}(B_i)) = P(X^{-1}(B)) = P_X(B).$$

21. The probabilities are all the same, $P(Y_1 = 0, Y_2 = 0) = P(Y_1 = 0, Y_2 = 1) = P(Y_1 = 1, Y_2 = 0) = P(Y_1 = 1, Y_2 = 1) = 1/4$. For instance, $P(Y_1 = 0, Y_2 = 1) = P(1/4 \leq X < 1/2) = 1/4$.
22. First note that

$$\begin{aligned} \{X \in B\} &= \{y \in (0, 1) : X(y) \in B\} \\ &= \begin{cases} B \cap (0, 1/2), & 1/2 \notin B; \\ [B \cap (0, 1/2)] \cup [1/2, 1), & 1/2 \in B. \end{cases} \end{aligned}$$

Let λ be Lebesgue measure, and let ν be counting measure on $\{1/2\}$. Then $\mu(B) = 0$ if and only if $\lambda(B) = 0$ and $1/2 \notin B$. But then $P_X(B) = P(X \in B) = P(B \cap (0, 1/2)) = \lambda(B \cap (0, 1/2)) = 0$, and hence P_X is absolutely continuous w.r.t. $\lambda + \nu$. From the equation above,

$$P(X \in B) = \lambda(B \cap (0, 1/2)) + \frac{1}{2}1_B(1/2).$$

If f is the density, this should equal $\int_B f d(\lambda + \nu)$, and $f = 1_{(0, 1/2)} + \frac{1}{2}1_{\{1/2\}}$ works because

$$\begin{aligned}
\int f 1_B d(\lambda + \nu) &= \int \left(1_{B \cap (0, 1/2)} + \frac{1}{2} 1_{\{1/2\} \cap B} \right) d(\lambda + \nu) \\
&= \lambda(B \cap (0, 1/2)) + \nu(B \cap (0, 1/2)) \\
&\quad + \frac{1}{2} \lambda(\{1/2\} \cap B) + \frac{1}{2} \nu(\{1/2\} \cap B) \\
&= \lambda(B \cap (0, 1/2)) + \frac{1}{2} 1_B(1/2).
\end{aligned}$$

23. Define $g(x) = x 1_{(-1, 1)}(x)$, so that $Y = g(X)$. Integrating against the density of X ,

$$Ef(Y) = Ef(g(X)) = \int f(g(x)) \phi(x) dx = \int_{-1}^1 f(x) \phi(x) dx + cf(0),$$

where $c = \int_{|x| > 1} \phi(x) dx = 2\Phi(-1)$. But integrating against the density p of Y ,

$$\begin{aligned}
Ef(Y) &= \int fp d(\lambda + \nu) = \int fp d\lambda + \int fp d\mu \\
&= \int f(x)p(x) dx + f(0)p(0).
\end{aligned}$$

These expressions must agree for any integrable function f . If $f = 1_{\{0\}}$, this gives $p(0) = 2\Phi(-1)$. And if $f(0) = 0$, we must have $\int f(x)p(x) dx = \int f(x)\phi(x) 1_{(-1, 1)}(x) dx$. This will hold if $p(x) = \phi(x)$ when $0 < |x| < 1$. So the density is $p(x) = 2\Phi(-1) 1_{\{0\}}(x) + \phi(x) 1_{(0, 1)}(|x|)$.

24. The problem is trivial if μ is finite (just divide μ by $\mu(X)$). If μ is infinite but σ -finite, there exist sets $A_1, A_2, \dots$ in $\mathcal{B}$ with $\bigcup_i A_i = X$ and $0 < \mu(A_i) < \infty$. Define truncated measures μ_i , as suggested, by $\mu_i(B) = \mu(B \cap A_i)$. (Routine calculations show that μ_i is indeed a measure.) Note that $\mu_i(X) = \mu(A_i)$, so each μ_i is a finite measure. Let $b_i = 1/2^i$ (or any other sequence of positive constants summing to one) and define $c_i = b_i/\mu(A_i)$. Then $P = \sum_i c_i \mu_i$ is a probability measure since $P(X) = \sum_i c_i \mu_i(X) = \sum_i [b_i/\mu(A_i)] \mu(A_i) = 1$. Suppose $P(N) = \sum_i c_i \mu_i(N) = 0$. Then $\mu_i(N) = \mu(N \cap A_i) = 0$ for all i . By Boole's inequality (Problem 1.8), $\mu(N) = \mu(\bigcup_i [N \cap A_i]) \leq \sum_i \mu(N \cap A_i) = 0$. This shows that any null set for P is a null set for μ , and μ is thus absolutely continuous with respect to P .
25. a) Suppose $f = 1_A$. Then $f(X(e)) = 1$ if and only if $X(e) \in A$, so $f \circ X = 1_B$, where $B = \{e : X(e) \in A\}$. Note that the definition of P_X has $P_X(A) = P(B)$. Now

$$\int f(X(e)) dP(e) = \int 1_B(e) dP(e) = P(B),$$

and

$$\int f(x) dP_X(x) = \int 1_A(x) dP_X(x) = P_X(A).$$

So the equation holds for indicator functions. Next, suppose that f is a simple function: $f = \sum_{i=1}^n c_i 1_{A_i}$. Because integration is linear,

$$\begin{aligned} \int f(X(e)) dP(e) &= \int \sum_{i=1}^n c_i 1_{A_i}(X(e)) dP(e) \\ &= \sum_{i=1}^n c_i \int 1_{A_i}(X(e)) dP(e) = \sum_{i=1}^n c_i \int 1_{A_i}(x) dP_X(x) \\ &= \int \sum_{i=1}^n c_i 1_{A_i}(x) dP_X(x) = \int f(x) dP_X(x). \end{aligned}$$

Finally, let f be an arbitrary nonnegative measurable function, and let f_n be nonnegative simple functions increasing to f . Then $f_n \circ X$ increase to $f \circ X$, and using the monotone convergence theorem twice,

$$\begin{aligned} \int f(X(e)) dP(e) &= \lim_{n \rightarrow \infty} \int f_n(X(e)) dP(e) \\ &= \lim_{n \rightarrow \infty} \int f_n(x) dP_X(x) \\ &= \int f(x) dP_X(x). \end{aligned}$$

b) Using the same general approach, since P_X has density p with respect to μ ,

$$\int 1_A dP_X = P_X(A) = \int_A p d\mu = \int 1_A p d\mu,$$

and the equation holds for indicator functions. Next, if f is a simple function, $f = \sum_{i=1}^n c_i 1_{A_i}$, linearity gives

$$\begin{aligned} \int f dP_X &= \int \sum_{i=1}^n c_i 1_{A_i} dP_X \\ &= \sum_{i=1}^n c_i \int 1_{A_i} dP_X \\ &= \sum_{i=1}^n c_i \int 1_{A_i} p d\mu \\ &= \int \sum_{i=1}^n c_i 1_{A_i} p d\mu \\ &= \int f p d\mu. \end{aligned}$$

For the general case, let f_n be nonnegative simple functions increasing to f . Then $f_n p$ increase to $f p$, and using the monotone convergence theorem twice,

$$\int f dP_X = \lim_{n \rightarrow \infty} \int f_n dP_X = \lim_{n \rightarrow \infty} \int f_n p d\mu = \int f p d\mu.$$

26. a) Integrating e^{-x} and differentiating x^α , integration by parts gives

$$\begin{aligned} \Gamma(\alpha + 1) &= \int_0^\infty x^\alpha e^{-x} dx \\ &= -x^\alpha e^{-x} \Big|_0^\infty + \int_0^\infty \alpha x^{\alpha-1} e^{-x} dx \\ &= \alpha \Gamma(\alpha). \end{aligned}$$

Using this repeatedly, $\Gamma(x+1) = x\Gamma(x) = x(x-1)\Gamma(x-1) = \cdots x(x-1)\cdots 1\Gamma(1)$. But $\Gamma(1) = \int_1^\infty e^{-x} dx = 1$, and so $\Gamma(x+1) = x!$, $x = 0, 1, \dots$

b) The change of variables $u = x/\beta$ (so $dx = \beta du$) gives

$$\int p(x) dx = \int_0^\infty \beta p(\beta u) du = \int_0^\infty \frac{1}{\Gamma(\alpha)} u^{\alpha-1} e^{-u} du = \frac{\Gamma(\alpha)}{\Gamma(\alpha)} = 1.$$

c) The same change of variables gives

$$\begin{aligned} EX^r &= \int x^r p(x) dx = \int_0^\infty \beta^{r+1} u^r p(\beta u) du \\ &= \frac{\beta^r}{\Gamma(\alpha)} \int_0^\infty u^{\alpha+r} e^{-u} du = \frac{\beta^r \Gamma(\alpha + r)}{\Gamma(\alpha)}. \end{aligned}$$

Using this, $EX = \beta\Gamma(\alpha+1)/\Gamma(\alpha) = \beta\alpha$, $EX^2 = \beta^2\Gamma(\alpha+2)/\Gamma(\alpha) = \beta(\alpha+1)\alpha$, and $\text{Var}(X) = EX^2 - (EX)^2 = \alpha\beta^2$.

27. Integrating against the density, $EX^p = \int_0^1 x^p dx = 1/(p+1)$. So $EX = 1/2$, $EX^2 = 1/3$, $\text{Var}(X) = EX^2 - (EX)^2 = 1/3 - 1/4 = 1/12$, $\text{Var}(X^2) = EX^4 - (EX^2)^2 = 1/5 - 1/9 = 4/45$ and $\text{Cov}(X, X^2) = EX^3 - (EX)(EX^2) = 1/4 - 1/6 = 1/12$. So the mean and covariance are

$$E \begin{pmatrix} X \\ X^2 \end{pmatrix} = \begin{pmatrix} 1/2 \\ 1/3 \end{pmatrix} \quad \text{and} \quad \text{Cov} \begin{pmatrix} X \\ X^2 \end{pmatrix} = \begin{pmatrix} 1/12 & 1/12 \\ 1/12 & 4/45 \end{pmatrix}.$$

28. The mean and covariance are

$$E \begin{pmatrix} X \\ I\{X > c\} \end{pmatrix} = \begin{pmatrix} 0 \\ 1 - \Phi(c) \end{pmatrix}$$

and

$$\text{Cov} \begin{pmatrix} X \\ I\{X > c\} \end{pmatrix} = \begin{pmatrix} 1 & \phi(c) \\ \phi(c) & \Phi(c)(1 - \Phi(c)) \end{pmatrix}.$$

32. By Fubini's theorem,

$$\begin{aligned}
\int_0^\infty h(t) dt &= \int_0^\infty E \frac{1 - \cos(tX)}{t^2} dt \\
&= \int_0^\infty \int \frac{1 - \cos(tx)}{t^2} dP_X(x) dt \\
&= \int \int_0^\infty \frac{1 - \cos(tx)}{t^2} dt dP_X(x) \\
&= \int \int_0^\infty |x| \frac{1 - \cos u}{u^2} du dP_X(x) \\
&= \int \frac{\pi}{2} |x| dP_X(x) \\
&= \frac{\pi}{2} E|X|.
\end{aligned}$$

33. The sum is $\sum_{n=1}^\infty c_n = 1$.

36. a) By independence of X and Y , $P(X + Y \leq s | Y = y) = P(X \leq s - y) = F_X(s - y)$, and so, $P(X + Y \leq s | Y) = F_X(s - Y)$. Then

$$F_S(s) = P(X + Y \leq s) = EP(X + Y \leq s | Y) = EF_X(s - Y).$$

b) Using the independence, for $y > 0$,

$$P(XY \leq w | Y = y) = P(X \leq w/y | Y = y) = F_X(w/y).$$

So, $P(XY \leq w | Y) = F_X(w/Y)$ almost surely, and

$$F_W(w) = P(XY \leq w) = EP(XY \leq w | Y) = EF_X(w/Y).$$

37. Note that $p_X(x - Y) = 0$ if $Y \geq x$. The change of variables $u = y/x$ gives

$$\begin{aligned}
p_S(x) &= E \frac{(x - Y)^{\alpha-1} e^{-(x-Y)}}{\Gamma(\alpha)} 1_{(0,x)}(Y) = \int_0^x \frac{(x - y)^{\alpha-1} y^{\beta-1} e^{-x}}{\Gamma(\alpha)\Gamma(\beta)} dy \\
&= \int_0^1 \frac{x^{\alpha+\beta-1} e^{-x} u^{\beta-1} (1 - u)^{\alpha-1}}{\Gamma(\alpha)\Gamma(\beta)} du = \frac{x^{\alpha+\beta-1} e^{-x}}{\Gamma(\alpha + \beta)},
\end{aligned}$$

for $x > 0$. So $X + Y \sim \Gamma(\alpha + \beta, 1)$.

38. a) The mean of the exponential distribution Q_λ is

$$\int x dQ_\lambda(x) = \int x q_\lambda(x) dx = \int_0^\infty x \lambda e^{-\lambda x} dx = \frac{1}{\lambda}.$$

So $E[Y|X = x] = 1/x$ and $E[Y|X] = 1/X$.

b) By smoothing,

$$EY = EE[Y|X] = E(1/X) = \sum_{x=1}^n \frac{1}{x} \frac{2x}{n(n+1)} = \frac{2}{n+1}.$$

39. a) $P[Y > y|X = k] = Q_k((y, \infty)) = \int_y^\infty k e^{-ku} du = e^{-ky}$. So $P(Y > y|X) = e^{-Xy}$.
 b) By smoothing,

$$P(Y > y) = EP(Y > y|X) = Ee^{-Xy} = \sum_{k=1}^n \frac{e^{-ky}}{n} = \frac{1 - e^{-ny}}{n(e^y - 1)}.$$

(The sum here is geometric.)

c) The density is

$$F'_Y(y) = \frac{d}{dy}(1 - P(Y > y)) = \frac{e^y(1 - e^{-ny})}{n(e^y - 1)^2} - \frac{e^{-ny}}{e^y - 1}.$$

B.2 Problems of Chapter 2

1. a) The mass functions can be written as $\exp[\log(1-p)x + \log(p)]$, which has exponential family form.
 b) The canonical parameter is $\eta = \log(1-p)$. Solving, $p = 1 - e^\eta$, and so the mass function in canonical form is $\exp[\eta x + \log(1 - e^\eta)]$, with $A(\eta) = -\log(1 - e^\eta)$.
 c) Since $T = X$, by (2.4), $EX = A'(\eta) = e^\eta/(1 - e^\eta) = (1-p)/p$.
 d) The joint mass functions are

$$\prod_{i=1}^n [p(1-p)^{x_i}] = p^n (1-p)^{T(x)},$$

where $T(x) = \sum_i x_i$. With the same definition for η , this can be rewritten as

$$\exp[\log(1-p)T(x) + n \log(p)] = \exp[\eta T(x) + n \log(1 - e^\eta)].$$

Now $A(\eta) = -n \log(1 - e^\eta)$, and so

$$ET = A'(\eta) = \frac{ne^\eta}{1 - e^\eta} = \frac{n(1-p)}{p},$$

and

$$\text{Var}(T) = A''(\eta) = \frac{ne^\eta}{(1 - e^\eta)^2} = \frac{n(1-p)}{p^2}.$$

2. From the definition,

$$e^{A(\eta)} = \iint h(x, y) e^{\eta xy} dx dy = \frac{1}{\sqrt{2\pi}} \int \exp\left[-\frac{1}{2}(1 - \eta^2)y^2\right] dy.$$

This integral is finite if and only if $|\eta| < 1$, and so $\Xi = (-1, 1)$. Doing the integral, $e^{A(\eta)} = 1/\sqrt{1 - \eta^2}$ and the densities are $\exp[\eta xy + \log(1 - \eta^2)/2] h(x, y)$.

4. The parameter space is $\Xi = (-1, \infty)$, and the densities are

$$p_\eta(x) = \frac{1}{2}(\eta+1)(\eta+2)(\eta+3)x^\eta(1-x)^2, \quad x \in (0, 1).$$

5. The integral (below) defining $A(\eta)$ is finite if and only if $\eta \geq 0$, and so $\Xi = [0, \infty)$. To evaluate the integral, let $y = \sqrt{x}$. Then

$$\begin{aligned} e^{A(\eta)} &= \int_0^\infty e^{-\eta x - 2\sqrt{x}} \frac{dx}{\sqrt{x}} = 2 \int_0^\infty e^{-\eta y^2 - 2y} dy \\ &= \frac{2\sqrt{\pi}e^{1/\eta}}{\sqrt{\eta}} \int_0^\infty \frac{\exp\left[-\frac{(y+1/\eta)^2}{2[1/(2\eta)]}\right]}{\sqrt{2\pi[1/(2\eta)]}} dy. \end{aligned}$$

If $Y \sim N(-1/\eta, 1/(2\eta))$, then the integral here is $P(Y > 0) = \Phi(-\sqrt{2/\eta})$. So

$$A(\eta) = \log \sqrt{4\pi} + 1/\eta - \log \sqrt{\eta} + \log \Phi(-\sqrt{2/\eta}),$$

and $p_\eta(x) = \exp[-2\sqrt{x} - \eta x - A(\eta)]/\sqrt{x}$, $x > 0$. Then

$$E_\eta X = -E_\eta T = -A'(\eta) = \frac{1}{\eta^2} + \frac{1}{2\eta} - \frac{\phi(\sqrt{2/\eta})}{\Phi(-\sqrt{2/\eta})\sqrt{2\eta^3}}.$$

6. The parameter space is $\Xi = \mathbb{R}^2$, and the densities are

$$p_\eta(x) = \frac{e^{\eta_1 x + \eta_2 x^2}}{1 + e^{\eta_1 + \eta_2} + e^{2\eta_1 + 4\eta_2}}, \quad x = 0, 1, 2.$$

7. The joint densities are

$$\prod_{i=1}^n [1 - e^{\alpha + \beta t_i}] [e^{\alpha + \beta t_i}]^{x_i} = \exp \left[\alpha \sum_{i=1}^n x_i + \beta \sum_{i=1}^n x_i t_i + \sum_{i=1}^n \log [1 - e^{\alpha + \beta t_i}] \right],$$

forming an exponential family with $T_1 = \sum_{i=1}^n X_i$ and $T_2 = \sum_{i=1}^n t_i X_i$.

8. The joint densities are

$$\begin{aligned} &\prod_{i=1}^n \frac{1}{\sqrt{2\pi}} \exp \left[-\frac{1}{2}(x_i - \alpha - \beta t_i)^2 \right] \\ &= \frac{1}{\sqrt{2\pi}^n} \exp \left[\alpha \sum_{i=1}^n x_i + \beta \sum_{i=1}^n t_i x_i - \frac{1}{2} \sum_{i=1}^n x_i^2 - \frac{1}{2} \sum_{i=1}^n (\alpha + \beta t_i)^2 \right], \end{aligned}$$

which is a two-parameter exponential family with $T_1 = \sum_{i=1}^n X_i$ and $T_2 = \sum_{i=1}^n t_i X_i$.

9. Since $P(X_i = x_i) = \exp(\alpha x_i + \beta t_i x_i) / (1 + \exp(\alpha + \beta t_i))$, the joint densities are

$$\prod_{i=1}^n \frac{\exp(\alpha x_i + \beta t_i x_i)}{1 + \exp(\alpha + \beta t_i)} = \frac{\exp[\alpha \sum_{i=1}^n x_i + \beta \sum_{i=1}^n t_i x_i]}{\prod_{i=1}^n (1 + \exp(\alpha + \beta t_i))},$$

which form a two-parameter exponential family with $T_1 = \sum_{i=1}^n X_i$ and $T_2 = \sum_{i=1}^n t_i X_i$.

15. Differentiating the identity

$$e^{B(\theta)} = \int \exp\{\eta(\theta) \cdot T(x)\} h(x) d\mu(x)$$

with respect to θ_i under the integral sign, gives

$$e^{B(\theta)} \frac{\partial B(\theta)}{\partial \theta_i} = \int \sum_{j=1}^s \frac{\partial \eta_j(\theta)}{\partial \theta_i} T_j(x) \exp\{\eta(\theta) \cdot T(x)\} h(x) d\mu(x).$$

Division by $e^{B(\theta)}$ then gives

$$\frac{\partial B(\theta)}{\partial \theta_i} = \sum_{j=1}^s \frac{\partial \eta(\theta)_j}{\partial \theta_i} E_\theta T_j, \quad i = 1, \dots, s.$$

(This also follows from the chain rule because $B(\theta) = A(\eta(\theta))$ and $E T_j = \partial A(\eta)/\partial \eta_j$.) These equations can be written as $\nabla B(\theta) = D\eta(\theta)' E_\theta T$, where $D\eta(\theta)$ denotes an $s \times s$ matrix with (i, j) th entry $\partial \eta_i(\theta)/\partial \theta_j$. Solving, $E_\theta T = [D\eta(\theta)']^{-1} \nabla B(\theta)$.

17. a) Let $f_n(k) = f(k)$ for $k \leq n$, and $f_n(k) = 0$ for $k > n$. Then $f_n \rightarrow f$ pointwise, and $|f_n| \leq |f|$. Note that f_n is a simple function and that $\int f_n d\mu = \sum_{k=1}^n f(k)$. The dominated convergence theorem gives $\int f_n d\mu \rightarrow \int f d\mu$, which is the desired result.

b) Define f_n as in part (a). Then $f_n \uparrow f$, and so $\sum_{k=1}^n f(k) = \int f_n d\mu \rightarrow \int f d\mu$ by the monotone convergence theorem.

c) Begin the g -sequence taking positive terms from the f -sequence until the sum exceeds K . Then, take negative terms from the f sequence until the sum is below K . Then take positive terms again, and so on. Because the summands tend to zero, the partial sums will have limit K .

19. a) Since $\int (p - p_n) d\mu = \int (p - p_n)^+ d\mu - \int (p - p_n)^- d\mu = 0$, we have $\int (p - p_n)^+ d\mu = \int (p - p_n)^- d\mu$. Using $|p_n - p| = |p - p_n| = (p - p_n)^+ + (p - p_n)^-$,

$$\begin{aligned} \int |p_n - p| d\mu &= \int (p - p_n)^+ d\mu + \int (p - p_n)^- d\mu \\ &= 2 \int (p - p_n)^+ d\mu. \end{aligned}$$

But $|(p - p_n)^+| \leq p$, which is an integrable function, and $(p(x) - p_n(x))^+ \rightarrow 0$. So by dominated convergence

$$\int |p_n - p| d\mu = 2 \int (p - p_n)^+ d\mu \rightarrow 0.$$

b) Since $P_n(A) = \int 1_A p_n d\mu$ and $P(A) = \int 1_A p d\mu$,

$$\begin{aligned} |P_n(A) - P(A)| &= \left| \int 1_A (p_n - p) d\mu \right| \\ &\leq \int 1_A |p_n - p| d\mu \leq \int |p_n - p| d\mu. \end{aligned}$$

22. Because $p_\theta(x) = \phi(x) \exp[\theta x - \log \Phi(\theta) - \theta^2/2]$, we have a canonical exponential family with $A(\theta) = \log \Phi(\theta) + \theta^2/2$. So $M_X(u) = \exp[A(\theta+u) - A(\theta)] = \Phi(\theta+u) \exp(u\theta + u^2/2)/\Phi(\theta)$. Also, $E_\theta X = A'(\theta) = \theta + \phi(\theta)/\Phi(\theta)$ and $\text{Var}_\theta(X) = A''(\theta) = 1 - \theta\phi(\theta)/\Phi(\theta) - \phi^2(\theta)/\Phi^2(\theta)$.
23. The exponential family $N(0, \sigma^2)$ has densities

$$\frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{x^2}{2\sigma^2}\right] = \frac{1}{\sqrt{2\pi}} \exp\left[\eta x^2 + \frac{1}{2} \log(-2\eta)\right],$$

where $\eta = -1/(2\sigma^2)$. So $A(\eta) = -\frac{1}{2} \log(-2\eta)$ and $K_T(u) = -\frac{1}{2} \log[-2(\eta+u)] + \frac{1}{2} \log(-2\eta)$, which simplifies to $-\frac{1}{2} \log(1-2u)$ when $\eta = -1/2$ (or $\sigma^2 = 1$). Then

$$\begin{aligned} K'_T(u) &= \frac{1}{1-2u}, & K''_T(u) &= \frac{2}{(1-2u)^2}, \\ K'''_T(u) &= 8(1-2u)^3, & K''''_T(u) &= \frac{48}{(1-2u)^4}, \end{aligned}$$

and so the first four cumulants of $T \sim Z^2$ are 1, 2, 8, and 48.

24. The first four cumulants of $T = XY$ are 0, 1, 0, and 4.
25. From the last part of Problem 2.1 with $n = 1$, the first two cumulants are $\kappa_1 = A'(\eta) = (1-p)/p$ and $\kappa_2 = A''(\eta) = (1-p)/p^2$. Because $p = p(\eta) = 1 - e^\eta$, $p' = -e^\eta = p - 1$. Using this,

$$\kappa_3 = A'''(\eta) = \frac{-2p'}{p^3} + \frac{p'}{p^2} = \frac{2}{p^3} - \frac{3}{p^2} + \frac{1}{p},$$

and

$$\kappa_4 = A''''(\eta) = \frac{6p'}{p^4} - \frac{6p'}{p^3} + \frac{p'}{p^2} = \frac{6}{p^4} - \frac{12}{p^3} + \frac{7}{p^2} - \frac{1}{p}.$$

26. The third cumulant is

$$\kappa_3 = np(1-p)(1-2p),$$

and the third moment is

$$EX^3 = np(1-p)(1-2p) + 3n^2p^2(1-p) + n^3p^3.$$

27. a) For notation, let $f_{ij}(u) = \partial^{i+j} f(u) / (\partial u_1^i \partial u_2^j)$, and let M and K be the moment generating function and cumulant generating function for T , so that $K = \log M$. Taking derivatives,

$$K_{10} = \frac{M_{10}}{M}, \quad K_{11} = \frac{M_{11}M - M_{10}M_{01}}{M^2},$$

and

$$K_{21} = \frac{M_{21}M^2 - M_{20}M_{01}M - 2M_{11}M_{10}M + 2M_{10}^2M_{01}}{M^3}.$$

At zero we get

$$\kappa_{2,1} = ET_1^2T_2 - (ET_1^2)(ET_2) - 2(ET_1T_2)(ET_1) + 2(ET_1)^2(ET_2).$$

b) Taking one more derivative,

$$K_{22} = \frac{M_{22}M^3 - 2M_{21}M_{01}M^2 - 2M_{12}M_{10}M^2 - M_{20}M_{02}M^2 + 2M_{20}M_{01}^2M + 2M_{02}M_{10}^2M - 2M_{11}M_{11}M^2 + 8M_{11}M_{10}M_{01}M - 6M_{10}^2M_{01}^2}{M^4}.$$

Since $M_{10}(0) = ET_1 = 0$ and $M_{01}(0) = ET_2 = 0$, at zero we get

$$\kappa_{22} = ET_1^2T_2^2 - (ET_1^2)(ET_2^2) - 2(ET_1T_2)^2.$$

28. Taking $\eta = (-\lambda, \alpha)$, X has density

$$\frac{\exp[\eta_1 T_1(X) + \eta_2 T_2(x) - \log \Gamma(\eta_2) + \eta_2 \log(-\eta_1)]}{x}, \quad x > 0.$$

These densities form a two-parameter exponential family with cumulant generating function $A(\eta) = \log \Gamma(\eta_2) - \eta_2 \log(-\eta_1)$. The cumulants of T are derivatives of A :

$$\begin{aligned}
\kappa_{10} &= \frac{\partial A(\eta)}{\partial \eta_1} = -\frac{\eta_2}{\eta_1} = \frac{\alpha}{\lambda}, \\
\kappa_{01} &= \frac{\partial A(\eta)}{\partial \eta_2} = \psi(\eta_2) - \log(-\eta_1) = \psi(\alpha) - \log \lambda, \\
\kappa_{20} &= \frac{\partial^2 A(\eta)}{\partial \eta_1^2} = \frac{\eta_2}{\eta_1^2} = \frac{\alpha}{\lambda^2}, \\
\kappa_{11} &= \frac{\partial^2 A(\eta)}{\partial \eta_1 \partial \eta_2} = -\frac{1}{\eta_1} = \frac{1}{\lambda}, \\
\kappa_{02} &= \frac{\partial^2 A(\eta)}{\partial \eta_2^2} = \psi'(\eta_2) = \psi'(\alpha), \\
\kappa_{30} &= \frac{\partial^3 A(\eta)}{\partial \eta_1^3} = -\frac{2\eta_2}{\eta_1^3} = \frac{2\alpha}{\lambda^3}, \\
\kappa_{21} &= \frac{\partial^3 A(\eta)}{\partial \eta_1^2 \partial \eta_2} = \frac{1}{\eta_1^2} = \frac{1}{\lambda^2}, \\
\kappa_{12} &= \frac{\partial^3 A(\eta)}{\partial \eta_1 \partial \eta_2^2} = 0, \\
\kappa_{03} &= \frac{\partial^3 A(\eta)}{\partial \eta_2^3} = \psi''(\eta_2) = \psi''(\alpha).
\end{aligned}$$

B.3 Problems of Chapter 3

2. The marginal density of X_i is $t_i \theta x_i^{t_i \theta} / x$, $x \in (0, 1)$. Multiplying these together, the joint density is

$$\theta^n \left(\prod_{i=1}^n \frac{t_i}{x_i} \right) \left(\prod_{i=1}^n x_i^{t_i} \right)^\theta.$$

By the factorization theorem, $T = \prod_{i=1}^n X_i^{t_i}$ is sufficient. An equivalent sufficient statistic is $\sum_{i=1}^n t_i \log X_i$.

3. The joint densities are

$$\exp \left[\mu \sum_{i=1}^n \frac{x_i}{\sigma_i^2} - \sum_{i=1}^n \frac{x_i^2 + \mu^2}{2\sigma_i^2} - \sum_{i=1}^n \log \sigma_i - n \log \sqrt{2\pi} \right],$$

and $T = \sum_{i=1}^n X_i / \sigma_i^2$ is sufficient by the factorization theorem. The weighted average $T / \sum_{i=1}^n \sigma_i^{-2}$ is a natural estimator for θ .

4. By independence, the joint mass functions are

$$P(X_1 = x_1, \dots, X_n = x_n) = p_1^{n_1(x)} p_2^{n_2(x)} p_3^{n_3(x)},$$

where $p_i = P(\{i\})$, $i = 1, 2, 3$, and $n_i(x) = \#\{j : x_j = i\}$. Since $n_1(x) + n_2(x) + n_3(x) = n$, we can write the joint mass functions as $p_1^{n_1(x)} p_2^{n_2(x)} p_3^{n-n_1(x)-n_2(x)}$, and $T = (n_1, n_2)$ is sufficient by the factorization theorem.

6. a) The joint densities are

$$\exp \left[(\alpha - 1) \sum_{i=1}^n \log x_i + (\beta - 1) \sum_{i=1}^n \log(1 - x_i) + n \log \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} \right],$$

a full rank exponential family with

$$T = \left(\sum_{i=1}^n \log X_i, \sum_{i=1}^n \log(1 - X_i) \right)$$

a minimal sufficient statistic.

- b) Now the joint densities are

$$\exp \left[(\beta - 1) \sum_{i=1}^n \log [x_i^2(1 - x_i)] + \sum_{i=1}^n \log x_i + n \log \frac{\Gamma(3\beta)}{\Gamma(2\beta)\Gamma(\beta)} \right],$$

a full rank exponential family with minimal sufficient statistic

$$\sum_{i=1}^n \log [x_i^2(1 - x_i)] = 2T_1 + T_2.$$

- c) The densities, parameterized by β , are

$$p_\beta(x) = \exp \left[(\beta^2 - 1)T_1(x) + (\beta - 1)T_2(x) + n \log \frac{\Gamma(\beta + \beta^2)}{\Gamma(\beta)\Gamma(\beta^2)} \right].$$

Suppose $p_\beta(x) \propto_\beta p_\beta(y)$. Then

$$\frac{p_2(x)}{p_1(x)} = \frac{p_2(y)}{p_1(y)} \quad \text{and} \quad \frac{p_3(x)}{p_1(x)} = \frac{p_3(y)}{p_1(y)}.$$

Taking the logarithm of these and using the formula for p_β ,

$$3T_1(x) + T_2(x) + n \log 20 = 3T_1(y) + T_2(y) + n \log 20,$$

and

$$8T_1(x) + 2T_2(x) + n \log 495 = 8T_1(y) + 2T_2(y) + n \log 495.$$

These equations imply $T(x) = T(y)$, and T is minimal sufficient by Theorem 3.11.

7. The statistic $T = (\sum_{i=1}^n X_i, \sum_{i=1}^n t_i X_i)$, is minimal sufficient.

8. a) The statistic (N_{11}, N_{12}, N_{21}) is minimal sufficient. (The statistic $(N_{11}, N_{12}, N_{21}, N_{22})$ is also minimal sufficient.)
 b) With the constraint, $(N_{11} + N_{12}, N_{11} + N_{21})$ is minimal sufficient.
9. a) The joint density is zero unless $x_i > \theta$, $i = 1, \dots, n$, that is, unless $M(x) = \min\{x_1, \dots, x_n\} > \theta$. Using this, the joint density can be written as

$$p_\theta(x) = c^n(\theta) I\{M(x) > \theta\} \prod_{i=1}^n f(x_i),$$

and $M(X)$ is sufficient by the factorization theorem.

- b) If $p_\theta(x) \propto_\theta p_\theta(y)$, then the region where the two functions are zero must agree, and $M(x)$ must equal $M(y)$. So M is minimal sufficient by Theorem 3.11.
10. The joint densities are $p_\theta(x) = 2^{-n} \prod (1 + \theta x_i) = 2^{-n} \prod (1 + \theta x_{(i)})$, where $x_{(1)} \leq \dots \leq x_{(n)}$ are the ordered values. Note that p_θ is a polynomial in θ with degree n , with roots $-1/x_{(i)}$. Suppose $p_\theta(x) \propto_\theta p_\theta(y)$. Then these polynomials must have the same roots, and we must have $x_{(i)} = y_{(i)}$, $i = 1, \dots, n$. So the order statistics are minimal sufficient by Theorem 3.11.
16. a) Let $T(x) = \max\{x_1, \dots, x_n\}$, and $M(x) = \min\{x_1, \dots, x_n\}$. Then the joint density will be positive if and only if $M(x) > 0$ and $T(x) < \theta$. Introducing suitable indicator functions, the joint density can be written $\prod_{i=1}^n (2x_i) I\{M(x) > 0\} I\{T(x) < \theta\} / \theta^{2n}$. So $T = T(X)$ is sufficient by the factorization theorem.
- b) For $t \in (0, \theta)$, $P(X_i \leq t) = \int_0^t 2x dx / \theta^2 = t^2 / \theta^2$. So $P(T \leq t) = P(X_1 \leq t, \dots, X_n \leq t) = P(X_1 \leq t) \times \dots \times P(X_n \leq t) = t^{2n} / \theta^{2n}$. Taking the derivative of this with respect to t , T has density $2nt^{2n-1} / \theta^{2n}$, $t \in (0, \theta)$.
- c) Suppose $E_\theta f(T) = c$, for all $\theta > 0$. Then $\int_0^\theta f(t) 2nt^{2n-1} dt / \theta^{2n} = c$, which implies $\int_0^\theta t^{2n-1} f(t) dt = c\theta^{2n} / (2n)$, for all $\theta > 0$. Taking a derivative with respect to θ , $\theta^{2n-1} f(\theta) = c\theta^{2n-1}$, for a.e. θ , and hence $f(t) = c$, for a.e. t .
17. a) If $y > 0$, then $P(Y \leq y) = P(\lambda X \leq y) = \int_0^{y/\lambda} \lambda e^{-\lambda x} dx = 1 - e^{-y}$. So Y has density $d(1 - e^{-y})/dy = e^{-y}$, $y > 0$, the standard exponential density.
- b) The joint densities are $\lambda^n \exp\{-n\lambda \bar{x}\}$, a full rank exponential family with $T = \bar{X}$ a complete sufficient statistic. Let $Y_i = \lambda X_i$, so that regardless of the value of λ , $Y_1, \dots, Y_n$ are i.i.d. from the standard exponential distribution. Then $(X_1^2 + \dots + X_n^2) / \bar{X}^2 = (Y_1^2 + \dots + Y_n^2) / \bar{Y}^2$ is ancillary, and independence follows by Basu's theorem.
29. $f(x) = 1/(1+x)$ is bounded and convex on $(0, \infty)$.
30. Because $\eta_0 < \eta < \eta_1$, $\eta = \gamma\eta_0 + (1-\gamma)\eta_1$ for some $\gamma \in (0, 1)$, and because the exponential function is convex,

$$e^{\eta T(x)} < \gamma e^{\eta_0 T(x)} + (1-\gamma) e^{\eta_1 T(x)}.$$

Multiplying by $h(x)$ and integrating against μ ,

$$\int e^{\eta T(x)} h(x) d\mu(x) < \gamma \int e^{\eta_0 T(x)} h(x) d\mu(x) + (1-\gamma) \int e^{\eta_1 T(x)} h(x) d\mu(x).$$

From the definition of Ξ , the upper bound here is finite, and η must then also lie in Ξ .

31. Suppose X is absolutely continuous with density f , and define $Y = g(X)/f(X)$. Then

$$EY = \int \frac{g(x)}{f(x)} f(x) dx = \int g(x) dx = 1.$$

The function $h(y) = -\log y = \log(1/y)$ is strictly convex on $(0, \infty)$ (its second derivative is $1/y^2$). So by Jensen's inequality,

$$Eh(Y) = E \log \left[\frac{f(X)}{g(X)} \right] = \int \log \left[\frac{f(x)}{g(x)} \right] f(x) dx \geq h(EY) = -\log 1 = 0.$$

The inequality is strict unless Y is constant a.e. If Y is constant a.e., then $Y = EY = 1$ and $f(x) = g(x)$ a.e.

B.4 Problems of Chapter 4

1. a) The joint densities form a two-parameter exponential family with $T = (T_x, T_y) = (\sum_{i=1}^m X_i, \sum_{j=1}^n Y_j)$ as a complete sufficient statistic. Since T_x has a gamma distribution,

$$ET_x^{-1} = \frac{\lambda_x^m}{\Gamma(m)} \int_0^\infty t^{m-2} e^{-\lambda_x t} dt = \frac{\lambda_x}{m-1}.$$

Also, $ET_y = n/\lambda_y$. So $(m-1)T_y/(nT_x)$ is unbiased for λ_x/λ_y and must be UMVU since it is a function of T .

b) Integrating against the gamma density,

$$ET_x^{-2} = \frac{\lambda_x^m}{\Gamma(m)} \int_0^\infty t^{m-3} e^{-\lambda_x t} dt = \frac{\lambda_x^2}{(m-1)(m-2)}.$$

Also, $ET_y^2 = (ET_y)^2 + \text{Var}(T_y) = n(n+1)/\lambda_x^2$. So

$$\begin{aligned} E \left(d \frac{T_y}{T_x} - \frac{\lambda_x}{\lambda_y} \right)^2 &= d^2 E \frac{T_y^2}{T_x^2} - 2d \frac{\lambda_x}{\lambda_y} E \frac{T_y}{T_x} + \frac{\lambda_x^2}{\lambda_y^2} \\ &= \left(d^2 \frac{n(n+1)}{(m-1)(m-2)} - 2d \frac{n}{m-1} + 1 \right) \frac{\lambda_x^2}{\lambda_y^2}, \end{aligned}$$

which is minimized taking $d = (m-2)/(n+1)$. So the best multiple of $\bar{Y}/\bar{X}$ is $(m-2)T_y/[(n+1)T_x] = \frac{n(m-2)}{(n+1)m} \bar{Y}/\bar{X}$.

c) Since $\delta = I\{X_1 > 1\}$ is evidently unbiased, the UMVU estimator must

be $E[\delta|T]$, and by independence this must be $P(X_1 > 1|T_x]$. The joint density of X_1 and $S = X_2 + \cdots + X_m$ is $g(x, s) = \lambda_x^m s^{m-2} e^{-\lambda_x(s+x)} / (m-2)!$, $s > 0$ and $x > 0$. From this, the joint density of X_1 and T_x is $f(x, t) = \lambda_x^m (t-x)^{m-2} e^{-\lambda_x t} / (m-2)!$, $0 < x < t$. Dividing by the marginal density of T_x , $\lambda_x^m t^{m-1} e^{-\lambda_x t} / (m-1)!$, the conditional density of X given $T = t$ is $f(x|t) = (m-1)(1-x/t)^{m-2}/t$, $0 < x < t$. Integrating this conditional density, $P(X_1 > 1|T_x] = I\{T_x \geq 1\}(1 - 1/T_x)^{m-1}$.

2. a) The joint densities are

$$\frac{\exp \left[\frac{\mu_x \sum_{i=1}^n x_i}{\sigma^2} + \frac{\mu_y \sum_{j=1}^m y_j}{2\sigma^2} - \frac{2 \sum_{i=1}^n x_i^2 + \sum_{j=1}^m y_j^2}{4\sigma^2} - \frac{2n+m}{4\sigma^2} \right]}{(2\pi\sigma^2)^{n/2} (4\pi\sigma^2)^{m/2}},$$

a full rank exponential family with complete sufficient statistic $T = (\sum_{i=1}^n X_i, \sum_{j=1}^m Y_j, 2 \sum_{i=1}^n X_i^2 + \sum_{j=1}^m Y_j^2)$.

b) Expanding the squares and simplifying, $2(n-1)S_x^2 + (m-1)S_y^2 = 2 \sum_{i=1}^n X_i^2 + \sum_{j=1}^m Y_j^2 - 2n\bar{X}^2 - m\bar{Y}^2$ is a function of T with mean $2(n+m-2)\sigma^2$. So $S_p^2 = [2(n-1)S_x^2 + (m-1)S_y^2]/(2n+2m-4)$ is UMVU for σ^2 .

c) Because $E(\bar{X} - \bar{Y})^2 = (\mu_x - \mu_y)^2 + \sigma^2/n + 2\sigma^2/m$, $(\bar{X} - \bar{Y})^2 - (1/n + 2/m)S_p^2$ is UMVU for $(\mu_x - \mu_y)^2$.

d) With the additional constraint, $(2 \sum_{i=1}^n X_i + 3 \sum_{j=1}^m Y_j, 2 \sum_{i=1}^n X_i^2 + \sum_{j=1}^m Y_j^2)$ is complete sufficient. The first statistic here has mean $(2n+9m)\mu_x$, so $(2 \sum_{i=1}^n X_i + 3 \sum_{j=1}^m Y_j)/(2n+9m)$ is the UMVU estimator of μ_x .

3. The joint mass functions are

$$\frac{\lambda^{x_1+\cdots+x_n} e^{-n\lambda}}{x_1! \times \cdots \times x_n!}.$$

These densities form a full rank exponential family with $T = X_1 + \cdots + X_n$ as a complete sufficient statistic. Since T has a Poisson distribution with mean $n\lambda$, $\delta(T)$ will be an unbiased estimator of $\cos \lambda$ if

$$\sum_{t=0}^{\infty} \frac{\delta(t)(n\lambda)^t e^{-n\lambda}}{t!} = \cos \lambda$$

or if

$$\begin{aligned} \sum_{t=0}^{\infty} \frac{\delta(t)n^t}{t!} \lambda^t &= e^{n\lambda} \cos \lambda = \frac{e^{(n+i)\lambda} + e^{(n-i)\lambda}}{2} \\ &= \sum_{t=0}^{\infty} \frac{(n+i)^t + (n-i)^t}{2t!} \lambda^t. \end{aligned}$$

Equating coefficients of λ^t in these expansions,

$$\delta(t) = \frac{1}{2} \left[\left(1 + \frac{i}{n}\right)^t + \left(1 - \frac{i}{n}\right)^t \right] = \left(1 + \frac{1}{n^2}\right)^{t/2} \cos(t\omega),$$

where $\omega = \arctan(1/n)$.

4. The joint densities are

$$\frac{1}{(2\pi)^{n/2}} \exp \left[\alpha \sum_{i=1}^n t_i x_i + \beta \sum_{i=1}^n t_i^2 x_i - \frac{1}{2} \sum_{i=1}^n x_i^2 - \frac{1}{2} \sum_{i=1}^n (\alpha t_i + \beta t_i^2)^2 \right].$$

These densities form a full rank exponential family with

$$T = \left(\sum_{i=1}^n t_i X_i, \sum_{i=1}^n t_i^2 X_i \right)$$

a complete sufficient statistic. Now

$$ET_1 = \alpha \sum_{i=1}^n t_i^2 + \beta \sum_{i=1}^n t_i^3 \quad \text{and} \quad ET_2 = \alpha \sum_{i=1}^n t_i^3 + \beta \sum_{i=1}^n t_i^4.$$

Using these,

$$\frac{T_1 \sum_{i=1}^n t_i^4 - T_2 \sum_{i=1}^n t_i^3}{\sum_{i=1}^n t_i^2 \sum_{i=1}^n t_i^4 - (\sum_{i=1}^n t_i^3)^2} \quad \text{and} \quad \frac{T_2 \sum_{i=1}^n t_i^2 - T_1 \sum_{i=1}^n t_i^3}{\sum_{i=1}^n t_i^2 \sum_{i=1}^n t_i^4 - (\sum_{i=1}^n t_i^3)^2}$$

are unbiased estimators for α and β . Since they are functions of the complete sufficient statistic T , they are UMVU.

5. a) Expanding the quadratic, $S^2 = (\sum_{i=1}^n X_i^2 - n\bar{X}^2)/(n-1)$. If we let $\mu(\theta) = E_\theta X_i$, then $E_\theta X_i^2 = \mu^2(\theta) + \sigma^2(\theta)$, and $E_\theta \bar{X} = \mu^2(\theta) + \sigma^2(\theta)/n$. So

$$E_\theta S^2 = \frac{1}{n-1} \left(\sum_{i=1}^n [\mu^2(\theta) + \sigma^2(\theta)] - n \left[\mu^2(\theta) + \frac{1}{n} \sigma^2(\theta) \right] \right) = \sigma^2(\theta).$$

- b) If X_i is Bernoulli, then $X_i = X_i^2$ and

$$S^2 = \frac{\sum_{i=1}^n X_i - n\bar{X}^2}{n-1} = \frac{n\bar{X}(1-\bar{X})}{n-1}.$$

The joint mass functions form a full rank exponential family with $\bar{X}$ as a complete sufficient statistic. Since δ is unbiased and is a function of $\bar{X}$, δ is UMVU.

- c) Again we have a full rank exponential family with $\bar{X}$ as a complete sufficient statistic. Because $EX_i = 1/\theta$ and $\text{Var}(X_i) = 1/\theta^2$, $E\bar{X}^2 = \theta^{-2} +$

θ^{-2}/n , and $n\bar{X}^2/(n+1)$ is an unbiased estimator of σ^2 . This estimator is UMVU because it is a function of the complete sufficient statistic. Next, by symmetry $E_\theta[X_1^2|\bar{X} = c] = \cdots = E_\theta[X_n^2|\bar{X} = c]$. The UMVU estimator must equal $E_\theta[\delta|\bar{X}]$, and therefore

$$\frac{nc^2}{n+1} = E_\theta \left[\frac{\sum_{i=1}^n X_i^2 - n\bar{X}^2}{n-1} \mid \bar{X} = c \right] = \frac{nE_\theta[X_i^2|\bar{X} = c] - nc^2}{n-1}.$$

From this,

$$E_\theta[X_i^2|\bar{X} = c] = \frac{2nc^2}{n+1}.$$

6. Because $\delta + cU$ is unbiased and δ is UMVU, for any θ ,

$$\text{Var}_\theta(\delta + cU) = \text{Var}_\theta(\delta) + 2c \text{Cov}_\theta(\delta, U) + c^2 \text{Var}_\theta(U) \geq \text{Var}_\theta(\delta).$$

So $h(c) = c^2 \text{Var}_\theta(U) + 2c \text{Cov}_\theta(\delta, U) \geq 0$. Since $h(0) = 0$, this will hold for all c only if $h'(0) = 2 \text{Cov}_\theta(\delta, U) = 0$.

7. Suppose δ is unbiased for $g_1(\theta) + g_2(\theta)$, and that $\text{Var}_\theta(\delta) < \infty$. Then $U = \delta - \delta_1 - \delta_2$ is an unbiased estimator of zero. By Problem 4.6, $\text{Cov}_\theta(U, \delta_1 + \delta_2) = \text{Cov}_\theta(U, \delta_1) + \text{Cov}_\theta(U, \delta_2) = 0$. Since these variables are uncorrelated,

$$\begin{aligned} \text{Var}_\theta(\delta) &= \text{Var}_\theta(U + \delta_1 + \delta_2) \\ &= \text{Var}_\theta(U) + \text{Var}_\theta(\delta_1 + \delta_2) \geq \text{Var}_\theta(\delta_1 + \delta_2). \end{aligned}$$

8. If $M(x) = \min x_i$, then the joint densities are

$$p_\theta(x) = \frac{\theta^n}{\prod_{i=1}^n x_i^2} I\{M(x) > \theta\},$$

and M is sufficient by the factorization theorem. Next, for $x > \theta$,

$$\begin{aligned} P_\theta(M > x) &= P_\theta(X_1 > x, \dots, X_n > x) \\ &= P_\theta(X_1 > x) \times \cdots \times P_\theta(X_n > x) = (\theta/x)^n. \end{aligned}$$

So M has cumulative distribution function $1 - (\theta/x)^n$, $x > \theta$, and density $n\theta^n/x^{n+1}$, $x > \theta$. If $\delta(M)$ is an unbiased estimator of $g(\theta)$, then

$$\int_\theta^\infty \delta(x) \frac{n\theta^n}{x^{n+1}} dx = g(\theta)$$

or

$$\int_\theta^\infty \frac{n\delta(x)}{x^{n+1}} dx = \frac{g(\theta)}{\theta^n}.$$

Taking a derivative with respect to θ ,

$$\frac{n\delta(x)}{x^{n+1}} = \frac{ng(x)}{x^{n+1}} - \frac{g'(x)}{x^n}.$$

In particular, if $g(\theta) = c$ for all $\theta > 0$, this calculation shows that $\delta(x) = c$, and so M is complete. In general, $\delta(M) = g(M) - Mg'(M)/n$ is the UMVU estimator of $g(\theta)$.

10. If we assume δ is unbiased and can be written as a power series $\delta(x) = c_0 + c_1x + \cdots$, then by Fubini's theorem we anticipate

$$\begin{aligned} E_\theta \delta(X) &= \int_0^\infty \left(\sum_{n=0}^\infty c_n x^n \right) \theta e^{-\theta x} dx \\ &= \sum_{n=0}^\infty \int_0^\infty c_n x^n \theta e^{-\theta x} dx = \sum_{n=0}^\infty \frac{n! c_n}{\theta^n}. \end{aligned}$$

The form here is a power series in $1/\theta$. Writing

$$\frac{1}{1+\theta} = \frac{1/\theta}{1+1/\theta} = - \sum_{n=1}^\infty (-1)^n \theta^{-n},$$

by matching coefficients for powers of $1/\theta$, $c_0 = 0$ and $c_n = (-1)^{n+1}/n!$, $n = 1, 2, \dots$. This gives $\delta = 1 - e^{-X}$. The steps in this derivation only work if $\theta > 1$, but it is easy to show directly that δ is unbiased. Because the densities form a full rank exponential family, X is complete, and δ is UMVU.

11. The joint mass functions form a full rank exponential family with $T = X_1 + \cdots + X_3$ complete sufficient. The estimator $\delta = I\{X_1 = X_2 = 0\}$ is unbiased. By Theorem 4.4, $\eta(t) = E[\delta(X)|T = t] = P(X_1 = X_2 = 0|T = t)$ is UMVU. To calculate η we need $P_\theta(T = t)$. This event occurs if and only if trial $t + 3$ is a success, and there are exactly two successes in the first $t + 2$ trials. Thus

$$P_\theta(T = t) = \binom{t+2}{2} \theta^3 (1-\theta)^t.$$

Since $P_\theta(X_1 = X_2 = 0, T = t) = P(X_1 = X_2 = 0, X_3 = t) = \theta^3(1-\theta)^t$,

$$\eta(t) = \frac{P_\theta(X_1 = X_2 = 0, T = t)}{P_\theta(T = t)} = \frac{2}{(t+1)(t+2)}.$$

12. a) Since

$$E_\theta X = \int_{-1}^1 \frac{1}{2}(x + \theta x^2) dx = \frac{\theta}{3},$$

$3X$ is unbiased for θ .

b) Integrating against the density,

$$b = E_\theta |X| = \int_{-1}^1 \frac{1}{2}(1 + \theta x)|x| dx = \frac{1}{2},$$

for all $\theta \in [-1, 1]$.

c) Since

$$E_{\theta}X^2 = \int_{-1}^1 \frac{1}{2}(x^2 + \theta x^3) dx = \frac{1}{3},$$

$\text{Var}_{\theta}(3X) = 3 - \theta^2$. Also, $\text{Var}_{\theta}(|X|) = 1/3 - 1/4 = 1/12$. Finally,

$$E_{\theta}3X|X| = \int_{-1}^1 \frac{3}{2}(x + \theta x^2)|x| dx = \frac{3\theta}{4},$$

and so $\text{Cov}_{\theta}(3X, |X|) = 3\theta/4 - \theta/2 = \theta/4$. So

$$\begin{aligned}\text{Var}_{\theta_0}(3X + c|X|) &= \text{Var}_{\theta_0}(3X) + 2c\text{Cov}_{\theta_0}(3X, |X|) + c^2\text{Var}_{\theta_0}(|X|) \\ &= (3 - \theta_0^2) + \frac{c\theta_0}{2} + \frac{c^2}{12},\end{aligned}$$

minimized when $c = -3\theta_0$. Since the variance of this estimator is smaller than the variance of $3X$ when $\theta = \theta_0$, $3X$ cannot be UMVU.

24. Since $\bar{X}$ and S^2 are independent, using (4.10),

$$\begin{aligned}Et &= E \frac{\sqrt{n}\bar{X}}{S} = \sqrt{n}(E\bar{X})(ES^{-1}) = \delta \frac{\sqrt{n-1}\Gamma((n-2)/2)}{\sqrt{2}\Gamma((n-1)/2)}, \\ Et^2 &= n(E\bar{X}^2)(ES^{-2}) = (1 + \delta^2)\frac{n-1}{n-3},\end{aligned}$$

and

$$\text{Var}(t) = \frac{n-1}{n-3} + \delta^2 \left(\frac{n-1}{n-3} - \frac{(n-1)\Gamma^2((n-2)/2)}{2\Gamma^2((n-1)/2)} \right).$$

28. a) Since $g(\theta) = \theta$, $g(\theta + \Delta) - g(\theta) = \Delta$, so the lower bound is

$$\frac{\Delta^2}{E_{\theta} \left(\frac{p_{\theta+\Delta}(X)}{p_{\theta}(X)} - 1 \right)^2}.$$

Because we need $p_{\theta+\Delta} = 0$ whenever $p_{\theta} = 0$, Δ must be negative. Also, $\theta + \Delta$ must be positive, so $\Delta \in (-\theta, 0)$. To evaluate the expectation, note that $p_{\theta+\Delta}(X)/p_{\theta}(X)$ will be $\theta^n/(\theta + \Delta)^n$ if $M = \max\{X_1, \dots, X_n\} < \theta + \Delta$, which happens with probability $(\theta + \Delta)^n/\theta^n$ under P_{θ} . Otherwise, $p_{\theta+\Delta}(X)/p_{\theta}(X)$ will be zero. After a bit of algebra, the lower bound is found to be

$$\frac{\Delta^2}{[\theta^n/(\theta + \Delta)^n] - 1} = \frac{c^2\theta^2/n^2}{(1 - c/n)^{-n} - 1}$$

b) $g_n(c) = c^2/[(1 - c/n)^{-n} - 1] \rightarrow c^2/(e^c - 1) = g(c)$.

c) Setting derivatives to zero, the value c_0 maximizing g over $c \in (0, \infty)$ is the unique positive solution of the equation $e^{-c} = 1 - c/2$. This gives $c_0 = 1.59362$, and an approximate lower bound of $g(c_0)\theta^2/n^2 = 0.64761\theta^2/n^2$.

30. a) We have $\log p_\theta(X) = -\sum_{i=1}^n (X_i - \alpha - \beta t_i)^2 / 2 - n \log \sqrt{2\pi}$, and so

$$I(\alpha, \beta) = -E \nabla^2 \log p_\theta(X) = \begin{pmatrix} n & \sum_{i=1}^n t_i \\ \sum_{i=1}^n t_i & \sum_{i=1}^n t_i^2 \end{pmatrix}.$$

b) If $g(\theta) = \alpha$, then $\nabla g(\theta) = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and the lower bound is

$$\begin{aligned} (1, 0) I^{-1}(\alpha, \beta) \begin{pmatrix} 1 \\ 0 \end{pmatrix} &= \frac{(1, 0) \begin{pmatrix} \sum_{i=1}^n t_i^2 & -\sum_{i=1}^n t_i \\ -\sum_{i=1}^n t_i & n \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix}}{n \sum_{i=1}^n t_i^2 - (\sum_{i=1}^n t_i)^2} \\ &= \frac{\sum_{i=1}^n t_i^2}{n \sum_{i=1}^n t_i^2 - (\sum_{i=1}^n t_i)^2} \\ &= \frac{1/n}{1 - (\sum_{i=1}^n t_i/n)^2 / (\sum_{i=1}^n t_i^2/n)}. \end{aligned}$$

c) Now $I(\alpha) = n$, and so the lower bound for the variance is $1/n$.

d) The bound in (b) is larger. This is clear from the final expression, with equality only if $\sum_{i=1}^n t_i = 0$.

e) If $g(\theta) = \alpha\beta$, then $\nabla g(\theta) = \begin{pmatrix} \beta \\ \alpha \end{pmatrix}$ and the lower bound is

$$\frac{(\beta, \alpha) \begin{pmatrix} \sum_{i=1}^n t_i^2 & -\sum_{i=1}^n t_i \\ -\sum_{i=1}^n t_i & n \end{pmatrix} \begin{pmatrix} \beta \\ \alpha \end{pmatrix}}{n \sum_{i=1}^n t_i^2 - (\sum_{i=1}^n t_i)^2} = \frac{\beta^2 \sum_{i=1}^n t_i^2 - 2\alpha\beta \sum_{i=1}^n t_i + n\alpha^2}{n \sum_{i=1}^n t_i^2 - (\sum_{i=1}^n t_i)^2}.$$

31. This is a location model, and so

$$I(\theta) = \int \frac{[f'(x)]^2}{f(x)} dx = \int \frac{4x^2}{\pi(1+x^2)^3} dx = \frac{1}{2}.$$

If $\xi = \theta^3$, $\theta = \xi^{1/3} = h(\xi)$, and the information for ξ is $I^*(\xi) = I[h(\xi)] \times [h'(\xi)]^2 = 1/(18\xi^{4/3})$.

32. Since $p_\theta(x) = \theta^{2x} e^{-\theta^2} / x!$,

$$I(\theta) = \text{Var}_\theta \left(\frac{\partial \log p_\theta(X)}{\partial \theta} \right) = \text{Var}_\theta \left(\frac{2X}{\theta} - 2\theta \right) = 4.$$

33. Since the exponential distributions form a canonical exponential family with $A(\lambda) = -\log \lambda$, the Fisher information for λ is $\tilde{I}(\lambda) = A''(\lambda) = 1/\lambda^2$. Using (4.18),

$$I(\theta) = \frac{\tilde{I}(\lambda)}{[h'(\lambda)]^2} = \frac{1}{[\lambda h'(\lambda)]^2}.$$

From this, $I(\theta)$ will be constant if $h(\lambda) = \log \lambda$.

34. a) Multiplying the conditional densities,

$$\begin{aligned}\log p_{\theta, \sigma}(X) &= -\frac{(1-\rho^2)(X_1-\theta)^2}{2\sigma^2} - \frac{1}{2\sigma^2} \sum_{j=1}^{n-1} (X_{j+1} - \rho X_j - (1-\rho)\theta)^2 \\ &\quad - \frac{n}{2} \log(2\pi) - n \log \sigma + \log \sqrt{1-\rho^2}.\end{aligned}$$

Taking derivatives,

$$\begin{aligned}-\frac{\partial^2}{\partial \theta^2} \log p_{\theta, \sigma}(X) &= \frac{1-\rho^2}{\sigma^2} + \frac{(n-1)(1-\rho)^2}{\sigma^2}, \\ -\frac{\partial^2}{\partial \theta \partial \sigma} \log p_{\theta, \sigma}(X) &= \frac{2(1-\rho^2)\epsilon_1}{\sigma^3} + \frac{2(1-\rho)}{\sigma^3} \sum_{j=1}^{n-1} \eta_{j+1},\end{aligned}$$

and

$$-\frac{\partial^2}{\partial \sigma^2} \log p_{\theta, \sigma}(X) = \frac{3(1-\rho^2)(X_1-\theta)^2}{\sigma^4} + \frac{3}{\sigma^4} \sum_{j=1}^{n-1} \eta_{j+1}^2 - \frac{n}{\sigma^2},$$

where $\epsilon_j = X_j - \theta$ and $\eta_{j+1} = \epsilon_{j+1} - \rho\epsilon_j$. The conditional distribution of η_{j+1} given $X_1 = x_1, \dots, X_j = x_j$, or equivalently, given ϵ_1 and $\eta_2, \dots, \eta_j$, is $N(0, \sigma^2)$. From this, $\eta_2, \dots, \eta_n$ are i.i.d. from $N(0, \sigma^2)$, and these variables are independent of $\epsilon_1 \sim N(0, \sigma^2/(1-\rho^2))$. Using this, it is easy to take expectations of these logarithmic derivatives, giving

$$I(\theta, \sigma) = \begin{pmatrix} \frac{1-\rho^2 + (n-1)(1-\rho)^2}{\sigma^2} & 0 \\ 0 & \frac{2n}{\sigma^2} \end{pmatrix}.$$

b) The lower bound is $\sigma^2/[1-\rho^2 + (n-1)(1-\rho)^2]$.

c) Since $\epsilon_2 = \rho\epsilon_1 + \eta_2$, $\text{Var}(\epsilon_2) = \rho^2\sigma^2/(1-\rho^2) + \sigma^2 = \sigma^2/(1-\rho^2)$. Further iteration gives $\text{Var}(X_i) = \text{Var}(\epsilon_i) = \sigma^2/(1-\rho^2)$. If $i > j$, then

$$X_i = \rho X_{i-1} + \eta_i = \dots = \rho^{i-j} X_j + \rho^{i-j+1} \eta_{j+1} + \dots + \rho \eta_{i-1} + \eta_i,$$

and from this it is easy to see that

$$\text{Cov}(X_i, X_j) = \frac{\rho^{|i-j|}\sigma^2}{1-\rho^2}.$$

Noting that $\sum_{j=1}^{n-1} \rho^j = (\rho - \rho^n)/(1-\rho)$ and

$$\sum_{i=1}^{n-1} j \rho^j = \rho \frac{d}{d\rho} \sum_{j=1}^{n-1} \rho^j = \frac{\rho - \rho^{n+1}}{(1-\rho)^2} - \frac{n\rho^n}{1-\rho},$$

$$\begin{aligned}
\text{Var}(\bar{X}) &= \frac{\sigma^2}{n^2(1-\rho^2)} \sum_{i=1}^n \sum_{j=1}^n \rho^{|i-j|} \\
&= \frac{\sigma^2}{n^2(1-\rho^2)} \left[n + 2 \sum_{j=1}^{n-1} (n-j) \rho^j \right] \\
&= \frac{\sigma^2}{n(1-\rho)^2} - \frac{2\rho\sigma^2(1-\rho^n)}{n^2(1-\rho)^3(1+\rho)}.
\end{aligned}$$

B.5 Problems of Chapter 5

1. a) The joint mass functions are

$$\binom{m}{x} \binom{n}{y} \exp \left[x \log \frac{\theta}{1-\theta} + y \log \frac{\theta^2}{1-\theta^2} + n \log \theta + 2m \log \theta \right].$$

These mass functions form a curved exponential family with minimal sufficient statistic (X, Y) .

b) $E_\theta[X^2 + X] = m(m-1)\theta^2$, and $E_\theta Y = n\theta^2$. So $E_\theta[nX^2 + nX - m(m-1)Y] = 0$ for all $\theta \in (0, 1)$.

2. a) The joint densities are

$$ph(p) \exp \left[x \log \left(\frac{p}{1-p} \right) + y \log \left(\frac{h(p)}{1-h(p)} \right) \right].$$

These form a curved exponential family unless the canonical parameters are linearly related, that is, unless

$$\log \left(\frac{h(p)}{1-h(p)} \right) = a + b \log \left(\frac{p}{1-p} \right)$$

for some constants a and b . Solving for $h(p)$, this is the same as the equation stated in the problem.

b) If $h(p) = p/2$, we have a curved family with (X, Y) minimal sufficient, but not complete, because $E_p(X - 2Y) = 0$ for all $p \in (0, 1)$. For an example where (X, Y) is complete, note that $E_p g(X, Y) = ph(p)g(1, 1) + p(1-h(p))g(1, 0) + (1-p)h(p)g(0, 1) + (1-p)(1-h(p))g(0, 0)$. We need to find a function h with $ph(p)$, $p(1-h(p))$, and $(1-p)h(p)$ linearly independent. One choice that works is $h(p) = p^2$. Then

$$\begin{aligned}
E_p g(X, Y) &= p^3 [g(1, 1) - g(1, 0) - g(0, 1) + g(0, 0)] \\
&\quad + p^2 [g(0, 1) - g(0, 0)] + p [g(1, 0) - g(0, 0)] + g(0, 0).
\end{aligned}$$

If this is zero for all $p \in (0, 1)$, then the coefficients of the various powers of p must vanish, and it is easy to see that this can only happen if $g(0, 0) = g(1, 0) = g(0, 1) = g(1, 1) = 0$. Thus $g(X, Y) = 0$ and (X, Y) is complete.

6. a) Consider the event $X = 4$ and $Y = y$. This happens if and only if A wins exactly 3 of the first $3 + y$ games and then wins the next game. The chance of 3 wins in $3 + y$ trials is $\binom{3+y}{3}\theta^3(1-\theta)^y$. The outcome of the next game is independent of this event, and so

$$P(X = 4, Y = y) = \binom{3+y}{3}\theta^4(1-\theta)^y, \quad y = 0, \dots, 3.$$

Similarly,

$$P(X = x, Y = 4) = \binom{3+x}{3}\theta^x(1-\theta)^4, \quad x = 0, \dots, 3.$$

The joint mass functions have form $h(x, y) \exp[x \log \theta + y \log(1 - \theta)]$. Because the relationship between the canonical parameters $\log \theta$ and $\log(1 - \theta)$ is nonlinear, this exponential family is curved.

- b) Let f be an arbitrary function, and suppose

$$\begin{aligned} h(\theta) &= E_{\theta} f(X, Y) \\ &= \sum_{x=0}^3 f(x, 4) \binom{3+x}{3} \theta^x (1-\theta)^4 + \sum_{y=0}^3 f(4, y) \binom{3+y}{3} \theta^4 (1-\theta)^y \\ &= 0, \end{aligned}$$

for all $\theta \in (0, 1)$. This function h is a polynomial in θ . Letting θ tend to zero, the constant term in this polynomial is $f(0, 4)$, and so $f(0, 4)$ must be zero. If $f(0, 4)$ is zero, then the linear term (dividing by θ and letting θ tend to zero) is $4f(1, 4)$, so $f(1, 4)$ must be zero. Similarly $f(2, 4) = f(3, 4) = 0$. Then

$$\frac{h(\theta)}{\theta^4} = \sum_{y=0}^3 f(4, y) \binom{3+y}{3} (1-\theta)^y = 0.$$

Because this is a polynomial in $1 - \theta$, the coefficients of powers of $1 - \theta$ must vanish, giving $f(4, 0) = \dots = f(4, 3) = 0$. Thus $f(X, Y) = 0$ almost surely, demonstrating that T is complete.

- c) Let δ be an indicator that team A wins the first game. Then δ is unbiased for θ , and the UMVU estimator must be $E(\delta|X, Y) = P(\delta = 1|X, Y)$. Arguments similar to those used deriving the joint mass function give

$$\begin{aligned} P(\delta = 1|X = 4, Y = y) &= \frac{P(\delta = 1, X = 4, Y = y)}{P(X = 4, Y = y)} \\ &= \frac{\theta^2 \binom{2+y}{2} \theta^2 (1-\theta)^y}{\binom{3+y}{3} \theta^4 (1-\theta)^y} = \frac{3}{3+y}, \end{aligned}$$

and

$$P(\delta = 1 | X = x, Y = 4) = \frac{\theta(1 - \theta) \binom{x+2}{x-1} \theta^{x-1} (1 - \theta)^3}{\binom{3+x}{3} \theta^x (1 - \theta)^4} = \frac{x}{3+x}.$$

The UMVU is $(X - I\{Y = 4\}) / (X + Y - 1)$.

7. a) $P(T = 0) = P(X = 0) = e^{-\lambda}$, and for $k = 0, 1, \dots$,

$$\begin{aligned} P(T = k + 1) &= P(X + Y = k, X > 0) \\ &= P(X + Y = k) - P(X + Y = k, X = 0) \\ &= \frac{(2\lambda)^k e^{-2\lambda}}{k!} - \frac{\lambda^k e^{-2\lambda}}{k!}. \end{aligned}$$

c) Let N denote the sample size, so $N = 1$ if $X = 0$ and $N = 2$ if $X > 0$, and let $W = 0$ if $X = 0$ and $W = X + Y$ if $X > 0$. Using Theorem 5.4, the joint densities form an exponential family with canonical parameter $\eta = (\log \lambda, -\lambda)$ and sufficient statistic (W, N) . Because η does not satisfy a linear constraint, the exponential family is curved, and (W, N) is minimal sufficient.

b) There is a one-to-one relationship between T and (W, N) , so T is minimal sufficient. (This can also be shown directly.)

d) Suppose $E_\lambda g(T) = 0$ for all $\lambda > 0$. Then

$$e^{2\lambda} E_\lambda g(T) = e^\lambda g(0) + \sum_{k=0}^{\infty} \frac{g(k+1)(2^k - 1)}{k!} \lambda^k = 0.$$

The constant term in this power series is $g(0)$, so $g(0) = 0$. Setting the coefficient of λ^k to zero, $g(k+1) = 0$ for $k = 1, 2, \dots$. Since T is never 1, $g(T)$ must be zero. Thus T is complete.

13. a) Solving, $N_{22} = (n + D - R - C)/2$, $N_{11} = (R + C + D - n)/2$, $N_{12} = (R - C - D + n)/2$ and $N_{21} = (C - R - D + n)/2$. So the joint mass function can be written as

$$\begin{aligned} \binom{n}{n_{11}, \dots, n_{22}} \exp \left[r \log \sqrt{\frac{p_{11}p_{12}}{p_{21}p_{22}}} + c \log \sqrt{\frac{p_{11}p_{21}}{p_{12}p_{22}}} \right. \\ \left. + d \log \sqrt{\frac{p_{11}p_{22}}{p_{12}p_{21}}} + n \log \sqrt{\frac{p_{12}p_{21}p_{22}}{p_{11}}} \right], \end{aligned}$$

where $r = R(n_{11}, \dots, n_{22}) = n_{11} + n_{12}$, and c and d are defined similarly. These densities form a full rank exponential family: (R, C, D) cannot satisfy a linear constraint because there is a one-to-one linear association between it and (N_{11}, N_{12}, N_{21}) , and the three canonical parameters η_r, η_c , and η_d can vary freely over $\mathbb{R}^3$.

b) They are related by $\eta_d = \log \sqrt{\alpha}$.

c) Multiplying the marginal mass functions, the joint mass function has form

$$h(x) \exp \left[\sum_{i=1}^m r_i \eta_{r,i} + \sum_{i=1}^m c_i \eta_{c,i} + \log \sqrt{\alpha} \sum_{i=1}^m d_i - \sum_{i=1}^m A_i(\eta_i) \right].$$

These mass functions form a full rank $(2m + 1)$ -parameter exponential family with complete sufficient statistic

$$T = \left(R_1, \dots, R_m, C_1, \dots, C_m, \sum_{i=1}^m D_i \right).$$

16. a) Since N_{i+} and N_{+j} are independent with

$$N_{i+} \sim \text{Binomial}(n, p_{i+}) \text{ and } N_{+j} \sim \text{Binomial}(n, p_{+j}),$$

$$\begin{aligned} E(\hat{p}_{i+} \hat{p}_{+j})^2 &= \frac{1}{n^4} E N_{i+}^2 E N_{+j}^2 \\ &= \frac{1}{n^2} [n p_{i+}^2 + p_{i+}(1 - p_{i+})] [n p_{+j}^2 + p_{+j}(1 - p_{+j})]. \end{aligned}$$

Subtracting $p_{i+}^2 p_{+j}^2$,

$$\begin{aligned} \text{Var}(\hat{p}_{i+} \hat{p}_{+j}) &= \frac{p_{i+}(1 - p_{i+})p_{+j}^2 + p_{+j}(1 - p_{+j})p_{i+}^2}{n} \\ &\quad + \frac{p_{i+}(1 - p_{i+})p_{+j}(1 - p_{+j})}{n^2}. \end{aligned}$$

b) Unbiased estimates of $p_{i+}(1 - p_{i+})$, $p_{+j}(1 - p_{+j})$, p_{i+}^2 , and p_{+j}^2 are $N_{i+}(n - N_{i+})/(n^2 - n)$, $N_{+j}(n - N_{+j})/(n^2 - n)$, $N_{i+}(N_{i+} - 1)/(n^2 - n)$, and $N_{+j}(N_{+j} - 1)/(n^2 - n)$, respectively. From these, the UMVU estimator for the variance above is

$$\begin{aligned} &\frac{N_{i+}(n - N_{i+})N_{+j}(N_{+j} - 1) + N_{+j}(n - N_{+j})N_{i+}(N_{i+} - 1)}{n^3(n - 1)^2} \\ &\quad + \frac{N_{i+}(n - N_{i+})N_{+j}(n - N_{+j})}{n^4(n - 1)^2}. \end{aligned}$$

B.6 Problems of Chapter 6

2. a) Conditioning on X , for $z \geq 0$,

$$P(X^2 Y^2 \leq z) = EP(X^2 Y^2 \leq z | X) = E[F_Y(\sqrt{z/X^2}) - F_Y(-\sqrt{z/X^2})].$$

Taking a derivative with respect to z , the density is

$$E \frac{f_Y(\sqrt{z/X^2}) + f_Y(-\sqrt{z/X^2})}{2\sqrt{zX^2}}, \quad z \geq 0.$$

b) Differentiating (1.21), since $-Y$ has density $\lambda e^{\lambda x} I\{x < 0\}$, the density is

$$E \lambda e^{\lambda(z-X)} I\{X > z\} = \int_{z \vee 0}^{\infty} \lambda^2 e^{\lambda(z-2x)} dx = \frac{\lambda}{2} e^{-\lambda|z|}.$$

3. Since X and Y are positive and $x \in (0, 1)$, $X/(X+Y) \leq x$ if and only if $X \leq xY/(1-x)$. So

$$\begin{aligned} P\left(\frac{X}{X+Y} \leq x \mid Y = y\right) &= E\left(I\left\{X \leq \frac{xY}{1-x}\right\} \mid Y = y\right) \\ &= F_X\left(\frac{xy}{1-x}\right). \end{aligned}$$

Thus $P(X/(X+Y) \leq x \mid Y) = F_X(xY/(1-x))$, and the desired identity follows by smoothing.

4. The change of variables $u = y/(1-x)$ in the integral against the density of Y gives,

$$\begin{aligned} p_V(x) &= \frac{1}{\Gamma(\alpha)\Gamma(\beta)} \int_0^\infty \frac{y}{(1-x)^2} \left(\frac{xy}{1-x}\right)^{\alpha-1} y^{\beta-1} e^{-y/(1-x)} dy \\ &= \frac{x^{\alpha-1}(1-x)^{\beta-1}}{\Gamma(\alpha)\Gamma(\beta)} \int_0^\infty u^{\alpha+\beta-1} e^{-u} du \\ &= \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1}(1-x)^{\beta-1}. \end{aligned}$$

5. a) For $x \in (0, 1)$,

$$p_X(x) = \int p(x, y) dy = \int_x^1 2 dy = 2(1-x).$$

Similarly, for $y \in (0, 1)$,

$$p_Y(y) = \int p(x, y) dx = \int_0^y 2 dx = 2y.$$

- b) For $y \in (x, 1)$,

$$p_{Y|X}(y|x) = \frac{p(x, y)}{p_X(x)} = \frac{2}{2(1-x)} = \frac{1}{1-x}.$$

- c) Because

$$E[Y|X = x] = \int_x^1 \frac{y}{1-x} dy = \frac{1+x}{2},$$

$$E[Y|X] = (1+X)/2.$$

d) Integrating against the joint density,

$$EXY = \iint xyp(x, y) dx dy = \int_0^1 \int_0^y 2xy dx dy = \int_0^1 y^3 dy = \frac{1}{4}.$$

e) By smoothing,

$$\begin{aligned} EXY &= EE[XY|X] = E\{XE[Y|X]\} = E\left[X\frac{1}{2}(1+X)\right] \\ &= \int_0^1 2(1-x)\frac{1}{2}x(1+x) dx = \int_0^1 [x - x^3] dx = \frac{1}{4}. \end{aligned}$$

6. a) For $x \in (0, 1)$,

$$p_X(x) = \sum_{y_1=0}^{\infty} \sum_{y_2=0}^{\infty} x^2(1-x)^{y_1+y_2} = 1.$$

b) Integrating the joint density,

$$\begin{aligned} p_Y(y_1, y_2) &= \int_0^1 x^2(1-x)^{y_1+y_2} dx \\ &= \frac{\Gamma(3)\Gamma(y_1+y_2+1)}{\Gamma(y_1+y_2+4)} = \frac{2(y_1+y_2)!}{(y_1+y_2+3)!}, \end{aligned}$$

and so

$$p_{X|Y}(x|y) = \frac{p(x, y_1, y_2)}{p_Y(y_1, y_2)} = \frac{(y_1+y_2+3)!}{2(y_1+y_2)!} x^2(1-x)^{y_1+y_2}.$$

c) Using the formula in the hint,

$$E(X|Y = y) = \int_0^1 \frac{(y_1+y_2+3)!}{2(y_1+y_2)!} x^3(1-x)^{y_1+y_2} dx = \frac{3}{y_1+y_2+4}$$

and

$$\begin{aligned} E(X^2|Y = y) &= \int_0^1 \frac{(y_1+y_2+3)!}{2(y_1+y_2)!} x^4(1-x)^{y_1+y_2} dx \\ &= \frac{12}{(y_1+y_2+4)(y_1+y_2+5)}. \end{aligned}$$

So,

$$E(X|Y) = \frac{3}{Y_1 + Y_2 + 4},$$

and

$$E(X^2|Y) = \frac{12}{(Y_1 + Y_2 + 4)(Y_1 + Y_2 + 5)}.$$

d) Since

$$\begin{aligned} EX &= \frac{1}{2} = EE(X|Y) = E\left[\frac{3}{Y_1 + Y_2 + 4}\right], \\ E\left[\frac{1}{Y_1 + Y_2 + 4}\right] &= \frac{1}{6}. \end{aligned}$$

11. Given $M = m$ and $Z = z$, the conditional distribution for (X, Y) must concentrate on the two points (m, z) and (z, m) . By symmetry, a natural guess is

$$\begin{aligned} \frac{1}{2} &= P(X = m, Y = z | M = m, Z = z) \\ &= P(X = z, Y = m | M = m, Z = z). \end{aligned}$$

To see that this is correct we need to check that smoothing works. Let h be an arbitrary function and define $g(x, y) = h(x, y, x \vee y, x \wedge y)$ so that $h(X, Y, M, Z) = g(X, Y)$. Then

$$\begin{aligned} E[h(X, Y, M, Z) | M, Z] &= E[g(X, Y) | M, Z] \\ &= \frac{1}{2}g(M, Z) + \frac{1}{2}g(Z, M). \end{aligned}$$

So smoothing works if $Eg(X, Y) = E[\frac{1}{2}g(M, Z) + \frac{1}{2}g(Z, M)]$, that is, if

$$\begin{aligned} &\iint g(x, y)f(x)f(y) dx dy \\ &= \iint \frac{1}{2}[g(x \vee y, x \wedge y) + g(x \wedge y, x \vee y)]f(x)f(y) dx dy \\ &= \iint \frac{1}{2}[g(x, y) + g(y, x)]f(x)f(y) dx dy. \end{aligned}$$

This holds because $\iint g(x, y) dx dy = \iint g(y, x) dx dy$, and the stated conditional distribution is correct.

12. From the example of Section 3.2, T and $U = X/(X + Y)$ are independent with $U \sim \text{Unif}(0, 1)$. Then

$$\begin{aligned} E[f(X, Y) | T = t] &= E[f(TU, T(1 - U)) | T = t] \\ &= \int_0^1 f(tu, t(1 - u)) du. \end{aligned}$$

(It is easy to check that smoothing works by viewing all expectations as integrals against the joint density of T and U .)

14. a) Integrating against y ,

$$p_X(x) = \int p(x, y) dy = \int_0^x e^{-x} dy = xe^{-x}, \quad x > 0,$$

and integrating against x ,

$$p_Y(y) = \int p(x, y) dx = \int_y^\infty e^{-x} dx = e^{-y}, \quad y > 0.$$

b) Integration against the marginal density gives

$$EY = \int_0^\infty ye^{-y} dy = 1 \quad \text{and} \quad EY^2 = \int_0^\infty y^2 e^{-y} dy = 2.$$

c) Dividing the joint density by the marginal density,

$$p_{Y|X}(y|x) = p(x, y)/p_X(x) = \frac{1}{x}, \quad y \in (0, x).$$

Integrating against this conditional density

$$E[Y|X = x] = \int_0^x \frac{y}{x} dy = \frac{x}{2}$$

and

$$E[Y^2|X = x] = \int_0^x \frac{y^2}{x} dy = \frac{x^2}{3}.$$

So $E[Y|X] = X/2$, and $E[Y^2|X] = X^2/3$.

d) Integrating against the marginal density of X ,

$$EE[Y|X] = \frac{1}{2}EX = \frac{1}{2} \int_0^\infty x^2 e^{-x} dx = 1,$$

and

$$EE[Y^2|X] = \frac{1}{3}EX^2 = \frac{1}{3} \int_0^\infty x^3 e^{-x} dx = 2.$$

B.7 Problems of Chapter 7

1. The likelihood is $p_\theta(x) = \theta^{T(x)} e^{-n\theta} / \prod_1^n x_i!$, where $T(x) = \sum_{i=1}^n x_i$. So the Bayes estimator is

$$\begin{aligned} \delta(x) &= \frac{\int_0^\infty \theta^{p+1} p_\theta(x) \lambda(\theta) d\theta}{\int_0^\infty \theta^p p_\theta(x) \lambda(\theta) d\theta} \\ &= \frac{\int_0^\infty \theta^{T(x)+p+1} e^{-(n+\eta)\theta} d\theta}{\int_0^\infty \theta^{T(x)+p} e^{-(n+\eta)\theta} d\theta} = \frac{T(x) + p + 1}{n + \eta}. \end{aligned}$$

2. The marginal density of X is

$$\begin{aligned}
 q(x) &= \int p_\theta(x) \lambda(\theta) d\theta \\
 &= \int_x^\infty \frac{1}{\theta(1+\theta)^2} d\theta \\
 &= \int_x^\infty \left[\frac{1}{\theta} - \frac{1}{1+\theta} - \frac{1}{(1+\theta)^2} \right] d\theta \\
 &= \log\left(\frac{1+x}{x}\right) - \frac{1}{1+x}.
 \end{aligned}$$

So $p(\theta|x) = 1/[\theta(1+\theta)^2 q(x)]$, $\theta > x$, and

$$\begin{aligned}
 E[|\Theta - d| \mid X = x] &= \int_x^\infty \frac{|\theta - d|}{\theta(1+\theta)^2 q(x)} d\theta \\
 &= \int_x^d \frac{d - \theta}{\theta(1+\theta)^2 q(x)} d\theta + \int_d^\infty \frac{\theta - d}{\theta(1+\theta)^2 q(x)} d\theta.
 \end{aligned}$$

Note that these integrals are like the integral for q when d is in the numerator and are easy to integrate when θ is in the numerator. After a bit of algebra,

$$E[|\Theta - d| \mid X = x] = d - \frac{2dq(d)}{q(x)} - \frac{1}{(1+x)q(x)} + \frac{2}{(1+d)q(x)}.$$

Since $q'(d) = -1/[d(1+d)^2]$, the derivative of this expression is $1 - 2q(d)/q(x)$. This function is strictly increasing from -1 to 1 as d varies from x to infinity. So it will have a unique zero, and this zero determines the Bayes estimator: $q(\delta_A(X)) = q(X)/2$. From this equation,

$$P[\delta_A(X) < \Theta \mid X = x] = \int_{\delta_A(x)}^\infty \frac{1}{\theta(1+\theta)^2 q(x)} d\theta = \frac{q(\delta_A(x))}{q(x)} = \frac{1}{2}.$$

3. Completing squares, the conditional density is proportional to

$$\begin{aligned}
 &\lambda(\theta)p_\theta(y) \\
 &\propto_\theta \exp \left[-\frac{\theta_1^2}{2\tau_1^2} - \frac{\theta_2^2}{2\tau_2^2} - \frac{n\theta_1^2}{2\sigma^2} - \frac{\theta_2^2}{2\sigma^2} \sum_{i=1}^n x_i^2 + \frac{\theta_1}{\sigma^2} \sum_{i=1}^n y_i + \frac{\theta_2}{\sigma^2} \sum_{i=1}^n x_i y_i \right] \\
 &\propto_\theta \exp \left[-\frac{\left[\theta_1 - \frac{\sum_{i=1}^n y_i}{n + \sigma^2/\tau_1^2} \right]^2}{2(n/\sigma^2 + 1/\tau_1^2)^{-1}} - \frac{\left[\theta_2 - \frac{\sum_{i=1}^n x_i y_i}{\sum_{i=1}^n x_i^2/\sigma^2 + 1/\tau_2^2} \right]^2}{2(\sum_{i=1}^n x_i^2/\sigma^2 + 1/\tau_2^2)^{-1}} \right].
 \end{aligned}$$

So given the data, Θ_1 and Θ_2 are independent normal variables. The Bayes estimates are the posterior means,

$$E[\Theta_1|X, Y] = \frac{\sum_{i=1}^n Y_i}{n + \sigma^2/\tau_1^2} \text{ and } E[\Theta_2|X, Y] = \frac{\sum_{i=1}^n x_i Y_i}{\sum_{i=1}^n x_i^2 + \sigma^2/\tau_2^2}.$$

4. a) Since $\lambda'(\theta) = [\alpha - \beta A'(\theta)]\lambda(\theta)$,

$$0 = \int_{\Omega} \lambda'(\theta) d\theta = E[\alpha - \beta A'(\Theta)].$$

So $EA'(\Theta) = \alpha/\beta$.

b) The joint density (or likelihood) is proportional to $e^{\theta n \bar{T} - n A(\theta)}$. Multiplying by $\lambda_{\alpha, \beta}$, the conditional density of Θ given $X = x$ is proportional to

$$e^{(\alpha + n \bar{T})\theta - (\beta + n)A(\theta)} \propto_{\theta} \lambda_{\alpha + n \bar{T}, \beta + n}.$$

So $\Theta|X = x \sim \Lambda_{\alpha + n \bar{T}, \beta + n}$. Using the result from part (a), the Bayes estimator of $A'(\Theta)$ is

$$E[A'(\Theta)|X] = \frac{\alpha + n \bar{T}}{\beta + n} = \frac{\beta}{\beta + n} \frac{\alpha}{\beta} + \frac{n}{\beta + n} \bar{T},$$

where the last equality expresses this estimator as a weighted average of $EA'(\Theta) = \alpha/\beta$ and $\bar{T}$.

c) Since $p_{\theta}(x) = \theta e^{-\theta x}$, $x > 0$, we should take $T(x) = -x$, and $A(\theta) = -\log \theta$. Then

$$\lambda_{\alpha, \beta}(\theta) \propto_{\theta} e^{\alpha \theta + \beta \log \theta} = \theta^{\beta} e^{\alpha \theta}, \quad \theta > 0.$$

For convergence, α should be negative. This density is proportional to a gamma density, and so $\Lambda_{\alpha, \beta}$ is the gamma distribution with shape parameter $\beta + 1$ and failure rate $-\alpha$. Since $1/\theta = -A'(\theta)$, the Bayes estimator, using results from part (b), is

$$-E[A'(\Theta)|X] = -\frac{\alpha + n \bar{T}}{\beta + n} = \frac{|\alpha| + n \bar{X}}{\beta + n}.$$

6. a) The joint density is $\lambda(\theta)p_{\theta}(x) = f_{\theta}(x)/2$. Integrating (summing) out θ , the marginal density of X is $q(x) = [f_0(x) + f_1(x)]/2$. So the conditional density of Θ given X is

$$\lambda(\theta|x) = \frac{\lambda(\theta)p_{\theta}(x)}{q(x)} = \frac{f_{\theta}(x)}{f_0(x) + f_1(x)}, \quad \theta = 0, 1.$$

This is the mass function for a Bernoulli distribution with success probability $p = p(x) = f_1(x)/(f_0(x) + f_1(x))$. The Bayes estimator under squared error loss is the mean of this conditional distribution,

$$E(\Theta|X) = \frac{f_1(X)}{f_0(X) + f_1(X)}.$$

- b) From Theorem 7.1, the Bayes estimator should minimize the posterior risk. For zero-one loss the estimator should be one if $p(X) > 1/2$, and zero if $p(X) < 1/2$. If $p(X) = 1/2$, either is optimal. Equivalently, δ should be one if $f_1(X) > f_0(X)$, and should be zero if $f_1(X) < f_0(X)$.
7. Using Theorem 7.1, the Bayes estimator $\delta(x)$ should be the value d minimizing

$$E\left[\frac{(d - \Theta)^2}{d} \mid X = x\right] = d - 2E[\Theta \mid X = x] + \frac{E[\Theta^2 \mid X = x]}{d}.$$

Setting the derivative to zero, $\delta = \sqrt{E[\Theta^2 \mid X]}$. If $T(x) = x_1 + \cdots + x_n$, then since $\lambda(\theta \mid x) \propto \lambda(\theta)p_\theta(x) \propto \theta^n e^{-[1+T(x)]\theta}$,

$$E[\Theta^2 \mid X] = \frac{\int_0^\infty \theta^{n+2} e^{-(1+T)\theta} d\theta}{\int_0^\infty \theta^n e^{-(1+T)\theta} d\theta} = \frac{(n+1)(n+2)}{(1+T)^2}.$$

So the Bayes estimator is $\sqrt{n^2 + 3n + 2}/(1+T)$.

16. a) Let f and F be the density and cumulative distribution functions for the standard Cauchy distribution, so the respective density and cumulative distribution functions of X (or Y) are $f(x-\theta)$ and $F(x-\theta)$. By smoothing,

$$P(A \leq x) = EP(X \leq 2x - Y \mid Y) = EF(2x - Y - \theta).$$

Taking d/dx , A has density

$$\begin{aligned} 2Ef(2x - Y - \theta) &= \int \frac{2 dy}{\pi^2 [(2x - y - \theta)^2 + 1] [y^2 + 1]} \\ &= \frac{1}{\pi [(x - \theta)^2 + 1]}. \end{aligned}$$

So A and X have the same density.

- b) We have $|A - \theta| = |X - \theta|$ along two lines: $Y - \theta = X - \theta$ and $Y - \theta = -3(X - \theta)$. These lines divide the (X, Y) plane into four regions: two where $|A - \theta| < |X - \theta|$ (boundaries for these two regions form obtuse angles), and two where $|A - \theta| > |X - \theta|$. Similarly, the lines $Y - \theta = X - \theta$ and $Y - \theta = -(X - \theta)$ divide the plane into four regions. But these regions are symmetric, and so the chance (X, Y) lies any of them is $1/4$. The region where $|A - \theta| < |X - \theta|$ contains two of the symmetric regions, and so $P(|A - \theta| < |X - \theta|) > 1/2$.

B.8 Problems of Chapter 8

1. By the covariance inequality (equation (4.11)) $|\text{Cov}(X_i, X_j)| \leq \sigma^2$. Of course, by the independence, this covariance must be zero if $|i - j| \geq m$. So for any i , $\sum_{j=1}^n \text{Cov}(X_i, X_j) \leq (2m - 1)\sigma^2$. Thus

$$\begin{aligned}
 E(\bar{X}_n - \xi)^2 &= \frac{1}{n^2} \text{Var} \left(\sum_{i=1}^n X_i \right) = \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n \text{Cov}(X_i, X_j) \\
 &\leq \frac{1}{n^2} \sum_{i=1}^n (2m-1) \sigma^2 = \frac{(2m-1) \sigma^2}{n} \rightarrow 0.
 \end{aligned}$$

2. Since $\log(1-u)/u \rightarrow -1$ as $u \rightarrow 0$ and

$$P(M_n \leq x) = P(X_i \leq x, i = 1, \dots, n) = (1 - e^{-\lambda x})^n,$$

if $\epsilon > 0$,

$$\log P \left(\frac{\log n}{M_n} \geq \lambda + \epsilon \right) = n^{\epsilon/(\lambda+\epsilon)} \frac{\log[1 - e^{-\lambda \log(n)/(\lambda+\epsilon)}]}{e^{-\lambda \log(n)/(\lambda+\epsilon)}} \rightarrow -\infty.$$

So $P(\log n/M_n \geq \lambda + \epsilon) \rightarrow 0$. Similarly, if $\epsilon \in (0, \lambda)$,

$$\log P \left(\frac{\log n}{M_n} > \lambda - \epsilon \right) = n^{-\epsilon/(\lambda-\epsilon)} \frac{\log[1 - e^{-\lambda \log(n)/(\lambda-\epsilon)}]}{e^{-\lambda \log(n)/(\lambda-\epsilon)}} \rightarrow 0.$$

So $P(\log n/M_n > \lambda - \epsilon) \rightarrow 1$, and $P(\log n/M_n \leq \lambda - \epsilon) \rightarrow 0$. Hence $\log(n)/M_n$ is consistent.

3. Because M_n lies between 0 and θ , $n(\hat{\theta} - \theta)$ lies between $-n\theta$ and θ . So $P_\theta(n(\hat{\theta} - \theta) \leq y)$ is one if $y \geq \theta$. If $y \leq \theta$, then for n sufficiently large,

$$\begin{aligned}
 P_\theta(n(\hat{\theta} - \theta) \leq y) &= P_\theta \left(M_n \leq \theta - \frac{\theta - y}{n+1} \right) \\
 &= \left(1 - \frac{1 - y/\theta}{n+1} \right)^n \rightarrow e^{y/\theta-1}.
 \end{aligned}$$

So $Y_n \rightarrow Y$ where Y has cumulative distribution function H_θ given by $H_\theta(y) = \min\{1, e^{y/\theta-1}\}$.

4. a) By the central limit theorem, $\sqrt{n}(\hat{p} - p) \Rightarrow N[0, p(1-p)]$, and so by the delta method, Proposition 8.14, with $f(x) = x^2$, $\sqrt{n}(\hat{p}_n^2 - p^2) \Rightarrow N[0, 4p^3(1-p)]$.
 b) $T_n = X_1 + \dots + X_n$ is complete sufficient and $4X_1X_2X_3(1-X_4)$ is unbiased. The UMVU estimator is $\delta_n(T) = E[4X_1X_2X_3(1-X_4)|T]$, given explicitly by

$$\begin{aligned}
 \delta_n(t) &= \frac{4P(X_1 = X_2 = X_3 = 1, X_4 = 0, X_5 + \dots + X_n = t-3)}{P(T = t)} \\
 &= \frac{4t(t-1)(t-2)(n-t)}{n(n-1)(n-2)(n-3)}.
 \end{aligned}$$

- c) The difference $n\delta_n - n\hat{\sigma}_n^2$ is

$$\begin{aligned} & \frac{4T(T-1)(T-2)(n-T)}{(n-1)(n-2)(n-3)} - \frac{4T^3(n-T)}{n^3} \\ &= \frac{4T(n-T)[T^2(6n^2-11n+6)-3Tn^3+2n^3]}{n^3(n-1)(n-2)(n-3)} \\ &= \frac{4n^3\hat{p}_n(1-\hat{p}_n)}{(n-1)(n-2)(n-3)} \left(\hat{p}_n^2 \frac{6n^2-11n+6}{n^2} - 3\hat{p}_n + \frac{2}{n} \right), \end{aligned}$$

which converges in probability to $4p(1-p)(6p^2-3p) = 27/32$. By the central limit theorem, $\sqrt{n}(\hat{p}_n - p) \Rightarrow Z \sim N(0, 3/16)$. Now $\hat{\sigma}^2 = f(\hat{p}_n)$ with $f(p) = 4p^3(1-p)$, and since $f'(3/4) = 0$, a two-term Taylor approximation is necessary to derive the limiting distribution using the delta method. With a suitable intermediate value p_n , lying between p and $\hat{p}_n$,

$$n(\hat{\sigma}^2 - \sigma^2) = n(f(\hat{p}_n) - f(p)) = \frac{1}{2}[\sqrt{n}(\hat{p}_n - p)]^2 f''(p_n).$$

Because $p_n \xrightarrow{p} p$ and f'' is continuous, $f''(p_n) \xrightarrow{p} f''(p) = -9$. Using Theorem 8.13, $n(\hat{\sigma}^2 - \sigma^2) \Rightarrow -9Z^2/2$ and $n(\delta_n - \sigma^2) \Rightarrow 27/32 - 9Z^2/2$.

5. If $\epsilon > 0$,

$$P(X_i \geq \theta + \epsilon) = \int_{\theta+\epsilon}^{\infty} (x - \theta)e^{\theta-x} dx = (1 + \epsilon)e^{-\epsilon} < 1,$$

and so

$$P(M_n \geq \theta + \epsilon) = P(X_i \geq \theta + \epsilon, i = 1, \dots, n) = [(1 + \epsilon)e^{-\epsilon}]^n \rightarrow 0.$$

Since $M_n \geq \theta$, M_n is consistent. Next, for $x > 0$,

$$P(\sqrt{n}(M_n - \theta) > x) = \left[\left(1 + \frac{x}{\sqrt{n}} \right) e^{-x/\sqrt{n}} \right]^n.$$

To evaluate the limit, we use the facts that if $c_n \rightarrow c$, then $(1 + c_n/n)^n \rightarrow e^c$, and that $[(1+u)e^{-u} - 1]/u^2 \rightarrow -1/2$, which follows from Taylor expansion or l'Hôpital's rule. Since

$$n \left[\left(1 + \frac{x}{\sqrt{n}} \right) e^{-x/\sqrt{n}} - 1 \right] = x^2 \frac{(1 + x/\sqrt{n})e^{-x/\sqrt{n}} - 1}{(x/\sqrt{n})^2} \rightarrow -x^2/2,$$

$P(\sqrt{n}(M_n - \theta) > x) \rightarrow e^{-x^2/2}$. So, $\sqrt{n}(M_n - \theta) > x \Rightarrow Y$, where Y has cumulative distribution function $P(Y \leq y) = 1 - e^{-y^2/2}$, $y > 0$ (a Weibull distribution).

6. Let F denote the cumulative distribution function of Y , and let $y > 0$ be a continuity point of F . If $y + \epsilon$ is also a continuity point of F , then since

$$\{A_n Y_n \leq y\} \subset \{Y_n \leq y + \epsilon\} \cup \{A_n \leq y/(y + \epsilon)\},$$

we have

$$P(A_n Y_n \leq y) \leq P(Y_n \leq y + \epsilon) + P(A_n \leq y/(y + \epsilon)) \rightarrow F(y + \epsilon).$$

From this, $\limsup P(A_n Y_n \leq y) \leq F(y + \epsilon)$. Because F is continuous at y and ϵ can be arbitrarily small (F can have at most a countable number of discontinuities), $\limsup P(A_n Y_n \leq y) \leq F(y)$. Similarly, if $y - \epsilon$ is positive and a continuity point of F , then

$$P(A_n Y_n \leq y) \geq P(Y_n \leq y - \epsilon) - P(A_n < 0 \text{ or } A_n \geq y/(y - \epsilon)) \rightarrow F(y - \epsilon).$$

From this, $\liminf P(A_n Y_n \leq y) \geq F(y - \epsilon)$, and since ϵ can be arbitrarily small, $\liminf P(A_n Y_n \leq y) \geq F(y)$. Thus

$$\lim P(A_n Y_n \leq y) = F(y).$$

Similar arguments show that

$$\lim P(A_n Y_n \leq y) = F(y)$$

when y is negative or zero with F continuous at y .

16. The log-likelihood $l(\alpha, \beta, \sigma^2)$ is

$$\begin{aligned} & -\frac{1}{2\sigma^2} \sum_{i=1}^n Y_i^2 + \frac{\alpha}{\sigma^2} \sum_{i=1}^n Y_i + \frac{\beta}{\sigma^2} \sum_{i=1}^n x_i Y_i - \frac{n\alpha^2}{2\sigma^2} \\ & - \frac{\beta^2}{2\sigma^2} \sum_{i=1}^n x_i^2 - \frac{\alpha\beta}{\sigma^2} \sum_{i=1}^n x_i - n \log \sqrt{2\pi\sigma^2}. \end{aligned}$$

With any fixed value for σ^2 this is a quadratic function of α and β , maximized when both partial derivatives are zero. From the form, the answer is the same regardless of the value for σ^2 . This gives the following equations for $\hat{\alpha}$ and $\hat{\beta}$: $-2 \sum_{i=1}^n Y_i + 2n\hat{\alpha} + 2\hat{\beta} \sum_{i=1}^n x_i = 0$ and $-2 \sum_{i=1}^n x_i Y_i + 2\hat{\beta} \sum_{i=1}^n x_i^2 + 2\hat{\alpha} \sum_{i=1}^n x_i = 0$. Solving, $\hat{\beta} = (\sum_{i=1}^n x_i Y_i - n\bar{x}\bar{Y}) / (\sum_{i=1}^n x_i^2 - n\bar{x}^2)$ and $\hat{\alpha} = \bar{Y} - \hat{\beta}\bar{x}$. Next, $\hat{\sigma}^2$ must maximize $l(\hat{\alpha}, \hat{\beta}, \sigma^2) = -\sum_{i=1}^n e_i^2 / (2\sigma^2) - n \log \sqrt{2\pi\sigma^2}$, where $e_i = Y_i - \hat{\alpha} - \hat{\beta}x_i$ (called the i th residual). The derivative here with respect to σ^2 is $\sum_{i=1}^n e_i^2 / (2\sigma^4) - n / (2\sigma^2)$ which has a unique zero when σ^2 is $\hat{\sigma}^2 = \sum_{i=1}^n e_i^2 / n$. Note that the function goes to $-\infty$ as $\sigma^2 \downarrow 0$ or as $\sigma^2 \rightarrow \infty$. So this value must give the maximum, and $\hat{\sigma}^2$ is thus the maximum likelihood estimator of σ^2 .

17. The likelihood is

$$\begin{aligned} & \frac{1}{\sqrt{2\pi}^n} \exp \left[-\frac{X_1^2}{2} - \frac{1}{2} \sum_{i=1}^{n-1} (X_{i+1} - \rho X_i)^2 \right] \\ & \propto \exp \left[\rho \sum_{i=1}^{n-1} X_i X_{i+1} - \frac{\rho^2}{2} \sum_{i=1}^{n-1} X_i^2 \right]. \end{aligned}$$

Maximizing the quadratic function in the exponential, the maximum likelihood estimator is

$$\hat{\rho} = \sum_{i=1}^{n-1} X_i X_{i+1} \bigg/ \sum_{i=1}^{n-1} X_i^2.$$

19. By the central limit theorem, $\sqrt{n}(\bar{X}_n - 1/\theta) \Rightarrow N(0, 1/\theta^2)$. Using the delta method with $f(x) = 1/x$, $\sqrt{n}(1/\bar{X}_n - \theta) \Rightarrow N(0, \theta^2)$. Next, by the central limit theorem, $\sqrt{n}(\hat{p}_n - e^{-\theta}) \Rightarrow N[0, e^{-\theta}(1 - e^{-\theta})]$, and so by the delta method with $f(x) = -\log x$, $\sqrt{n}(-\log \hat{p}_n - \theta) \Rightarrow N(0, e^\theta - 1)$. So the asymptotic relative efficiency is $\theta^2/(e^\theta - 1)$.
20. By the central limit theorem, $\sqrt{n}(\bar{X}_n - \theta) \Rightarrow N(0, \theta)$, and so by the delta method with $f(x) = x(x+1)$, $\sqrt{n}[\bar{X}_n(\bar{X}_n + 1) - \theta(\theta + 1)] \Rightarrow N(0, 4\theta^3 + 4\theta^2 + \theta)$. Next, let $Z = (X_1 - \theta)/\sqrt{\theta} \sim N(0, 1)$. Then $\text{Var}(X_1^2) = \text{Var}(\theta Z^2 + 2\theta^{3/2}Z + \theta^2) = \text{Var}(\theta Z^2) + \text{Var}(2\theta^{3/2}Z) + 2\text{Cov}(\theta Z^2, 2\theta^{3/2}Z) = 2\theta^2 + 4\theta^3 + 0$, and by the central limit theorem, $\sqrt{n}(\delta_n - \theta(\theta + 1)) \Rightarrow N(0, 4\theta^3 + 2\theta^2)$. So the asymptotic relative efficiency of $\bar{X}_n(\bar{X}_n + 1)$ with respect to δ_n is $(4\theta^3 + 2\theta^2)/(4\theta^3 + 4\theta^2 + \theta)$.
21. Since $\text{Var}(X_i^2) = 2\sigma^4$ and $\hat{\sigma}^2 = \sum_{i=1}^n X_i^2/n$, by the central limit theorem, $\sqrt{n}(\hat{\sigma}^2 - \sigma^2) \Rightarrow N(0, 2\sigma^4)$. By the delta method with f the square root function, $\sqrt{n}(\hat{\sigma} - \sigma) \Rightarrow N(0, \sigma^2/2)$. By Theorem 8.18, $\sqrt{n}(Q_n - c\sigma) \Rightarrow N[0, 3\sigma^2/(16\phi^2(c))]$. So $\sqrt{n}(\tilde{\sigma} - \sigma) \Rightarrow N[0, 3\sigma^2/(16c^2\phi^2(c))]$. The asymptotic relative efficiency of $\tilde{\sigma}$ with respect to $\hat{\sigma}$ is $8c^2\phi^2(c)/3 = 0.1225$ ($c = 0.6745$).
24. a) For $x > 0$, $P(|X_i| \leq x) = \Phi(x/\sigma) - \Phi(-x/\sigma) = 2\Phi(x/\sigma) - 1$. So $|X_i|$ has density $2\phi(x/\sigma)/\sigma$, $x > 0$, and median $\Phi^{-1}(3/4)\sigma = 0.6745\sigma$. Hence $\tilde{\sigma} = cM \xrightarrow{p} 0.6745c\sigma$. This estimator will be consistent if $c = 1/0.6745 = 1.4826$.
- b) By (8.5),

$$\sqrt{n}(M - 0.6745\sigma) \Rightarrow N[0, \sigma^2/((16\phi^2(0.6745)))] = N(0, 0.6189\sigma^2),$$

and so

$$\sqrt{n}(\tilde{\sigma} - \sigma) \Rightarrow N(0, 1.3604\sigma^2).$$

c) The log-likelihood is

$$l_n(\sigma) = -\frac{1}{2\sigma^2} \sum_{i=1}^n X_i^2 - n \log(\sqrt{2\pi}\sigma).$$

Setting $l'_n(\sigma)$ to zero, the maximum likelihood estimator is $\hat{\sigma} = \sum_{i=1}^n X_i^2/n$. The summands in $\hat{\sigma}^2$ have common variance $2\sigma^4$, and by the central limit theorem,

$$\sqrt{n}(\hat{\sigma}^2 - \sigma^2) \Rightarrow N(0, 2\sigma^4).$$

Using the delta method,

$$\sqrt{n}(\hat{\sigma} - \sigma) \Rightarrow N(0, \sigma^2/2).$$

d) Dividing the variances, the asymptotic relative efficiency is

$$\frac{\sigma^2/2}{1.3604\sigma^2} = 0.3675.$$

B.9 Problems of Chapter 9

8. Define $S_x^2 = \sum_{i=1}^m (X_i - \bar{X})^2 / (m-1)$ and $S_y^2 = \sum_{j=1}^n (Y_j - \bar{Y})^2 / (n-1)$. Then $(m-1)S_x^2/\sigma_x^2 \sim \chi_{m-1}^2$ and $(n-1)S_y^2/\sigma_y^2 \sim \chi_{n-1}^2$ are independent, and $F = \sigma_y^2 S_x^2 / (\sigma_x^2 S_y^2)$ has an F distribution with $m-1$ and $n-1$ degrees of freedom. The distribution of F does not depend on unknown parameters. Therefore it is a pivot. If c_1 and c_2 are the $(\alpha/2)$ th and $(1-\alpha/2)$ th quantiles for this F distribution, then $P(c_1 < F < c_2) = 1 - \alpha$. This event, $c_1 < F < c_2$, occurs if and only if

$$\frac{\sigma_x}{\sigma_y} \in \left(\sqrt{\frac{S_x^2}{c_2 S_y^2}}, \sqrt{\frac{S_x^2}{c_1 S_y^2}} \right),$$

and this is the desired $1 - \alpha$ confidence interval.

9. a) The likelihood function is $\theta^{-n} I\{X_{(n)} \leq \theta\}$, maximized at $\hat{\theta} = X_{(n)}$.
 b) For $x \in (0, 1)$, $P(\hat{\theta}/\theta \leq x) = P(X_{(n)} \leq x\theta) = x^n$, and so $\hat{\theta}/\theta$ is a pivot. The $(\alpha/2)$ th and $(1 - \alpha/2)$ th quantiles for this distribution are $(\alpha/2)^{1/n}$ and $(1 - \alpha/2)^{1/n}$. So

$$\begin{aligned} 1 - \alpha &= P[(\alpha/2)^{1/n} < \hat{\theta}/\theta < (1 - \alpha/2)^{1/n}] \\ &= P[\theta \in (\hat{\theta}/(1 - \alpha/2)^{1/n}, \hat{\theta}/(\alpha/2)^{1/n})]. \end{aligned}$$

Hence $(\hat{\theta}/(1 - \alpha/2)^{1/n}, \hat{\theta}/(\alpha/2)^{1/n})$ is the desired $1 - \alpha$ confidence interval.

10. For $t > 0$,

$$P(\theta X_i \leq t) = P(X_i \leq t/\theta) = \int_0^{t/\theta} \theta e^{-\theta x} dx = 1 - e^{-t},$$

and so θX_i has a standard exponential distribution. So θT has a gamma distribution with density $x^{n-1} e^{-x} / \Gamma(n)$, $x > 0$. If $\gamma_{\alpha/2}$ and $\gamma_{1-\alpha/2}$ are the upper and lower $(\alpha/2)$ th quantiles for this distribution, then

$$P(\gamma_{1-\alpha/2} < \theta T < \gamma_{\alpha/2}) = P(\theta \in (\gamma_{1-\alpha/2}/T, \gamma_{\alpha/2}/T)) = 1 - \alpha,$$

which shows that $(\gamma_{1-\alpha/2}/T, \gamma_{\alpha/2}/T)$ is a $1 - \alpha$ confidence interval for θ .

11. a) The cumulative distribution function of $Y = (X - \theta)/\sigma$ is

$$\begin{aligned} F_Y(y) &= P\left(\frac{X - \theta}{\sigma} \leq y\right) = P(X \leq \theta + \sigma y) \\ &= \int_{-\infty}^{\theta + \sigma y} g\left(\frac{x - \theta}{\sigma}\right) \frac{dx}{\sigma} = \int_{-\infty}^y g(u) du. \end{aligned}$$

From this, Y has density g .

b) Let $Y_i = (X_i - \theta)/\sigma$, $i = 1, 2$. Then $X_i = \theta + \sigma Y_i$, and using this, $W = (Y_1 + Y_2)/|Y_1 - Y_2|$. Since Y_1 and Y_2 are independent, (Y_1, Y_2) has joint density $g(y_1)g(y_2)$, which does not depend on θ or σ . Because W is a function of Y_1 and Y_2 , its distribution does not depend on θ or σ .

c) Let $q_{\alpha/2}$ and $q_{1-\alpha/2}$ denote the upper and lower $(\alpha/2)$ th quantiles for the distribution of W . Then

$$\begin{aligned} 1 - \alpha &= P\left(q_{1-\alpha/2} < \frac{X_1 + X_2 - 2\theta}{|X_1 - X_2|} < q_{\alpha/2}\right) \\ &= P\left[\theta \in \left(\bar{X} - \frac{1}{2}|X_1 - X_2|q_{\alpha/2}, \bar{X} + \frac{1}{2}|X_1 - X_2|q_{1-\alpha/2}\right)\right], \end{aligned}$$

and the interval in this expression is a $1 - \alpha$ confidence interval for θ .

d) The variable $V = |X_1 - X_2|/\sigma = |Y_1 - Y_2|$ is a pivot. If $q_{\alpha/2}$ and $q_{1-\alpha/2}$ are the upper and lower $(\alpha/2)$ th quantiles for the distribution of V , then

$$\begin{aligned} 1 - \alpha &= P\left(q_{1-\alpha/2} < \frac{|X_1 - X_2|}{\sigma} < q_{\alpha/2}\right) \\ &= P\left[\sigma \in \left(\frac{|X_1 - X_2|}{q_{\alpha/2}}, \frac{|X_1 - X_2|}{q_{1-\alpha/2}}\right)\right], \end{aligned}$$

and the interval in this expression is a $1 - \alpha$ confidence interval for σ .

12. By the addition law,

$$\begin{aligned} P_\theta(g(\theta) \in S_1 \cap S_2) &= P_\theta(g(\theta) \in S_1) + P_\theta(g(\theta) \in S_2) \\ &\quad - P_\theta(g(\theta) \in S_1 \cup S_2) \\ &\geq P_\theta(g(\theta) \in S_1) + P_\theta(g(\theta) \in S_2) - 1 \\ &\geq 1 - 2\alpha. \end{aligned}$$

13. a) Multiplying the marginal density of X_i times the conditional density of Y_i given X_i , the joint density of (X_i, Y_i) is $\exp\{-x^2/2 - (y - x\theta)^2/2\}/(2\pi)$. So the joint density for the entire sample is $\prod_{i=1}^n [\exp\{-x_i^2/2 - (y_i - x_i\theta)^2/2\}/(2\pi)]$, and the log-likelihood function is

$$l_n(\theta) = -\frac{1}{2} \sum_{i=1}^n X_i^2 - \frac{1}{2} \sum_{i=1}^n (Y_i - X_i\theta)^2 - n \log(2\pi).$$

This is a quadratic function of θ , maximized when

$$l'_n(\theta) = -2 \sum_{i=1}^n X_i(Y_i - X_i\theta) = 0.$$

This gives $\hat{\theta} = \sum_{i=1}^n X_i Y_i / \sum_{i=1}^n X_i^2$.

b) Fisher information is

$$I(\theta) = -E_{\theta} \frac{\partial^2 \log f_{\theta}(X, Y)}{\partial \theta^2} = E_{\theta} X^2 = 1.$$

c) As $n \rightarrow \infty$, $\sqrt{n}(\hat{\theta} - \theta) \Rightarrow N(0, 1)$.

d) $(\hat{\theta} \pm z_{\alpha/2}/\sqrt{n})$.

e) The observed Fisher information is $-l''_n(\hat{\theta}) = \sum_{i=1}^n X_i^2$, and the associated confidence interval is $(\hat{\theta} \pm z_{\alpha/2}/[\sum_{i=1}^n X_i^2]^{1/2})$. The main difference is that now the width of the interval varies according to the observed information.

f) Given $X_1 = x_1, \dots, X_n = x_n$, the variables $Y_1, \dots, Y_n$ are conditionally independent with $N(x_i\theta, 1)$ as the marginal distribution for Y_i . So, given $X_1 = x_1, \dots, X_n = x_n$, $\sum_{i=1}^n X_i Y_i \sim N(\theta \sum_{i=1}^n x_i^2, \sum_{i=1}^n x_i^2)$. From this, the conditional distribution of $[\sum_{i=1}^n X_i^2]^{1/2}(\hat{\theta} - \theta)$ given $X_1 = x_1, \dots, X_n = x_n$ is $N(0, 1)$. By smoothing,

$$\begin{aligned} P\left[\left(\sum_{i=1}^n X_i^2\right)^{1/2}(\hat{\theta} - \theta) \leq x\right] &= EP\left[\left(\sum_{i=1}^n X_i^2\right)^{1/2}(\hat{\theta} - \theta) \leq x \mid X_1, \dots, X_n\right] \\ &= E\Phi(x) = \Phi(x). \end{aligned}$$

So $[\sum_{i=1}^n X_i^2]^{1/2}(\hat{\theta} - \theta) \sim N(0, 1)$, and using this it is easy to show that the coverage probability for the interval in part (e) is exactly $1 - \alpha$.

15. Since the Fisher information for a single Bernoulli observation is $I(p) = 1/[p(1-p)]$, the first two confidence regions/intervals are

$$\begin{aligned} \text{CI}_1 &= \left\{ p : \frac{\sqrt{n}|\hat{p} - p|}{\sqrt{p(1-p)}} < z_{\alpha/2} \right\} \\ &= \left(\frac{\hat{p} + z_{\alpha/2}^2/(2n)}{1 + z_{\alpha/2}^2/n} \pm \frac{z_{\alpha/2}\sqrt{\hat{p}(1-\hat{p}) + z_{\alpha/2}^2/(4n)}}{\sqrt{n}(1 + z_{\alpha/2}^2/n)} \right), \end{aligned}$$

and

$$\text{CI}_2 = \left(\hat{p} \pm z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \right).$$

Since $l_n(p) = \log \binom{n}{X} + n\hat{p} \log p + n(1-\hat{p}) \log(1-p)$, the observed Fisher information is $-l''_n(\hat{p}) = n/[\hat{p}(1-\hat{p})]$. Using this, CI_3 , based on the observed

Fisher information, is the same as CI_2 . Finally, the profile confidence region is

$$CR_4 = \{p : \hat{p} \log[\hat{p}/p] + (1 - \hat{p}) \log[(1 - \hat{p})/(1 - p)] < z_{\alpha/2}^2/(2n)\}.$$

This region is an interval, but it can only be found numerically. With the stated data, the confidence intervals are

$$\begin{aligned} CI_1 &= (0.2189, 0.3959), \\ CI_2 &= CI_3 = (0.2102, 0.3898), \end{aligned}$$

and

$$CI_4 = (0.2160, 0.3941).$$

20. a) The joint densities are $p^{x_1}(1-p)p^{2x_2}(1-p^2)$, an exponential family with sufficient statistic $T = X_1 + 2X_2$. The maximum likelihood estimator solves

$$l'(p) = -\frac{(3+T)p^2 + p - T}{p(1-p^2)} = 0,$$

which gives

$$\hat{p} = \hat{p}(T) = \frac{-1 + \sqrt{1 + 4T(3+T)}}{6 + 2T}.$$

- b) Since

$$\begin{aligned} P(Y = y) &= \sum_{x=0}^y P(X_1 = y - x, X_2 = x) \\ &= p^y(1-p)(1-p^2)(1+p+\cdots+p^y) \\ &= p^y(1-p^2)(1-p^{1+y}), \end{aligned}$$

we have

$$P(X_2 = x|Y = y) = \frac{P(X_1 = y - x, X_2 = x)}{P(Y = y)} = \frac{p^x(1-p)}{1-p^{1+y}}$$

and

$$\begin{aligned} e(y, p) &= E(T|Y = y) = y + E(X_2|Y = y) \\ &= y + \frac{1-p}{1-p^{1+y}} \sum_{x=0}^y xp^x = y + \frac{p(1-p^y) - p(1-p)yp^y}{(1-p)(1-p^{1+y})}. \end{aligned}$$

The algorithm then evolves with $\hat{p}_j = \hat{p}(T_j)$ and $T_{j+1} = e(y, \hat{p}_j)$.

- c) The iterates are $T_1 = 124/21 = 5.9048$, $\hat{p}_1 = 0.76009$, $T_2 = 6.7348$, and $\hat{p}_2 = 0.78198$.

21. a) The joint densities are $\exp[\theta T(x) - n \log[(2 \sinh \theta)/\theta]]$, where $T(x) = x_1 + \cdots + x_n$. This is an exponential family, and the maximum likelihood estimator $\hat{\theta}_x$ is the unique solution of the equation

$$\frac{T}{n} = \frac{A'(\theta)}{n} = \coth \theta - \frac{1}{\theta}.$$

b) Data $Y_1, \dots, Y_n$ are i.i.d. Bernoulli variables with success probability $p = P(X_i > 0) = (e^\theta - 1)/(e^\theta - e^{-\theta})$. Noting that $1 - p = (1 - e^{-\theta})/(e^\theta - e^{-\theta}) = p/e^\theta$, $\theta = \log[p/(1-p)]$, and the relation between θ and p is one-to-one. Naturally, the maximum likelihood estimator of p based on $Y_1, \dots, Y_n$ is $\hat{p} = (Y_1 + \cdots + Y_n)/n$, and so $\hat{\theta}_y = \log[\hat{p}/(1-\hat{p})]$.

c) From the independence, $E[X_j|Y = y] = E[X_j|Y_j = y_j]$, which is

$$\begin{aligned} \frac{\int_0^1 \theta x e^{\theta x} dx}{\int_0^1 \theta e^{\theta x} dx} &= 1 - \frac{1}{\theta} + \frac{1}{e^\theta - 1}, & \text{if } y_j = 1, \\ \frac{\int_{-1}^0 \theta x e^{\theta x} dx}{\int_{-1}^0 \theta e^{\theta x} dx} &= -\frac{1}{\theta} + \frac{1}{e^\theta - 1}, & \text{if } y_j = 0. \end{aligned}$$

So $E[T|Y] = \sum_{i=1}^n Y_i - n/\theta + n/(e^\theta - 1)$.

d) Because

$$\sum_{i=1}^n Y_i = n\hat{p} = \frac{n(e^{\hat{\theta}_y} - 1)}{e^{\hat{\theta}_y} - e^{-\hat{\theta}_y}},$$

if we start the algorithm at $\hat{\theta}_y$, then

$$\frac{T_1}{n} = \frac{e^{\hat{\theta}_y} - 1}{e^{\hat{\theta}_y} - e^{-\hat{\theta}_y}} - \frac{1}{\hat{\theta}_y} + \frac{1}{e^{\hat{\theta}_y} - 1} = \coth(\hat{\theta}_y) - \frac{1}{\hat{\theta}_y}.$$

From the equation in part (a), the next estimate, $\hat{\theta}_1$, will also be $\hat{\theta}_y$.

e) The iterates are $T_1 = 1/2$, $\hat{\theta}_1 = 0.30182$, $T_2 = 0.62557$, and $\hat{\theta}_2 = 0.37892$.

27. a) Since $\sqrt{n}(\hat{\theta}_n - \theta) \Rightarrow Y \sim N(0, I^{-1}(\theta))$ and $\sqrt{n}(\hat{\eta}_n - \eta) = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \cdot \sqrt{n}(\hat{\theta}_n - \theta)$, we have $\sqrt{n}(\hat{\eta}_n - \eta) \Rightarrow \begin{pmatrix} 1 \\ 0 \end{pmatrix} \cdot Y \sim N(0, \tau^2)$ with

$$\tau^2 = (1, 0)I^{-1}(\theta)\begin{pmatrix} 1 \\ 0 \end{pmatrix} = [I^{-1}(\theta)]_{11} = \frac{I_{22}}{I_{11}I_{22} - I_{12}^2}.$$

b) The limiting variance is $\nu^2 = 1/E[\partial \log f_\theta(X)/\partial \eta]^2 = 1/I_{11}$. So $\nu^2 \geq \tau^2$ if and only if $I_{11}I_{22}/(I_{11}I_{22} - I_{12}^2) \leq 1$, which always holds, and $\nu^2 = \tau^2$ if and only if $I_{12} = 0$.

c) If $I(\cdot)$ is continuous, then

$$\hat{\tau}^2 = \frac{I_{22}(\hat{\theta})}{I_{11}(\hat{\theta})I_{22}(\hat{\theta}) - I_{12}(\hat{\theta})^2}$$

is a consistent estimator of τ^2 . Then $\sqrt{n}(\hat{\eta}_n - \eta)/\hat{\tau} \Rightarrow N(0, 1)$, and using this asymptotic pivot, $(\hat{\eta}_n \pm z_{\alpha/2}\hat{\tau}/\sqrt{n})$ is an asymptotic $1 - \alpha$ confidence interval for η .

d) The observed Fisher information divided by n , $\tilde{I} = -\nabla^2 l(\hat{\theta}_n)/n$, is a consistent estimator of $I(\theta)$, and so

$$\tilde{\tau}^2 = \frac{\tilde{I}_{22}}{\tilde{I}_{11}\tilde{I}_{22} - \tilde{I}_{12}^2}$$

is a consistent estimator of τ^2 . Continuing as in part (c), $(\hat{\eta}_n \pm z_{\alpha/2}\tilde{\tau}/\sqrt{n})$ is an asymptotic $1 - \alpha$ confidence interval for η .

29. a) The joint density of W, X, Y is

$$f_{\alpha, \beta}(w, x, y) = q(w, x) \frac{\exp\left[-\frac{1}{2}(y - \alpha w - \beta x)^2\right]}{\sqrt{2\pi}},$$

and so the Fisher information $I(\alpha, \beta)$ is

$$-E\nabla^2 \log f_{\alpha, \beta}(W, X, Y) = E \begin{pmatrix} W^2 & WX \\ WX & X^2 \end{pmatrix} = \begin{pmatrix} EW^2 & EWX \\ EWX & EX^2 \end{pmatrix}.$$

The gradient of the log-likelihood is

$$\nabla l(\alpha, \beta) = \begin{pmatrix} \sum_{i=1}^n W_i(Y_i - \alpha W_i - \beta X_i) \\ \sum_{i=1}^n X_i(Y_i - \alpha W_i - \beta X_i) \end{pmatrix}.$$

Setting this equal to zero, the maximum likelihood estimators are

$$\hat{\alpha} = \frac{(\sum_{i=1}^n X_i^2)(\sum_{i=1}^n W_i Y_i) - (\sum_{i=1}^n W_i X_i)(\sum_{i=1}^n X_i Y_i)}{(\sum_{i=1}^n X_i^2)(\sum_{i=1}^n W_i^2) - (\sum_{i=1}^n W_i X_i)^2}$$

and

$$\hat{\beta} = \frac{(\sum_{i=1}^n W_i^2)(\sum_{i=1}^n X_i Y_i) - (\sum_{i=1}^n W_i X_i)(\sum_{i=1}^n W_i Y_i)}{(\sum_{i=1}^n X_i^2)(\sum_{i=1}^n W_i^2) - (\sum_{i=1}^n W_i X_i)^2}.$$

Since $\sqrt{n}(\hat{\theta} - \theta) \Rightarrow N(0, I^{-1})$, $\sqrt{n}(\hat{\alpha} - \alpha) \Rightarrow N(0, [I(\alpha, \beta)^{-1}]_{1,1})$ with $[I(\alpha, \beta)^{-1}]_{1,1} = EX^2/(EX^2EW^2 - (EXW)^2)$.

b) If β is known, then $\hat{\alpha}$ solves $l'(\alpha) = \sum_{i=1}^n W_i(Y_i - \alpha W_i - \beta X_i) = 0$, which gives $\hat{\alpha} = (\sum_{i=1}^n W_i Y_i - \beta \sum_{i=1}^n W_i X_i) / \sum_{i=1}^n W_i^2$. The Fisher information is just EW^2 in this case, and so $\sqrt{n}(\hat{\alpha} - \alpha) \Rightarrow N(0, 1/EW^2)$. This is the same as the limiting distribution in part (a) when $EXW = 0$.

Otherwise the distribution in part (a) has larger variance.

30. By Taylor expansion about θ ,

$$g(\hat{\theta}_n) = g(\theta) + \nabla g(\tilde{\theta}_n) \cdot (\hat{\theta}_n - \theta),$$

where $\tilde{\theta}_n$ is an intermediate value on the line segment between $\hat{\theta}_n$ and θ . Since $\hat{\theta}_n$ is consistent, $\tilde{\theta}_n \xrightarrow{P_\theta} \theta$, and so $\nabla g(\tilde{\theta}_n) \xrightarrow{P_\theta} \nabla g(\theta)$. Because $\sqrt{n}(\hat{\theta}_n - \theta) \Rightarrow Z \sim N(0, I(\theta)^{-1})$, using Theorem 9.30,

$$\sqrt{n}(g(\hat{\theta}_n) - g(\theta)) = \nabla g(\tilde{\theta}_n) \cdot \sqrt{n}(\hat{\theta}_n - \theta) \Rightarrow \nabla g(\theta) \cdot Z \sim N(0, \nu(\theta)),$$

where $\nu(\theta) = \nabla g(\theta)' I(\theta)^{-1} \nabla g(\theta)$. This proves Proposition 9.31. To show that (9.13) is a $1 - \alpha$ asymptotic confidence interval, if $\hat{\nu} \xrightarrow{P_\theta} \nu(\theta)$, then $1/\sqrt{\hat{\nu}} \xrightarrow{P_\theta} 1/\sqrt{\nu(\theta)}$ by Proposition 8.5. If $Y \sim N(0, \nu(\theta))$, then using Theorem 8.13,

$$\frac{1}{\sqrt{\hat{\nu}}} \sqrt{n}(g(\hat{\theta}_n) - g(\theta)) \Rightarrow \frac{1}{\sqrt{\nu(\theta)}} Y \sim N(0, 1).$$

One natural estimate for ν is $\hat{\nu}_1 = \nu(\hat{\theta}_n)$. Another estimator, based on observed Fisher information, is

$$\hat{\nu}_2 = -n \nabla g(\hat{\theta}_n)' (\nabla^2 l(\hat{\theta}_n))^{-1} \nabla g(\hat{\theta}_n).$$

The asymptotic confidence intervals are $(g(\hat{\theta}) \pm z_{\alpha/2} \sqrt{\hat{\nu}_i/n})$, $i = 1$ or 2 .

31. The likelihood is

$$L(\theta) = \binom{n}{N_{11}, \dots, N_{22}} \theta_1^{N_{+1}} (1 - \theta_1)^{N_{+2}} \theta_2^{N_{1+}} (1 - \theta_2)^{N_{2+}},$$

and maximum likelihood estimators for θ_1 and θ_2 are $\hat{\theta}_1 = N_{+1}/n$ and $\hat{\theta}_2 = N_{1+}/n$. The observed Fisher information matrix is

$$\begin{aligned} -\nabla^2 l(\hat{\theta}) &= \begin{pmatrix} \frac{N_{+1}}{\hat{\theta}_1^2} + \frac{N_{+2}}{(1 - \hat{\theta}_1)^2} & 0 \\ 0 & \frac{N_{1+}}{\hat{\theta}_2^2} + \frac{N_{2+}}{(1 - \hat{\theta}_2)^2} \end{pmatrix} \\ &= \begin{pmatrix} \frac{n}{\hat{\theta}_1(1 - \hat{\theta}_1)} & 0 \\ 0 & \frac{n}{\hat{\theta}_2(1 - \hat{\theta}_2)} \end{pmatrix}. \end{aligned}$$

Since $-\nabla^2 l(\hat{\theta})/n \xrightarrow{P_\theta} I(\theta)$,

$$I(\theta) = \begin{pmatrix} \frac{1}{\theta_1(1 - \theta_1)} & 0 \\ 0 & \frac{1}{\theta_2(1 - \theta_2)} \end{pmatrix}.$$

In this example, $g(\theta) = \theta_1 \theta_2$ and the two estimates of $\nu(\theta)$ are the same:

$$(\hat{\theta}_2, \hat{\theta}_1) \begin{pmatrix} \hat{\theta}_1(1 - \hat{\theta}_1) & 0 \\ 0 & \hat{\theta}_2(1 - \hat{\theta}_2) \end{pmatrix} \begin{pmatrix} \hat{\theta}_2 \\ \hat{\theta}_1 \end{pmatrix} = \hat{\theta}_1 \hat{\theta}_2 (\hat{\theta}_1 + \hat{\theta}_2 - 2\hat{\theta}_1 \hat{\theta}_2).$$

The confidence intervals are both

$$\left(\hat{\theta}_1 \hat{\theta}_2 \pm z_{\alpha/2} \sqrt{\frac{\hat{\theta}_1 \hat{\theta}_2 (\hat{\theta}_1 + \hat{\theta}_2 - 2\hat{\theta}_1 \hat{\theta}_2)}{n}} \right).$$

B.10 Problems of Chapter 10

1. The joint density can be written as

$$\frac{\exp \left[-\frac{1}{2}(x_1 - \theta)^2 - \frac{1}{2} \sum_{j=1}^{n-1} (x_{j+1} - \theta - \frac{1}{2}(x_j - \theta))^2 \right]}{\sqrt{2\pi}^n},$$

so we have a location family. Ignoring terms that do not depend on θ , the likelihood is proportional to

$$\begin{aligned} & \exp \left[X_1 \theta - \frac{1}{2} \theta^2 + \frac{1}{2} \sum_{j=1}^{n-1} (X_{j+1} - \frac{1}{2} X_j) \theta - \frac{1}{8} (n-1) \theta^2 \right] \\ & = \exp \left[\frac{1}{4} T \theta - \frac{1}{8} (n+3) \theta^2 \right] \propto \exp \left[-\frac{n+3}{8} \left(\theta - \frac{T}{n+3} \right)^2 \right], \end{aligned}$$

where $T = (3X_1 + X_2 + \cdots + X_{n-1} + 2X_n)$. This likelihood is proportional to a normal density with mean $T/(n+3)$. Because the minimum risk equivariant estimator is the mean of the normalized likelihood, it must be $T/(n+3)$.

2. a) By dominated convergence, if c is a continuity point of F ,

$$g'(c) = E \frac{\partial}{\partial c} |X - c| = -E \text{Sign}(X - c) = P(X < c) - P(X > c),$$

which is zero if c is the median.

- b) Dominated convergence (when c is a continuity point of F) gives

$$g'(c) = E[-aI\{X > c\} + bI\{X < c\}] = bP(X < c) - aP(X > c),$$

which equals zero if $P(X < c) = a/(a+b)$. So g is minimized if c is the $(a/(a+b))$ th quantile of F .

4. An equivariant estimator must have form $X - c$, with minimal risk if c is chosen to minimize

$$E_{\theta}L(\theta, X - c) = E_0[a(X - c)^+ + b(c - X)].$$

From Problem 10.2, c should be the $(a/(a+b))$ th quantile of the P_0 -cumulative distribution of X , given by $F(t) = \frac{1}{2} \int_{-\infty}^t e^{-|x|} dx$. Solving $F(t) = a/(a+b)$, $c = \log[2a/(a+b)]$ if $a \leq b$, and $c = -\log[2b/(a+b)]$ if $a \geq b$.

B.11 Problems of Chapter 11

1. a) The joint density of Λ_i with X_i is $\lambda^{\alpha} e^{-\lambda(1+x)}/\Gamma(\alpha)$. So the marginal density of X_i is

$$\int_0^{\infty} \frac{\lambda^{\alpha} e^{-\lambda(1+x)}}{\Gamma(\alpha)} d\lambda = \frac{\alpha}{(1+x)^{1+\alpha}}, \quad x > 0.$$

b) Dividing the joint density by the marginal density of X_i , the conditional density of Λ_i given $X_i = x$ is

$$\frac{(1+x)^{1+\alpha} \lambda^{\alpha} e^{-\lambda(1+x)}}{\Gamma(1+\alpha)},$$

a gamma density with shape parameter $1 + \alpha$ and scale $1/(1+x)$. The Bayes estimator is

$$E(\Lambda_i | X_i) = \frac{1 + \alpha}{1 + X_i}.$$

c) From part (a), the joint density is

$$\prod_{i=1}^p \frac{\alpha}{(1+x_i)^{1+\alpha}} = \exp \left[-(1+\alpha) \sum_{i=1}^p \log(1+x_i) + p \log \alpha \right].$$

So the log-likelihood is $l(\alpha) = -(1+\alpha) \sum_{i=1}^p \log(1+X_i) + p \log \alpha$. Then $l'(\alpha) = -\sum_{i=1}^p \log(1+X_i) + p/\alpha$ which is zero when $\alpha = p / \sum_{i=1}^p \log(1+X_i)$. This is the maximum likelihood estimator.

d) The empirical Bayes estimator for Λ_i is

$$\frac{1 + p / \sum_{j=1}^p \log(1+X_j)}{1 + X_i}.$$

2. a) Direct calculation shows that

$$\Theta_i | X = x_i, Y = y_i \sim N \left(\frac{x_i y_i \tau^2}{1 + x_i^2 \tau^2}, \frac{\tau^2}{1 + x_i^2 \tau^2} \right).$$

The Bayes estimate is $X_i Y_i \tau^2 / (1 + X_i^2 \tau^2)$.

b) By smoothing, $EY_i^2 = EE[Y_i^2|X_i, \Theta_i] = E[X_i^2\theta_i^2 + 1] = 1 + \tau^2$. A simple estimator of τ^2 is

$$\hat{\tau}^2 = \frac{1}{p} \sum_{i=1}^p Y_i^2 - 1.$$

c) The empirical Bayes estimator is

$$\hat{\theta}_i = \frac{X_i Y_i \hat{\tau}^2}{1 + X_i^2 \hat{\tau}^2}.$$

3. a) and b) The joint density of Θ_i with X_i is $\lambda \theta^x e^{-(1+\lambda)\theta}/x!$, the marginal density of X_i is

$$\int_0^\infty \frac{\lambda \theta^x e^{-(1+\lambda)\theta}}{x!} d\theta = \frac{\lambda}{(1+\lambda)^{x+1}},$$

the conditional density of Θ_i given $X_i = x$ is

$$\frac{(1+\lambda)^{x+1} \theta^x e^{-(1+\lambda)\theta}}{x!},$$

and

$$E[\Theta_i|X_i = x] = \int_0^\infty \frac{(1+\lambda)^{x+1} \theta^{x+1} e^{-(1+\lambda)\theta}}{x!} d\theta = \frac{x+1}{1+\lambda}.$$

By independence, $E[\Theta_i|X] = E[\Theta_i|X_i] = (X_i + 1)/(1 + \lambda)$, which is the Bayes estimate of Θ_i under compound squared error loss.

c) The joint density of $X_1, \dots, X_p$ in the Bayesian model is

$$\prod_{i=1}^p \frac{\lambda}{(1+\lambda)^{x_i+1}} = \exp \left[-T(x) \log(1+\lambda) + p \log \left(\frac{\lambda}{1+\lambda} \right) \right],$$

where $T(x) = x_1 + \dots + x_p$. The maximum likelihood estimator $\hat{\lambda}$ of λ solves

$$0 = -\frac{T}{1+\lambda} + \frac{p}{\lambda} - \frac{p}{1+\lambda} = \frac{-\lambda T + p}{\lambda(1+\lambda)},$$

giving $\hat{\lambda} = p/T = 1/\bar{X}$.

d) The empirical Bayes estimator for θ_i is $(X_i + 1)/(1 + 1/\bar{X})$.

B.12 Problems of Chapter 12

1. By smoothing, $E_\theta \psi = P_\theta((X, U) \in S) = E_\theta P_\theta((X, U) \in S | X)$. This will equal $E_\theta \varphi(X)$ if we can choose S so that $P_\theta((X, U) \in S | X = x) = \varphi(x)$. One solution is $S = \{(x, u) : u < \varphi(x)\}$.

2. This is like the Neyman–Pearson lemma. If f_0 and f_1 are densities for $N(0, 1)$ and $N(0, 4)$, then we want to maximize $\int h(x)f_1(x) dx$ with $\int h(x)f_0(x) dx = 0$. Adding a Lagrange multiplier, let us try to maximize $\int h(x)[f_1(x) - kf_0(x)] dx$. Here there is a bit of a difference. Because h has range $[-M, M]$ instead of $[0, 1]$, an optimal function h^* will satisfy

$$h^*(x) = M, \text{ if } \frac{f_1(x)}{f_0(x)} > k, \text{ and } h^*(x) = -M, \text{ if } \frac{f_1(x)}{f_0(x)} < k.$$

The likelihood ratio is $\frac{1}{2}e^{3x^2/8}$, so equivalently,

$$h^*(x) = M, \text{ if } |x| > k', \text{ and } h^*(x) = -M, \text{ if } |x| < k'.$$

To satisfy the constraint, $k' = \Phi^{-1}(3/4) = 0.67449$. Then $Eh^*(2Z) = 0.47186M$. (You can also solve this problem applying the Neyman–Pearson lemma to the test function $\varphi = (h + M)/(2M)$.)

3. By smoothing,

$$P(Z_1/Z_2 \leq x) = EP(Z_1/Z_2 \leq x|Z_2) = E\Phi(x|Z_2|).$$

(By symmetry, this is true regardless of the sign of Z_2 .) Taking d/dx , the density of Z_1/Z_2 is

$$\begin{aligned} E|Z_2|\phi(x|Z_2|) &= \frac{1}{2\pi} \int |z|e^{-(z^2+x^2z^2)/2} dz \\ &= \frac{1}{\pi} \int_0^\infty ze^{-(1+x^2)z^2/2} dz = -\frac{e^{-(1+x^2)z^2/2}}{\pi(1+x^2)} \Big|_0^\infty = \frac{1}{\pi(1+x^2)}. \end{aligned}$$

4. The likelihood ratio is

$$L = \exp\left[(X_1^2 - X_2^2) \left(\frac{1}{2\sigma_1^2} - \frac{1}{2\sigma_2^2}\right)\right].$$

If we assume $\sigma_1^2 < \sigma_2^2$, then Neyman–Pearson likelihood ratio tests will reject H_0 if $X_1^2 - X_2^2 \geq k$, and for a symmetric test, k should be zero. Taking $Z_1 = X_1/\sigma_1$ and $Z_2 = X_2/\sigma_2$, the error probability under H_0 is

$$\begin{aligned} P(X_1^2 - X_2^2 \geq 0) &= P(\sigma_1^2 Z_1^2 \geq \sigma_2^2 Z_2^2) \\ &= P(|Z_2/Z_1| \leq \sigma_1/\sigma_2) = \frac{2}{\pi} \tan^{-1}(\sigma_1/\sigma_2). \end{aligned}$$

If $\sigma_1^2 > \sigma_2^2$, the error rate is $2 \tan^{-1}(\sigma_2/\sigma_1)/\pi$.

6. a) By a change of variables,

$$Eh(X^2/2) = \frac{1}{2} \int_0^2 h(x^2/2) dx = \int_0^2 \frac{h(y)}{2\sqrt{2y}} dy,$$

and we want to maximize this integral with the constraint $\int_0^2 h(y) dy = 0$. Introducing a Lagrange multiplier, consider maximizing

$$\int_0^2 \left[\frac{1}{2\sqrt{2}y} - k \right] h(y) dy$$

without constraint. An optimal solution will have $h^*(y) = M$ when $y < 1/(8k^2)$ and $h^*(y) = -M$ when $y > 1/(8k^2)$. This solution will satisfy the constraint if $k = 1/\sqrt{8}$. This gives an upper bound of $(\sqrt{2} - 1)M$, and $h^*(y) = M \text{Sign}(1 - y)$ as a function achieving the bound.

b) Introducing a Lagrange multiplier and proceeding in the same fashion, consider maximizing

$$\int_0^2 \left[\frac{1}{2\sqrt{2}y} - k \right] h(y) dy$$

without constraint. Now an optimal solution will be $h^*(y) = My \text{Sign}(c - y)$ with $c = 1/(8k^2)$. Then $Eh^*(X) = M(c^2 - 2)/2$ so h^* will satisfy the constraint if $c = \sqrt{2}$, giving an upper bound of

$$Eh^*(X^2/2) = \int_0^{2^{3/4}} \frac{Mx^2}{4} dx - \int_{2^{3/4}}^2 \frac{Mx^2}{4} dx = \frac{2}{3}(2^{1/4} - 1)M.$$

7. a) If we can interchange differentiation and integration, then

$$\begin{aligned} \beta'(\theta) &= \frac{d}{d\theta} \int_0^\infty \varphi(x) p_\theta(x) dx = \int_0^\infty \varphi(x) \frac{\partial}{\partial \theta} p_\theta(x) dx \\ &= \int_0^\infty \frac{1 - \theta x}{\theta(1 + \theta x)} \varphi(x) p_\theta(x) dx = E_\theta \left[\frac{1 - \theta X}{\theta(1 + \theta X)} \varphi(X) \right]. \end{aligned}$$

Dominated convergence can be used to justify the interchange. Note that $\partial p_\theta(x)/\partial \theta = (1 - \theta x)/(1 + \theta x)^3$. Let $h = h_n$ be a sequence of positive constants all less than θ converging to zero, and let $\xi = \xi_n(x)$ be intermediate values in $[\theta, 2\theta]$ chosen so that

$$\frac{\varphi(x)(p_{\theta+h}(x) - p_\theta(x))}{h} = \frac{\varphi(x)(1 - \xi x)}{(1 + \xi x)^3}.$$

These functions converge pointwise to $\varphi(x)\partial p_\theta(x)/\partial \theta$ and are uniformly bounded in magnitude by $(1 + 2\theta x)/(1 + \theta x)^3$, an integrable function.

b) Introducing a Lagrange multiplier k , consider unconstrained maximization of

$$\beta'_\varphi(1) - k\beta_\varphi(1) = E_1 \left(\frac{1 - X}{1 + X} - k \right) \varphi.$$

An optimal test function φ^* should equal one when $(1 - X)/(1 + X) > k$ and zero, otherwise. Because $(1 - X)/(1 + X)$ is a decreasing function of X , this gives $\varphi^* = I\{X < c\}$, where $k = (1 - c)/(1 + c)$. This test has level $P_1(X < c) = c/(1 + c)$. If $c = \alpha/(1 - \alpha)$, or, equivalently, $k = 1 - 2\alpha$, then φ^* has level α , satisfying the constraint and maximizing $\beta'_\varphi(1)$ among all tests with $\beta_\varphi(1) = \alpha$.

9. Let $v_\theta(x) = \partial \log p_\theta(x) / \partial \theta$. Then the test φ^* should maximize $\int \varphi v_{\theta_0} p_{\theta_0} d\mu$ subject to the constraint $\int \varphi p_{\theta_0} d\mu = \alpha$. Introducing a Lagrange multiplier and arguing as in the Neyman–Pearson lemma, φ^* should have form

$$\varphi^*(x) = \begin{cases} 1, & v_{\theta_0}(x) > k; \\ 0, & v_{\theta_0}(x) < k. \end{cases}$$

10. Since $\log p_\theta(x) = \sum_{i=1}^n \log f_\theta(x_i)$, $v_\theta(x) = \sum_{i=1}^n s_\theta(x_i)$, where $s_\theta(x_i) = \partial \log f_\theta(x_i) / \partial \theta$. Note that $s_\theta(X_1), \dots, s_\theta(X_n)$ are i.i.d., $E_\theta s_\theta(X_i) = 0$, and $\text{Var}_\theta[s_\theta(X_i)] = I(\theta)$, the Fisher information from a single observation. By the central limit theorem, under P_θ , $v_\theta(X)/\sqrt{n} \Rightarrow N(0, I(\theta))$ as $n \rightarrow \infty$. So the level of the test φ^* in Problem 12.9 is approximately

$$P_{\theta_0}(v_{\theta_0}(X)/\sqrt{n} > k/\sqrt{n}) \approx 1 - \Phi(k/\sqrt{nI(\theta_0)}),$$

which is α if $k = z_\alpha \sqrt{nI(\theta_0)}$.

14. The mass function for X is $p(1-p)^x$, $x \geq 0$, which is an exponential family with canonical parameter $\eta = \log(1-p)$. The two hypotheses can be expressed in terms of η as $H_0: \eta \leq -\log 2$ and $H_1: \eta > -\log 2$. With η as the parameter the densities have monotone likelihood ratios in $T = X$, and so there will be a uniformly most powerful test with form $\varphi = 0$ if $X < k$, $\varphi = 1$ if $X > k$, and $\varphi = \gamma$ if $X = k$. For a fair coin ($p = 1/2$), $P(X \leq 3) = 93.75\%$ and $P(X = 4) = 3.125\%$. If $k = 4$ and $\gamma = 3/5$ the test will have level $\alpha = 5\%$. The power if $p = 40\%$ is 10.8864%.
15. The joint density for the data is

$$\frac{\sqrt{1-\rho^2}}{2\pi} \exp[\rho xy - (x^2 + y^2)/2],$$

which is an exponential family with $T = XY$. So, the uniformly most powerful test will reject H_0 if $XY > k$. Adjusting k to achieve a given level α is a bit tricky. The null density of $T = XY$ is $K_0(|t|/2)/(2\pi)$, where K_0 is a Bessel function. Numerical calculations using this density show that $\alpha = 5\%$ when $k = 3.19$.

16. a) Let $h = \log g$. The family will have monotone likelihood ratios in x if

$$\log \left(\frac{p_{\theta_2}(x)}{p_{\theta_1}(x)} \right) = h(x - \theta_2) - h(x - \theta_1)$$

is nondecreasing in x whenever $\theta_2 > \theta_1$. A sufficient condition for this is that $h'(x - \theta_2) - h'(x - \theta_1) \geq 0$. Since $x - \theta_2 \leq x - \theta_1$, this will hold if h' is nonincreasing, which follows if $h''(x) = d^2 \log g(x) / dx^2 \leq 0$.

b) As in part (a), a sufficient condition will be that the derivative of $\log(p_{\theta_2}(x)/p_{\theta_1}(x))$ is at least zero whenever $\theta_2 > \theta_1$, and this derivative is

$$\frac{1}{x} \left(\frac{g'(x/\theta_2)x/\theta_2}{g(x/\theta_2)} - \frac{g'(x/\theta_1)x/\theta_1}{g(x/\theta_1)} \right).$$

Since $x/\theta_2 < x/\theta_1$, this will hold if the function $xg'(x)/g(x)$ is nonincreasing, and a sufficient condition for this is

$$\frac{d}{dx} \left(\frac{xg'(x)}{g(x)} \right) = \frac{g(x)g'(x) + xg(x)g''(x) - x[g'(x)]^2}{g^2(x)} \leq 0, \quad x > 0.$$

17. a) Define $F(t) = P_{\theta_0}(T \leq t)$. The uniformly most powerful level α test is $\varphi_\alpha(x) = I\{T(x) > k(\alpha)\}$ with $k(\alpha)$ chosen so that $F(k(\alpha)) = 1 - \alpha$. Suppose $\alpha_0 < \alpha_1$. Then since F is nondecreasing, $k(\alpha_0) > k(\alpha_1)$. So if $T(x) > k(\alpha_0)$, $T(x)$ also exceeds $k(\alpha_1)$, and hence $\varphi_{\alpha_1}(x) = 1$ whenever $\varphi_{\alpha_0}(x) = 1$. Thus $\varphi_{\alpha_1}(x) \geq \varphi_{\alpha_0}(x)$ for all x , and since α_0 and α_1 are arbitrary, $\varphi_\alpha(x)$ is nondecreasing in α .
- b) Because F is nondecreasing and continuous, if $t > k(\alpha)$ then $F(t) \geq F(k(\alpha)) = 1 - \alpha$, and so $P = \inf\{\alpha : t > k(\alpha)\} \geq \inf\{\alpha : F(t) \geq 1 - \alpha\} = 1 - F(t)$. But in addition, if $F(t) > F(k(\alpha)) = 1 - \alpha$, then $t > k(\alpha)$, and so $P = \inf\{\alpha : t > k(\alpha)\} \leq \inf\{\alpha : F(t) > 1 - \alpha\}$, which is again $1 - F(t)$. So the p-value must be $1 - F(t) = P_{\theta_0}(T > t)$.
- c) Let $F^\leftarrow$ denote the largest inverse function of F : $F^\leftarrow(c) = \sup\{t : F(t) = c\}$, $c \in (0, 1)$. Then $F(T) \leq x$ if and only if $T \leq F^\leftarrow(x)$ and $P_{\theta_0}(F(T) \leq x) = P_{\theta_0}(T \leq F^\leftarrow(x)) = F(F^\leftarrow(x)) = x$. So $F(T)$ and the p-value $1 - F(T)$ are both uniformly distributed on $(0, 1)$ under P_{θ_0} .
20. Suppose $\theta_2 > \theta_1$. Then the log-likelihood ratio is

$$l(x) = \log L(x, \theta_1, \theta_2) = \log \frac{p_{\theta_2}(x)}{p_{\theta_1}(x)} = \log \left(\frac{1 + \theta_2 x}{1 + \theta_1 x} \right)$$

and

$$l'(x) = \frac{\theta_2 - \theta_1}{(1 + \theta_2 x)(1 + \theta_1 x)}.$$

This is positive for $x \in (-1, 1)$. So $l(x)$ and $L(x, \theta_1, \theta_2)$ are increasing functions of x , and the family has monotone likelihood ratios in x .

21. Fix $0 < \theta_1 < \theta_2 < 1$. The likelihood ratio

$$\frac{p_{\theta_2}(x)}{p_{\theta_1}(x)} = \frac{\theta_2 + (1 - \theta_2)f(x)}{\theta_1 + (1 - \theta_1)f(x)} = \frac{1 - \theta_2}{1 - \theta_1} + \frac{(\theta_2 - \theta_1)/(1 - \theta_1)}{\theta_1 + (1 - \theta_1)f(x)}$$

is monotone decreasing in $f(x)$, so we can take $T(X) = -f(X)$.

25. a) The joint densities are $f_\lambda(x, u) = \lambda^x e^{-\lambda}/x!$, $u \in (0, 1)$, $x = 0, 1, \dots$. If $[t]$ denotes the greatest integer less than or equal to t , then with $T(x, u) = x + u$, these densities can also be written as $\lambda^{[T(x, u)]} e^{-\lambda}/x!$. (From this, we see that T is sufficient, but not minimal sufficient, because X is also sufficient, and T is not a function of X .) If $\lambda_1 > \lambda_0$, then the likelihood ratio $f_{\lambda_1}(x, u)/f_{\lambda_0}(x, u) = (\lambda_1/\lambda_0)^{[T(x, u)]} e^{\lambda_0 - \lambda_1}$ is an increasing function of T , so the joint densities have monotone likelihood ratios in T .
- b) Let $F_\lambda(t) = P_\lambda(T \leq t)$. For $n = 0, 1, 2, \dots$, $F_\lambda(n) = P_\lambda(X \leq n - 1)$, which can be found summing the Poisson mass function. For nonintegral

values t , $F_\lambda(t)$ is the linear interpolation of the value of F_λ at the two adjacent integers. So, F_λ is strictly increasing and continuous on $(0, \infty)$. The UMP test has form $\varphi = 1$ if $T \geq k$, $\varphi = 0$ if $T < k$. (Randomization on the boundary is unnecessary because T is continuous.) The constant k is chosen (uniquely) so that $F_{\lambda_0}(k) = 1 - \alpha$. For the particular case, $F_2(5) = 0.947347$ and $F_2(6) = 0.9834364$. By linear interpolation, $F_2(5.073519) = 95\%$, and we should reject H_0 if $T \geq 5.073519$.

c) From part (b), if $T = t$, we accept the null hypothesis that the true value of the parameter is λ if and only if $F_\lambda(t) < 1 - \alpha$. For fixed t , $F_\lambda(t)$ is continuous and strictly decreasing on $[0, \infty)$, and so there is a unique value λ_t such that $F_{\lambda_t}(t) = 1 - \alpha$. The confidence interval is (λ_t, ∞) . For data $X = 2$ and $U = 0.7$, the observed value of T is 2.7. As in part (a), $F_\lambda(2.7) = (1 + \lambda + 7\lambda^2/20)e^{-\lambda}$, which is 95% at $\lambda_{2.7} = 0.583407$.

28. As in Example 12.10, the uniformly most powerful test of $\theta = \theta_0$ versus $\theta > \theta_0$ will reject if $T = \max\{X_1, \dots, X_n\} > c$, with c chosen so that

$$P_{\theta_0}(T > c) = 1 - (c/\theta_0)^n = \alpha.$$

Solving, $c = \theta_0(1 - \alpha)^{1/n}$, and the acceptance region for this test is

$$A(\theta_0) = \{x : T(x) < \theta_0(1 - \alpha)^{1/n}\}.$$

The confidence interval S_1 dual to these tests is

$$S_1 = \{\theta : X \in A(\theta)\} = \{\theta : T < \theta(1 - \alpha)^{1/n}\} = (T(1 - \alpha)^{-1/n}, \infty).$$

Similarly, the uniformly most powerful test of $\theta = \theta_0$ versus $\theta < \theta_0$ will reject if $T = \max\{X_1, \dots, X_n\} < c$, with c chosen so that

$$P_{\theta_0}(T < c) = (c/\theta_0)^n = \alpha.$$

This gives $c = \theta_0\alpha^{1/n}$, $A(\theta_0) = \{x : T(x) > \theta_0\alpha^{1/n}\}$, and

$$S_2 = \{\theta : T > \theta\alpha^{1/n}\} = (0, T\alpha^{-1/n}).$$

By the result in Problem 9.12,

$$S = S_1 \cap S_2 = (T(1 - \alpha)^{-1/n}, T\alpha^{-1/n})$$

should have coverage probability at least $1 - 2\alpha$, which is 95% if we take $\alpha = 2.5\%$. (In fact, it is easy to see that the coverage probability is exactly 95%.)

31. a) By dominated convergence,

$$\begin{aligned} \beta'(\theta) &= \frac{d}{d\theta} \int \varphi(x) \frac{1}{\sqrt{2\pi}} e^{-(x-\theta)^2/2} dx \\ &= \int \varphi(x) \frac{1}{\sqrt{2\pi}} \frac{\partial}{\partial \theta} e^{-(x-\theta)^2/2} dx \\ &= \int \varphi(x) \frac{1}{\sqrt{2\pi}} (x - \theta) e^{-(x-\theta)^2/2} dx \\ &= E_\theta[(X - \theta)\varphi(X)]. \end{aligned}$$

The desired result follows setting $\theta = 0$.

b) Using part (a) and writing expectations as integrals, we wish to maximize

$$\int \varphi(x) f_2(x) dx$$

with constraints

$$\int \varphi(x) f_0(x) dx = \alpha \quad \text{and} \quad \int \varphi(x) f_1(x) dx = 0,$$

where

$$f_2(x) = \frac{e^{-(x-1)^2/2}}{\sqrt{2\pi}}, \quad f_0(x) = \frac{e^{-x^2/2}}{\sqrt{2\pi}},$$

and

$$f_1(x) = \frac{x e^{-x^2/2}}{\sqrt{2\pi}}.$$

By the generalization of the Neyman–Pearson lemma, there are constants k_0 and k_1 such that the optimal test function φ has form

$$\varphi(x) = \begin{cases} 1, & f_2(x) \geq k_0 f_0(x) + k_1 f_1(x); \\ 0, & \text{otherwise.} \end{cases}$$

Dividing by f_0 ,

$$\varphi(x) = \begin{cases} 1, & e^{x-1/2} \geq k_0 + k_1 x; \\ 0, & \text{otherwise.} \end{cases}$$

Because $e^{x-1/2}$ is convex, $\varphi(x) = 0$ if and only if $x \in [c_1, c_2]$. To satisfy the second constraint, c_2 must be $-c_1$, and then the first constraint gives $c = \Phi^{-1}(1 - \alpha/2)$ as the common magnitude. So the optimal test function is $\varphi(x) = I\{|x| \geq c\}$.

32. a) By dominated convergence, we should have

$$\beta'(\theta) = \int \varphi \frac{\partial p_\theta}{\partial \theta} d\mu = \int \varphi \frac{\partial \log p_\theta}{\partial \theta} p_\theta d\mu = E_\theta \varphi l'(\theta),$$

where $l(\theta)$ is the log-likelihood. Here

$$l'(\theta) = \sum_{i=1}^2 \left[\frac{1}{\theta} - \frac{2X_i}{1 + \theta X_i} \right].$$

b) Reasoning as in the Neyman–Pearson lemma, the locally most powerful test will reject H_0 if $l'(\theta_0)$ exceeds some critical value, that is, if

$$\sum_{i=1}^2 \left[\frac{1}{\theta_0} - \frac{2X_i}{1 + \theta_0 X_i} \right] \geq k(\theta_0).$$

To find $k(\theta_0)$ we need the P_{θ_0} -distribution of the sum here. Solving the inequality, for $|x| < 1/\theta_0$,

$$P_{\theta_0} \left(\frac{1}{\theta_0} - \frac{2X_i}{1 + \theta_0 X_i} < x \right) = P_{\theta_0} \left(X_i > \frac{1 - \theta_0 x}{\theta_0(1 + \theta_0 x)} \right).$$

Since $P_{\theta}(X_i > c) = \int_c^{\infty} [\theta/(1 + \theta x)^2] dx = 1/(1 + \theta c)$, this expression equals

$$\left(1 + \frac{1 - \theta_0 x}{1 + \theta_0 x} \right)^{-1} = \frac{1 + \theta_0 x}{2},$$

which is the cumulative distribution for the uniform distribution on $(-1/\theta_0, 1/\theta_0)$. The density of this distribution is $f(x) = \theta_0/2$ for $|x| < 1/\theta_0$. The density for the sum of two independent variables with this density is “triangular” in shape: $g(s) = \int f(x)f(s-x) dx = [\frac{1}{2}\theta_0 - \frac{1}{4}|s|\theta_0^2]^+$.

If $k \in (0, 2/\theta_0)$, $\int_k^{2/\theta_0} g(s) ds = (k\theta_0 - 2)^2/8$. Setting this to α , $k(\theta_0) = (2 - \sqrt{8\alpha})/\theta_0$ (provided $\alpha < 1/2$), which is $1.6/\theta_0$ when $\alpha = 5\%$.

c) The confidence interval is

$$\left\{ \theta : \frac{2}{\theta} - \frac{2X_1}{1 + \theta X_1} - \frac{2X_2}{1 + \theta X_2} < \frac{1.6}{\theta} \right\} \\ = \left(\frac{-0.8(X_1 + X_2) + \sqrt{0.64(X_1 + X_2)^2 + 0.8X_1X_2}}{2X_1X_2}, \infty \right).$$

33. We want to minimize $\int_0^{\infty} \varphi(x) 2e^{-2x} dx$ with

$$\int_0^{\infty} \varphi(x) e^{-x} dx = \int_0^{\infty} \varphi(x) 3e^{-3x} dx = 1/2.$$

By the generalized Neyman–Pearson lemma, there should be Lagrange multipliers k_1 and k_2 so that an optimal test φ^* is one if $-2e^{-2x} > k_1e^{-x} + 3k_2e^{-3x}$, and zero if the opposite inequality holds. Equivalently, with $c_1 = -k_1/2$ and $c_2 = -3k_2/2$,

$$\varphi^*(x) = \begin{cases} 1, & c_1e^x + c_2e^{-x} > 1; \\ 0, & c_1e^x + c_2e^{-x} < 1. \end{cases}$$

If c_1 and c_2 have opposite signs, or one of them equals zero, then the left-hand side of these inequalities will be a monotone function of x , and φ^* will be a one-sided test. But then its power function will be monotone, and we cannot satisfy the constraints for the power function. So c_1 and c_2 must both be positive. Then $c_1e^x + c_2e^{-x}$ is convex, and φ^* will be a two-sided test¹ with form $\varphi^*(x) = 1 - 1_{(b_1, b_2)}(x)$, with b_1 and b_2 adjusted so that

¹ Reversing the null and alternative hypotheses, an extended argument using similar ideas can show that the test $1 - \varphi^*$ is a uniformly most powerful level $\alpha = 1/2$

$$P_1(X \in (b_1, b_2)) = e^{-b_1} - e^{-b_2} = 1/2$$

and

$$P_3(X \in (b_1, b_2)) = e^{-3b_1} - e^{-3b_2} = 1/2.$$

Solving,

$$b_1 = \log\left(\frac{4}{\sqrt{5}+1}\right) = 0.2119 \text{ and } b_2 = \log\left(\frac{4}{\sqrt{5}-1}\right) = 1.1744.$$

37. a) Suppose f is integrable, $\int |f| d\mu < \infty$. Since $|\varphi_n^2 f| \leq |f|$ and $\varphi_n^2 f$ converges pointwise to $\varphi^2 f$, by dominated convergence $\int \varphi_n^2 f d\mu \rightarrow \int \varphi^2 f d\mu$. Because f is an arbitrary integrable function, $\varphi_n^2 \xrightarrow{w} \varphi^2$.
- b) A dominated convergence argument now fails because $1/\varphi_n$ can be arbitrarily large. Because of this, it need not be the case that $1/\varphi_n \xrightarrow{w} 1/\varphi$. For instance, suppose μ is Lebesgue measure on $(0, 1)$, and $\varphi_n(x)$ is $1/n$ if $x \in (0, 1/n)$ and is one otherwise. Then φ_n converges pointwise to the test function φ that is identically one. But if f is one, $\int (1/\varphi_n) f d\mu = 2 - 1/n \rightarrow 2$, instead of $\int (1/\varphi) f d\mu = 1$.
41. The equation $E_{\theta_0} \varphi = \alpha$ gives $\varphi(0) + 4\varphi(1) + 4\varphi(2) = 9\alpha$, and the equation $E_{\theta_0} X \varphi = \alpha E_{\theta_0} X$ gives $\varphi(1) + 2\varphi(2) = 3\alpha$. If $\varphi(1) = 0$, then these equations give $\varphi(0) = 3\alpha$ and $\varphi(2) = 3\alpha/2$. This is the solution if $\alpha \leq 1/3$. When $\alpha > 1/3$, then $\varphi(0) = 1$, $\varphi(1) = (3\alpha - 1)/2$, and $\varphi(2) = (1 + 3\alpha)/4$.
42. The joint densities are $\exp\{-T(x)/(2\sigma^2)\}/(4\pi^2\sigma^4)$, where $T(x) = x_1^2 + \dots + x_4^2$, an exponential family. The UMP unbiased test will reject if and only if $T \leq c_1$ or $T \geq c_2$ with c_1 and c_2 adjusted so that $P_1(T \leq c_1) + P_1(T \geq c_2) = 5\%$ and $E_1 T \varphi = 20\%$. The density of T when $\sigma = 1$ is $te^{-t/2}/4$, and after a bit of calculus these equations become

$$1 - (1 + c_1/2)e^{-c_1/2} + (1 + c_2/2)e^{-c_2/2} = 5\%$$

and

$$4 - (4 + 2c_1 + c_1^2/2)e^{-c_1/2} + (4 + 2c_2 + c_2^2/2)e^{-c_2/2} = 20\%.$$

Numerical solution of these equations gives $c_1 = 0.607$ and $c_2 = 12.802$.

44. a) The densities form an exponential family with $T = X$, and by Theorem 12.26 the uniformly most powerful unbiased test will be two-sided with the proper level and uncorrelated with X if $\theta = \theta_0$. Since X is continuous, we do not need to worry about randomization, and can take $\varphi^* = I\{X \notin (c_1, c_2)\}$. The constants c_1 and c_2 are determined by

test of $H_0 : \theta \leq 1$ or $\theta \geq 3$ versus $H_1 : \theta \in (1, 3)$. In general, if $\theta_1 < \theta_2$, $\alpha \in (0, 1)$, and the data come from a one-parameter exponential family with $\eta(\cdot)$ a monotone function, then there will be a two-sided test φ^* with $\beta_{\varphi^*}(\theta_1) = \beta_{\varphi^*}(\theta_2) = 1 - \alpha$, and $1 - \varphi^*$ will be a uniformly most powerful level α test of $H_0 : \theta \leq \theta_1$ or $\theta \geq \theta_2$ versus $H_1 : \theta \in (\theta_1, \theta_2)$.

$$1 - \alpha = P_{\theta_0}(X \in (c_1, c_2)) = \frac{e^{\theta_0 c_2} - e^{\theta_0 c_1}}{2 \sinh(\theta_0)},$$

and $\text{Cov}_{\theta_0}(X, \varphi^*) = -\text{Cov}_{\theta_0}(X, 1 - \varphi^*) = 0$, which becomes

$$\int_{c_1}^{c_2} \frac{x \theta_0 e^{\theta_0 x}}{2 \sinh(\theta_0)} dx = (1 - \alpha) E_{\theta_0} X,$$

or

$$\frac{c_2 e^{\theta_0 c_2} - c_1 e^{\theta_0 c_1}}{2 \sinh(\theta_0)} - \frac{1 - \alpha}{\theta_0} = (1 - \alpha) \left[\coth(\theta_0) - \frac{1}{\theta_0} \right].$$

b) When $\theta_0 = 0$, $p_{\theta_0}(x) = 1/2$, $x \in (-1, 1)$, the uniform density. The equations for c_1 and c_2 are

$$1 - \alpha = P_0(X \in (c_1, c_2)) = \frac{c_2 - c_1}{2}$$

and

$$\int_{c_1}^{c_2} \frac{1}{2} x dx = \frac{1}{4} (c_2^2 - c_1^2) = 0.$$

Solving, $c_2 = 1 - \alpha$ and $c_1 = -(1 - \alpha)$.

B.13 Problems of Chapter 13

1. a) Since densities must integrate to one, if $\theta \neq 0$ and $\phi \neq 0$,

$$\begin{aligned} A(\theta, \phi) &= \log \int_0^1 \int_0^1 (x + y) e^{\theta x + \phi y} dx dy \\ &= \log \left[\frac{\theta(e^\theta - 1)(\phi e^\phi + 1 - e^\phi) + \phi(e^\phi - 1)(\theta e^\theta + 1 - e^\theta)}{\theta^2 \phi^2} \right]. \end{aligned}$$

- b) The marginal density of X is

$$\int_0^1 p_{\theta, \phi}(x, y) dy = \frac{x\phi(e^\phi - 1) + \phi e^\phi + 1 - e^\phi}{\phi^2} e^{\theta x - A(\theta, \phi)}.$$

This has the form in Theorem 13.2, with the dominating measure λ_ϕ having density $[x\phi(e^\phi - 1) + \phi e^\phi + 1 - e^\phi]/\phi^2$ with respect to Lebesgue measure on $(0, 1)$.

- c) The conditional density is

$$\frac{p_{\theta, \phi}(x, y)}{\int_0^1 p_{\theta, \phi}(u, y) du} = \frac{\theta^2 (x + y) e^{\theta x}}{\theta e^\theta + 1 - e^\theta + y\theta(e^\theta - 1)}.$$

Again, this has the form from Theorem 13.2, now with a dominating measure ν_y that has density $x + y$ with respect to Lebesgue measure on $(0, 1)$.

d) The UMP unbiased test will reject H_0 if and only if $X \geq c(Y)$ with $c(\cdot)$ chosen so that $P_{\theta=0}(X \geq c(y) \mid Y = y) = \alpha$. When $\theta = 0$, the conditional density of X given $Y = y$ is $2(x+y)/(1+2y)$. So

$$\begin{aligned} P_{\theta=0}(X \geq c(y) \mid Y = y) &= \int_{c(y)}^1 \frac{2(x+y)}{1+2y} dx \\ &= \frac{1 - c^2(y) + 2y - 2yc(y)}{1+2y}. \end{aligned}$$

Solving, this will be α if $c(y) = \sqrt{y^2 + (1+2y)(1-\alpha)} - y$.

e) Now the UMP unbiased test will reject H_0 if and only if $X \leq c_1(Y)$ or $X \geq c_2(Y)$, with $c_1(\cdot)$ and $c_2(\cdot)$ adjusted so that

$$\begin{aligned} \int_0^{c_1(y)} \frac{2(x+y)}{1+2y} dx + \int_{c_2(y)}^1 \frac{2(x+y)}{1+2y} dx \\ = \frac{c_1^2(y) + 2yc_1(y) + 1 - c_2^2(y) + 2y - 2yc_2(y)}{1+2y} = \alpha, \end{aligned}$$

and

$$\int_0^{c_1(y)} \frac{2x(x+y)}{1+2y} dx + \int_{c_2(y)}^1 \frac{2x(x+y)}{1+2y} dx = \alpha \int_0^1 \frac{2x(x+y)}{1+2y} dx,$$

or

$$\frac{2c_1^3(y) + 3yc_1^2(y) + 2 - 2c_2^3(y) + 3y - 3yc_2^2(y)}{3+6y} = \alpha \frac{2+3y}{3+6y}.$$

Explicit solution of these equations for $c_1(y)$ and $c_2(y)$ does not seem possible.

3. a) Letting $\theta_1 = \lambda_x - \lambda_y$ and $\theta_2 = -\lambda_x$, the joint densities are

$$\frac{\lambda_x^{\alpha_x} x^{\alpha_x-1} \lambda_y^{\alpha_y} y^{\alpha_y-1}}{xy\Gamma(\alpha_x)\Gamma(\alpha_y)} e^{-\lambda_x x - \lambda_y y} = \frac{|\theta_2|^{\alpha_x} x^{\alpha_x} |\theta_1 + \theta_2|^{\alpha_y} y^{\alpha_y}}{xy\Gamma(\alpha_x)\Gamma(\alpha_y)} e^{\theta_1 y + \theta_2(x+y)},$$

which is a canonical exponential family with sufficient statistics $T_1 = Y$ and $T_2 = X + Y$. With this parameterization, we are testing $H_0 : \theta_1 \leq 0$ versus $H_1 : \theta_1 > 0$, and the UMP unbiased test will reject H_0 if and only if $T_1 > z(T_2)$, with $z(\cdot)$ chosen so that $P_{(0,\theta_2)}[T_1 > z(t_2) \mid T_2 = t_2] = \alpha$. To compute this conditional probability, first note that the joint density of T_1 and T_2 is (the Jacobian for the transformation is one)

$$\frac{|\theta_2|^{\alpha_x} (t_2 - t_1)^{\alpha_x-1} |\theta_1 + \theta_2|^{\alpha_y} t_1^{\alpha_y-1}}{\Gamma(\alpha_x)\Gamma(\alpha_y)} e^{\theta_1 t_1 + \theta_2 t_2}, \quad 0 < t_1 < t_2.$$

The conditional density when $\theta_1 = 0$ is

$$\begin{aligned}
\frac{(t_2 - t_1)^{\alpha_x - 1} t_1^{\alpha_y - 1}}{\int_0^{t_2} (t_2 - \tau)^{\alpha_x - 1} \tau^{\alpha_y - 1} d\tau} &= \frac{(t_2 - t_1)^{\alpha_x - 1} t_1^{\alpha_y - 1}}{t_2^{\alpha_x + \alpha_y - 1} \int_0^1 (1 - u)^{\alpha_x - 1} u^{\alpha_y - 1} du} \\
&= \frac{\Gamma(\alpha_x) \Gamma(\alpha_y) (t_2 - t_1)^{\alpha_x - 1} t_1^{\alpha_y - 1}}{\Gamma(\alpha_x + \alpha_y) t_2^{\alpha_x + \alpha_y - 1}}.
\end{aligned}$$

Here the first equality arises from the change of variables $u = \tau/t_2$. The change of variables $u = t_1/t_2$ now gives

$$\begin{aligned}
P_{(0, \theta_2)}[T_1 > a \mid T_2 = t_2] &= \int_a^{t_2} \frac{\Gamma(\alpha_x) \Gamma(\alpha_y) (t_2 - t_1)^{\alpha_x - 1} t_1^{\alpha_y - 1}}{\Gamma(\alpha_x + \alpha_y) t_2^{\alpha_x + \alpha_y - 1}} dt_1 \\
&= \int_{a/t_2}^1 \frac{\Gamma(\alpha_x) \Gamma(\alpha_y)}{\Gamma(\alpha_x + \alpha_y)} u^{\alpha_y - 1} (1 - u)^{\alpha_x - 1} du \\
&= 1 - F(a/t_2),
\end{aligned}$$

where F is the cumulative distribution function for the beta distribution with parameters α_y and α_x . If q is the upper α th quantile for this distribution, so $F(q) = 1 - \alpha$, then the probability will be α if $a = qt_2$. Thus the UMP unbiased test rejects H_0 if and only if $T_1 > qT_2$.

b) From the definition, $\sum_{i=1}^n (X_i/\sigma_x)^2 \sim \chi_n^2 = \Gamma(n/2, 1/2)$, and since the reciprocal of the “failure rate” is a scale parameter, $ns_x^2 \sim \Gamma(n/2, 1/(2\sigma_x^2))$. Similarly, $ms_y^2 \sim \Gamma(m/2, 1/(2\sigma_y^2))$. From part (a), the UMP unbiased test will reject H_0 if and only if $ns_x^2/(ns_x^2 + ms_y^2) > q$, if and only if $F > mq/(n(1 - q))$, where q is the upper α th quantile for $\beta(m/2, n/2)$. This is the usual F -test, and this derivation shows that the upper α th quantile of $F_{n,m}$ is $mq/(n(1 - q))$.

5. The joint densities are

$$(2\pi)^{-n/2} \exp \left[\beta x' y + \gamma w' y - \frac{1}{2} \|y\|^2 - \frac{1}{2} \|\beta x + \gamma w\|^2 \right].$$

Introducing new parameters $\theta = \beta - \gamma$ and $\eta = \gamma$, the joint densities become

$$(2\pi)^{-n/2} \exp \left[\theta x' y + \eta(w + x)' y - \frac{1}{2} \|y\|^2 - \frac{1}{2} \|\theta x + \eta x + \eta w\|^2 \right],$$

and we would like to test $H_0 : \theta \leq 0$ versus $H_1 : \theta > 0$. These densities form an exponential family with canonical sufficient statistics $U = x'Y$ and $T = (x + w)'Y$. By Theorem 13.6, a uniformly most powerful unbiased test will reject H_0 if $U > c(T)$ with $c(\cdot)$ chosen so that

$$P_{\theta=0}(U > c(t) \mid T = t) = \alpha.$$

When $\theta = 0$, U and T have a bivariate normal distribution with $EU = \eta x'(x + w)$, $ET = \eta \|x + w\|^2$, $\text{Var}(U) = \|x\|^2$, $\text{Var}(T) = \|x + w\|^2$, and $\text{Cov}(U, T) = x'(x + w)$. So, when $\theta = 0$,

$$U|T = t \sim N\left(\frac{x'(x+w)t}{\|x+w\|^2}, \frac{\|x\|^2\|w\|^2 - (x'w)^2}{\|x+w\|^2}\right),$$

and the uniformly most powerful unbiased test will reject H_0 if

$$U - \frac{x'(x+w)T}{\|x+w\|^2} > z_\alpha \sqrt{\frac{\|x\|^2\|w\|^2 - (x'w)^2}{\|x+w\|^2}}.$$

6. a) Taking $\theta_1 = \log \lambda_x - \log \lambda_y$ and $\theta_2 = \log \lambda_y$, the likelihood is

$$\frac{\exp[\theta_1 T_1 + \theta_2 T_2 - m e^{\theta_1 + \theta_2} - n e^{\theta_2}]}{\prod_{i=1}^m X_i! \prod_{j=1}^n Y_j!},$$

where $T_1 = \sum_{i=1}^m X_i$ and $T_2 = \sum_{i=1}^m X_i + \sum_{j=1}^n Y_j$. The UMP unbiased test has form

$$\varphi = \begin{cases} 1, & T_1 > z(T_2); \\ \gamma(T_2), & T_1 = z(T_2); \\ 0, & T_1 < z(T_2), \end{cases}$$

with z, γ chosen so that $P(T_1 > z(t_2) \mid T_2 = t_2) + \gamma(t_2)P(T_1 = z(t_2) \mid T_2 = t_2) = \alpha$ when $\lambda_x = \lambda_y$. Note that if $\lambda_x = \lambda_y$

$$\begin{aligned} P(T_1 = t_1 \mid T_2 = t_2) &= \frac{P(\sum_{i=1}^m X_i = t_1, \sum_{j=1}^n Y_j = t_2 - t_1)}{P(\sum_{i=1}^m X_i + \sum_{j=1}^n Y_j = t_2)} \\ &= \frac{(m\lambda)^{t_1} (n\lambda)^{t_2 - t_1} e^{-(m+n)\lambda} / [t_1! (t_2 - t_1)!]}{((m+n)\lambda)^{t_2} e^{-(m+n)\lambda} / t_2!} \\ &= \binom{t_2}{t_1} \left(\frac{m}{m+n}\right)^{t_1} \left(\frac{n}{m+n}\right)^{t_2 - t_1}, \end{aligned}$$

and so, when $\lambda_x = \lambda_y$, $T_1 \mid T_2 = t_2 \sim \text{Binomial}(t_2, m/(m+n))$.

b) If $\lambda_x = \lambda_y$, $P(T_1 = 9 \mid T_2 = 9) = (2/3)^9 = 2.6\%$ and $P(T_1 = 8 \mid T_2 = 9) = 3(2/3)^8 = 11.7\%$, so in this case $z(9) = 8$ and $\gamma(9) = 20.5\%$. So the chance of rejection is 20.5%.

c) Using normal approximation for the binomial distribution, the approximate test will reject H_0 if

$$T_1 > \frac{mT_2 + z_\alpha \sqrt{mnT_2}}{m+n}.$$

B.14 Problems of Chapter 14

1. a) The matrix X should be

$$\begin{pmatrix} 1 & w_1 & x_1 \\ \vdots & \vdots & \vdots \\ 1 & w_n & x_n \end{pmatrix}.$$

- b) Because $w_1 + \cdots + w_n = x_1 + \cdots + x_n = 0$, w and x are both orthogonal to a column of 1s, and X will be of full rank unless w and x are collinear. Algebraically, this would occur if $D = S_{xx}S_{ww} - S_{xw}^2$ is zero.
- c) Using the formula $\hat{\beta} = (X'X)^{-1}X'Y$, since

$$X'X = \begin{pmatrix} n & 0 & 0 \\ 0 & S_{ww} & S_{wx} \\ 0 & S_{wx} & S_{xx} \end{pmatrix}, \quad X'Y = \begin{pmatrix} n\bar{Y} \\ S_{wY} \\ S_{xY} \end{pmatrix},$$

and

$$(X'X)^{-1} = \begin{pmatrix} 1/n & 0 & 0 \\ 0 & S_{xx}/D & -S_{wx}/D \\ 0 & -S_{wx}/D & S_{ww}/D \end{pmatrix},$$

we have

$$\hat{\beta}_1 = \bar{Y}, \quad \hat{\beta}_2 = \frac{S_{xx}S_{wY} - S_{wx}S_{xY}}{S_{xx}S_{ww} - S_{xw}^2}, \quad \text{and} \quad \hat{\beta}_3 = \frac{S_{ww}S_{xY} - S_{wx}S_{wY}}{S_{xx}S_{ww} - S_{xw}^2}.$$

- d) The covariance of $\hat{\beta}$ is $\text{Cov}(\hat{\beta}) = (X'X)^{-1}\sigma^2$, with $(X'X)^{-1}$ given above.
- e) The UMVU estimator of σ^2 is

$$S^2 = \frac{1}{n-3} \sum_{i=1}^n (Y_i - \hat{\beta}_1 - w_i\hat{\beta}_2 - x_i\hat{\beta}_3)^2.$$

- f) The variance of $\hat{\beta}_1$ is σ^2/n , estimated by S^2/n . So the confidence interval for β_1 is

$$\left(\hat{\beta}_1 \pm \frac{S}{\sqrt{n}} t_{\alpha/2, n-3} \right).$$

Since $\hat{\beta}_3 - \hat{\beta}_2 = (0 \ -1 \ 1) \hat{\beta}$,

$$\begin{aligned} \text{Var}(\hat{\beta}_3 - \hat{\beta}_2) &= (0 \ -1 \ 1) \text{Cov}(\hat{\beta}) \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix} \\ &= (0 \ -1 \ 1) (X'X)^{-1} \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix} \sigma^2 \\ &= \frac{S_{xx} + S_{ww} + 2S_{xw}}{S_{xx}S_{ww} - S_{xw}^2} \sigma^2 \\ &= \frac{\sum_{i=1}^n (x_i + w_i)^2}{S_{xx}S_{ww} - S_{xw}^2} \sigma^2. \end{aligned}$$

Plugging in S^2 to estimate σ^2 , the confidence interval for $\beta_3 - \beta_2$ is

$$\left(\hat{\beta}_3 - \hat{\beta}_2 \pm s \sqrt{\frac{\sum_{i=1}^n (x_i + w_i)^2}{S_{xx}S_{ww} - S_{xw}^2}} t_{\alpha/2, n-3} \right).$$

4. a) The dimension r is $2m - 1$ because the rows of the design matrix corresponding to (i, j) pairs $(1, 1), \dots, (1, m)$ and $(2, 1), \dots, (m, 1)$ are linearly independent.
 b) The least squares estimators $\hat{\alpha}_i$ and $\hat{\gamma}_j$ are not unique in this problem (since $r < p$), but they still minimize

$$L = \sum_{i=1}^m \sum_{j=1}^m (Y_{ij} - \alpha_i - \gamma_j)^2,$$

and must satisfy normal equations, obtained setting $\partial L / \partial \alpha_i$, $i = 1, \dots, m$ and $\partial L / \partial \gamma_j$, $j = 1, \dots, m$ to zero. This gives

$$\hat{\alpha}_i + \hat{\gamma} = \bar{Y}_{i\cdot}, \quad i = 1, \dots, m,$$

and

$$\hat{\gamma}_j + \hat{\alpha} = \bar{Y}_{\cdot j}, \quad j = 1, \dots, m,$$

where

$$\hat{\alpha} = \frac{1}{m} \sum_{i=1}^m \hat{\alpha}_i, \quad \hat{\gamma} = \frac{1}{m} \sum_{j=1}^m \hat{\gamma}_j,$$

$$\bar{Y}_{i\cdot} = \frac{1}{m} \sum_{j=1}^m Y_{ij}, \quad i = 1, \dots, m,$$

and

$$\bar{Y}_{\cdot j} = \frac{1}{m} \sum_{i=1}^m Y_{ij}, \quad j = 1, \dots, m.$$

Averaging these equations over i or j ,

$$\hat{\alpha} + \hat{\gamma} = \bar{Y} = \frac{1}{m^2} \sum_{i=1}^m \sum_{j=1}^m Y_{ij}.$$

So the least squares estimator for ξ_{ij} is

$$\hat{\xi}_{ij} = \hat{\alpha}_i + \hat{\gamma}_j = \bar{Y}_{i\cdot} + \bar{Y}_{\cdot j} - \bar{Y},$$

$i = 1, \dots, m, j = 1, \dots, m$.

c) Since $e_{ij} = Y_{ij} - \hat{\xi}_{ij}$, the estimator of σ^2 is

$$S^2 = \frac{1}{(m-1)^2} \sum_{i=1}^m \sum_{j=1}^m (Y_{ij} - \bar{Y}_{i\cdot} - \bar{Y}_{\cdot j} + \bar{Y})^2.$$

d) Because

$$\bar{\xi}_{i\cdot} = \frac{1}{m} \sum_{j=1}^m \xi_{ij} = \alpha_i + \bar{\gamma}, \quad i = 1, \dots, m,$$

we have $\alpha_i - \alpha_j = \bar{\xi}_{i\cdot} - \bar{\xi}_{j\cdot}$, expressing this difference as a linear function of ξ . The least squares estimator of $\alpha_i - \alpha_j$ is $\hat{\alpha}_i - \hat{\alpha}_j = \bar{Y}_{i\cdot} - \bar{Y}_{j\cdot}$.

e) Let $Z_i = m^{1/2}(\bar{Y}_{i\cdot} - \alpha_i - \bar{\gamma})/\sigma \sim N(0, 1)$. These variables depend on different subsets of the Y_{ij} , and hence they are independent. Also, averaging $\hat{\xi}_{ij} = \bar{Y}_{i\cdot} + \bar{Y}_{\cdot j} - \bar{Y}$ over j gives $\bar{Y}_{i\cdot} = \hat{\xi}_{i\cdot}$, so the variables Z_i , $i = 1, \dots, m$, are functions of $\hat{\xi}$ and are independent of S^2 . Since $(m-1)^2 S^2 / \sigma^2 \sim \chi_{(m-1)^2}^2$, by Definition 14.14,

$$\frac{\max Z_i - \min Z_i}{S/\sigma} = \frac{\max(\bar{Y}_{i\cdot} - \alpha_i) - \min(\bar{Y}_{i\cdot} - \alpha_i)}{S/\sqrt{m}}$$

has the studentized range distribution with parameters m and $(m-1)^2$. If q is the upper α th quantile of this distribution, and $I_{ij} = (\hat{\alpha}_i - \hat{\alpha}_j \pm qS/\sqrt{m})$, then, proceeding as in the text,

$$\begin{aligned} P(\alpha_i - \alpha_j \in I_{ij}, \forall i \neq j) &= P\left(\frac{\max(\bar{Y}_{i\cdot} - \alpha_i) - \min(\bar{Y}_{i\cdot} - \alpha_i)}{S/\sqrt{m}} < q\right) \\ &= 1 - \alpha. \end{aligned}$$

So I_{ij} , $i \neq j$, are simultaneous confidence intervals for the differences $\alpha_i - \alpha_j$.

f) Under H_0 , the data are distributed as they are in one-way analysis of variance, so $q = m$, and the least squares estimator for the mean of Y_{ij} is $\bar{Y}_{\cdot j}$. Then

$$\begin{aligned} \|\hat{\xi} - \hat{\xi}_0\|^2 &= \sum_{i=1}^m \sum_{j=1}^m (\bar{Y}_{i\cdot} - \bar{Y})^2 = m \sum_{i=1}^m (\bar{Y}_{i\cdot} - \bar{Y})^2, \\ T &= \frac{m \sum_{i=1}^m (\bar{Y}_{i\cdot} - \bar{Y})^2}{(m-1)S^2}, \end{aligned}$$

and we should reject H_0 if $T > F_{\alpha, m-1, (m-1)^2}$.

g) The test statistic has a noncentral F distribution with noncentrality parameter

$$\delta^2 = \frac{\|\xi - P_0 \xi\|^2}{\sigma^2} = \frac{m \sum_{i=1}^m (\alpha_i - \bar{\alpha})^2}{\sigma^2}$$

(found easily from the prior results, since $\xi - P_0 \xi$ equals $\hat{\xi} - \hat{\xi}_0$ when $\epsilon = 0$) and degrees of freedom $m-1$ and $(m-1)^2$. The power is the probability this distribution assigns to the interval $(F_{\alpha, m-1, (m-1)^2}, \infty)$.

h) Taking $\psi = (\alpha_1 - \alpha_m, \dots, \alpha_{m-1} - \alpha_m)'$, we can write any contrast $\sum_{i=1}^m a_i \alpha_i$ as $\sum_{i=1}^{m-1} a_i \psi_i$. From part (d), the least squares estimator of this contrast is

$$\sum_{i=1}^{m-1} a_i (\bar{Y}_i - \bar{Y}_m) = \sum_{i=1}^m a_i \bar{Y}_i.$$

The variance of this estimator is $\sum_{i=1}^m a_i^2 \sigma^2 / m = \|a\|^2 \sigma^2 / m$, estimated by $\|a\|^2 S^2 / m$. So the desired simultaneous confidence intervals are

$$\left(\sum_{i=1}^m a_i \bar{Y}_i \pm \|a\| S \sqrt{\frac{m-1}{m} F_{\alpha, m-1, (m-1)^2}} \right).$$

13. a) Let $\omega_0 = \{v \in \omega : Av = 0\}$, with dimension $r - q$ because A has full rank. Choose $\xi_0 \in \omega$ so that $A\xi_0 = \psi_0$, and introduce $Y^* = Y - \xi_0 \sim N(\xi^*, \sigma^2 I)$, with $\xi^* = \xi - \xi_0 \in \omega$. Since $\psi = \psi_0$ if and only if $A\xi = A\xi_0$ if and only if $A\xi^* = 0$, the null hypothesis is $H_0 : \xi^* \in \omega_0$, tested using the usual statistic

$$T = \frac{(n - r) \|PY^* - P_0 Y^*\|^2}{q \|Y^* - PY^*\|^2},$$

where P is the projection onto ω and P_0 is the projection onto ω_0 . The level α test rejects H_0 if $T \geq F_{\alpha, q, n-r}$.

b) Because $\xi_0 \in \omega$, $P\xi_0 = \xi_0$ and $\|Y^* - PY^*\|^2 = \|Y - PY\|^2$. So $T = \|PY^* - P_0 Y^*\|^2 / (qS^2)$, where $S^2 = \|Y - PY\|^2 / (n - r)$ is the usual estimator of σ^2 . Next, note that $P - P_0$ is the projection onto $\omega \cap \omega_0^\perp$, for if $v = v_0 + v_1 + v_2$ with $v_0 \in \omega_0$, $v_1 \in \omega \cap \omega_0^\perp$, and $v_2 \in \omega^\perp$, then $(P - P_0)v = (v_0 + v_1) - v_0 = v_1$. Because $A = AP$, the rows of A all lie in ω , and so they must span $\omega \cap \omega_0^\perp$. Consequently, $P - P_0 = A'(AA')^{-1}A$, as in the derivation for (14.14). So

$$\begin{aligned} \|(P - P_0)Y^*\|^2 &= (Y - \xi_0)'(P - P_0)(Y - \xi_0) \\ &= (Y - \xi_0)'A'(AA')^{-1}A(Y - \xi_0) \\ &= (\hat{\psi} - \psi_0)'(AA')^{-1}(\hat{\psi} - \psi_0), \end{aligned}$$

and

$$T = \frac{(\hat{\psi} - \psi_0)'(AA')^{-1}(\hat{\psi} - \psi_0)}{qS^2}.$$

The confidence set consists of all values ψ_0 where we would accept the null hypothesis $\psi = \psi_0$, that is, $\{\psi_0 : T(\psi_0) < F_{\alpha, q, n-r}\}$. Using the last formula for T , this region is the ellipse given in the problem.

14. a) Because $\sum_{l=1}^c x_{kl} = 0$, the columns of the design matrix X are orthogonal, and $X'X$ is diagonal. The $(0, 0)$ entry will be $\sum_{k=1}^p \sum_{l=1}^c x_{kl}^2$, and the other diagonal entries will all equal c . Also, the zeroth entry of $X'Y$ will be $\sum_{k=1}^p \sum_{l=1}^c x_{kl} Y_{kl}$, and the k th entry will be $\sum_{l=1}^c Y_{kl}$, $1 \leq k \leq p$. Since $\hat{\beta} = (X'X)^{-1}X'Y$,

$$\hat{\beta}_k = \frac{1}{c} \sum_{l=1}^c Y_{kl}, \quad k = 1, \dots, p,$$

are the least squares estimators for $\beta_1, \dots, \beta_p$, and

$$\hat{\beta}_0 = \frac{\sum_{k=1}^p \sum_{l=1}^c x_{kl} Y_{kl}}{\sum_{k=1}^p \sum_{l=1}^c x_{kl}^2}$$

is the least squares estimator for β_0 . If $Z_k = \sqrt{p}(\hat{\beta}_k - \beta_k)/\sigma$, $k = 1, \dots, p$, then $Z_1, \dots, Z_c$ are i.i.d. $N(0, 1)$ independent of

$$S^2 = \frac{1}{pc - p - 1} \sum_{k=1}^p \sum_{l=1}^c (Y_{kl} - \hat{\beta}_k - \hat{\beta}_0 x_{kl})^2.$$

Because $(pc - p - 1)S^2/\sigma^2 \sim \chi_{pc-p-1}^2$,

$$M = \frac{\max |Z_k|}{S/\sigma} = \frac{\max |\hat{\beta}_k - \beta_k|}{S/\sqrt{c}}$$

has the studentized maximum modulus distribution with parameters p and $pc - p - 1$. If q is the upper α th quantile for this distribution, then

$$P[\beta_k \in (\hat{\beta}_k \pm Sq/\sqrt{c}), k = 1, \dots, p] = P(M < q) = 1 - \alpha,$$

and $(\hat{\beta}_k \pm sq/\sqrt{c})$, $k = 1, \dots, p$, are simultaneous $1 - \alpha$ confidence intervals for $\beta_1, \dots, \beta_p$.

b) Let $\bar{x} = \sum_{l=1}^c x_{kl}/c$, and write

$$Y_{kl} = \beta_k + \beta_0 \bar{x} + \beta_0 (x_{kl} - \bar{x}) + \epsilon_{kl} = \beta_k^* + \beta_0^* (x_{kl} - \bar{x}) + \epsilon_{kl},$$

where $\beta_k^* = \beta_k + \beta_0 \bar{x}$ and $\beta_0^* = \beta_0$. Then the design matrix X^* has orthogonal columns, and proceeding as in part (a), $\hat{\beta}_k^* = \bar{Y}_k = \sum_{l=1}^c Y_{kl}/c$, $k = 1, \dots, p$, and

$$\hat{\beta}_0^* = \frac{\sum_{k=1}^p \sum_{l=1}^c Y_{kl} (x_{kl} - \bar{x})}{\sum_{k=1}^p \sum_{l=1}^c (x_{kl} - \bar{x})^2}$$

are the least squares estimators of β_k^* , $k = 1, \dots, p$, and β_0^* . Since $Z_k = \sqrt{c}(\bar{Y}_k - \beta_k^*)/\sigma$, $k = 1, \dots, p$, are i.i.d. $N(0, 1)$, independent of

$$S^2 = \frac{1}{pc - p - 1} \sum_{k=1}^p \sum_{l=1}^c (Y_{kl} - \hat{\beta}_k^* - \hat{\beta}_0^* x_{kl})^2,$$

and since $(pc - p - 1)S^2/\sigma^2 \sim \chi_{pc-p-1}^2$,

$$R = \frac{\max Z_k - \min Z_k}{S/\sigma} = \frac{\max_{j,k} |\beta_k - \beta_j - \bar{Y}_k + \bar{Y}_j|}{S/\sqrt{c}}$$

has the studentized maximum range distribution with parameters p and $pc - p - 1$. If q is the upper α th quantile of this distribution,

$$P[\beta_k - \beta_j \in (\bar{Y}_k - \bar{Y}_j \pm qS/\sqrt{c}), \forall k, j] = P(R < q) = 1 - \alpha.$$

So $(\bar{Y}_k - \bar{Y}_j \pm qS/\sqrt{c})$, $1 \leq j < k \leq p$, are simultaneous confidence intervals for the differences $\beta_k - \beta_j$, $1 \leq j < k \leq p$.

15. a) Since $\hat{\beta}_i = \sum_{j=1}^{n_i} Y_{ij}/n_i$, the (i, j) th entry of PY is $\hat{\beta}_i$. The (i, j) th entry of P_0Y is $\bar{Y} = \sum_{i=1}^p \sum_{j=1}^{n_i} Y_{ij}/n$, where $n = \sum_{i=1}^p n_i$. So

$$\|PY - P_0Y\|^2 = \sum_{i=1}^p \sum_{j=1}^{n_i} (\hat{\beta}_i - \bar{Y})^2 = \sum_{i=1}^p n_i (\hat{\beta}_i - \bar{Y})^2$$

and

$$S^2 = \frac{\|Y - PY\|^2}{n - p} = \frac{1}{n - p} \sum_{i=1}^p \sum_{j=1}^{n_i} (Y_{ij} - \hat{\beta}_i)^2.$$

The F -statistic for the test is

$$T = \frac{\sum_{i=1}^p n_i (\hat{\beta}_i - \bar{Y})^2}{(p - 1)S^2},$$

and we reject H_0 if and only if $T \geq F_{p-1, n-p}(1 - \alpha)$.

- b) The estimate $a_1\hat{\beta}_1 + \cdots + a_p\hat{\beta}_p$ has variance $\sigma^2 \sum_{i=1}^p a_i^2/n_i$, estimated by $S^2 \sum_{i=1}^p a_i^2/n_i$. So the simultaneous confidence intervals are

$$\sum_{i=1}^p a_i \beta_i \in \left(\sum_{i=1}^p a_i \hat{\beta}_i \pm S \sqrt{(p - 1)F_{p-1, n-p}(1 - \alpha) \sum_{i=1}^p \frac{a_i^2}{n_i}} \right).$$

B.17 Problems of Chapter 17

3. a) As usual, let

$$\bar{Y}_{i\cdot} = \frac{1}{n} \sum_{j=1}^n Y_{ij}, \quad i = 1, \dots, p,$$

and

$$\bar{Y}_{\cdot j} = \frac{1}{p} \sum_{i=1}^p Y_{ij}, \quad j = 1, \dots, n.$$

Introduce

$$\begin{aligned}
SS_e &= \sum_{i=1}^p \sum_{j=1}^n (Y_{ij} - \bar{Y}_{i.} - \bar{Y}_{.j} + \bar{Y}_{..})^2 \\
&= \sum_{j=1}^n \|Y_j - \bar{Y}\|^2 - \frac{1}{p} \sum_{j=1}^n [\mathbf{1}'(Y_j - \bar{Y})]^2
\end{aligned}$$

and

$$SS_\beta = p \sum_{j=1}^n (\bar{Y}_{.j} - \bar{Y}_{..})^2 = \frac{1}{p} \sum_{j=1}^n [\mathbf{1}'(Y_j - \bar{Y})]^2,$$

sums of squares that would arise testing null hypotheses if the β_j were viewed as fixed constants. Let $P = \mathbf{1}\mathbf{1}'/p$, the projection onto the linear span of $\mathbf{1}$, and let $Q = I - P$. The covariance of Y_j is $\Sigma = \sigma^2 I + \tau^2 \mathbf{1}\mathbf{1}' = \sigma^2 Q + (\sigma^2 + p\tau^2)P$, with determinant $\sigma^{2(p-1)}(\sigma^2 + p\tau^2)$ (the eigenvector $\mathbf{1}$ has eigenvalue $\sigma^2 + p\tau^2$, and the other eigenvalues are all σ^2), and $\Sigma^{-1} = \sigma^{-2}Q + (\sigma^2 + p\tau^2)^{-1}P$. The log-likelihood is

$$\begin{aligned}
l(\alpha, \sigma, \tau) &= -\frac{1}{2} \sum_{j=1}^n (Y_j - \alpha)' \Sigma^{-1} (Y_j - \alpha) - np \log \sqrt{2\pi} - \frac{n}{2} \log |\Sigma| \\
&= -\frac{1}{2} \sum_{j=1}^n (Y_j - \bar{Y})' \Sigma^{-1} (Y_j - \bar{Y}) - \frac{n}{2} (\bar{Y} - \alpha)' \Sigma^{-1} (\bar{Y} - \alpha) \\
&\quad - np \log \sqrt{2\pi} - \frac{n(p-1)}{2} \log \sigma^2 - \frac{n}{2} \log(\sigma^2 + p\tau^2).
\end{aligned}$$

(The cross-product term drops out because $\sum_{j=1}^n (Y_j - \bar{Y}) = 0$.) This is maximized over α by $\hat{\alpha} = \bar{Y}$, regardless of the value of σ or τ . To maximize over σ and τ , introduce $\eta^2 = \sigma^2 + p\tau^2$ and note that

$$\begin{aligned}
(Y_j - \bar{Y})' \Sigma^{-1} (Y_j - \bar{Y}) &= \frac{\|Q(Y_j - \bar{Y})\|^2}{\sigma^2} + \frac{\|P(Y_j - \bar{Y})\|^2}{\eta^2} \\
&= \frac{\|Y_j - \bar{Y}\|^2 - [\mathbf{1}'(Y_j - \bar{Y})]^2/p}{\sigma^2} + \frac{[\mathbf{1}'(Y_j - \bar{Y})]^2}{p\eta^2}.
\end{aligned}$$

So

$$\begin{aligned}
l(\hat{\alpha}, \sigma, \tau) &= -\frac{\sum_1^n \|Y_j - \bar{Y}\|^2 - \sum_1^n [\mathbf{1}'(Y_j - \bar{Y})]^2/p}{2\sigma^2} - \frac{\sum_1^n [\mathbf{1}'(Y_j - \bar{Y})]^2}{2p\eta^2} \\
&\quad - n(p-1) \log \sigma - n \log \eta \\
&= -\frac{SS_e}{2\sigma^2} - \frac{SS_\beta}{2\eta^2} - n(p-1) \log \sigma - n \log \eta.
\end{aligned}$$

Setting derivatives to zero suggests that the maximum likelihood estimators should be

$$\hat{\eta}^2 = \frac{1}{n} SS_\beta \quad \text{and} \quad \hat{\sigma}^2 = \frac{SS_e}{np - n}.$$

But there is a bit of a problem here because we need $\hat{\eta}^2 \geq \hat{\sigma}^2$. If these formulas give $\hat{\eta}^2 < \hat{\sigma}^2$, then the likelihood is maximized with $\hat{\eta} = \hat{\sigma}$. The common value will maximize $l(\hat{\alpha}, \sigma, \sigma) = \sum_{j=1}^n \|Y_j - \bar{Y}\|^2 / \sigma^2 - np \log \sigma$. Setting the derivative to zero, the common value is $\tilde{\sigma}^2 = \sum_{j=1}^n \|Y_j - \bar{Y}\|^2 / (np)$, and this is also the maximum likelihood estimator for σ^2 under H_0 . Plugging these estimates into the likelihood,

$$2 \log \lambda = 2l(\hat{\alpha}, \hat{\sigma}, \hat{\tau}) - 2l(\hat{\alpha}, \tilde{\sigma}, 0) = 2n(p-1) \log(\tilde{\sigma}/\hat{\sigma}) + 2n \log(\tilde{\sigma}/\hat{\eta}).$$

Remark: H_0 would not be rejected when $\hat{\eta}^2 = \hat{\sigma}^2$, and when $\hat{\eta}^2 > \hat{\sigma}^2$ there is a monotonic relationship between λ and the F -statistic that would be used to test H_0 if β_j were viewed as constants. So the F -test and likelihood ratio test here would be the same in practice.

b) Under H_0 , $\alpha = \alpha_0 \mathbf{1}$ and

$$(\bar{Y} - \alpha)' \Sigma^{-1} (\bar{Y} - \alpha) = \frac{\bar{Y}' Q \bar{Y}}{\sigma^2} + \frac{(\mathbf{1}' \bar{Y} - p \alpha_0)^2}{p(\sigma^2 + p \tau^2)},$$

which is minimized when α_0 is $\mathbf{1}' \bar{Y} / p$. So

$$(\bar{Y} - \tilde{\alpha})' \Sigma^{-1} (\bar{Y} - \tilde{\alpha}) = \frac{SS_{\alpha}}{n \sigma^2}$$

and

$$l(\tilde{\alpha}, \sigma, \tau) = -\frac{SS_e + SS_{\alpha}}{2\sigma^2} - \frac{SS_{\beta}}{2\eta^2} - n(p-1) \log \sigma - n \log \eta,$$

where $SS_{\alpha} = \sum_{i=1}^p \sum_{j=1}^n (\bar{Y}_i - \bar{Y}_{..})^2$. Setting derivatives to zero, and keeping in mind that we must have $\hat{\eta} \geq \tilde{\sigma}$,

$$\tilde{\sigma}^2 = \frac{SS_e + SS_{\alpha}}{n(p-1)} \quad \text{and} \quad \hat{\eta}^2 = \frac{SS_{\beta}}{n},$$

when $(p-1)SS_{\beta} \geq SS_e + SS_{\alpha}$, and $\tilde{\sigma}^2 = \hat{\eta}^2 = (SS_e + SS_{\alpha} + SS_{\beta}) / (np)$ when $(p-1)SS_{\beta} < SS_e + SS_{\alpha}$. Plugging these values into the log-likelihood function, $2 \log \lambda = n(p-1) \log(\tilde{\sigma}/\hat{\sigma}) + n \log(\tilde{\sigma}/\hat{\eta})$. The estimators $\tilde{\sigma}$ and $\hat{\eta}$ depend on SS_{β} ; thus the F -statistic and the likelihood ratio statistic are not equivalent.

4. a) The likelihood function is $L(\theta_x, \theta_y) = \theta_x \theta_y \exp[-\theta_x X - \theta_y Y]$. Setting derivatives to zero, the maximum likelihood estimators are $\hat{\theta}_x = 1/X$ and $\hat{\theta}_y = 1/Y$, and so $\sup_{\Omega} L(\theta) = L(\hat{\theta}_x, \hat{\theta}_y) = e^{-2}/(XY)$. The maximum likelihood estimator $\hat{\theta}_y$ for θ_y under H_0 maximizes $L(2\theta_y, \theta_y) = 2\theta_y^2 \exp[-(2X+Y)\theta_y]$. A bit of calculus gives $\tilde{\theta}_y = 2/(2X+Y)$, and then $\sup_{\Omega_0} L(\theta) = L(\tilde{\theta}_x, \tilde{\theta}_y) = 8e^{-2}/(2X+Y)^2$. The likelihood ratio test statistic is $\lambda = (2X+Y)^2/(8XY)$.

b) Define $U = \theta_x X$ and $V = \theta_y Y$, so U and V are independent standard exponential variables. Under H_0 , λ can be expressed in terms of these variables as $\lambda = (U + V)^2 / (4UV)$. The significance level of the test is

$$\begin{aligned}\alpha &= P_{H_0}(\lambda \geq c) = P[(U + V)^2 \geq 4cUV] \\ &= P(U \geq mV) + P(V \geq mU),\end{aligned}$$

where $m = 2c - 1 + 2\sqrt{c^2 - c}$. By smoothing, $P(U \geq mV) = EP(U \geq mV|V) = Ee^{-mV} = \int_0^\infty e^{-(m+1)v} dv = 1/(m+1)$. So $\alpha = 2/(m+1)$, and to achieve a specified level α , m should be $2/\alpha - 1$. From this, $c = (m+1)^2/(4m) = 1/[\alpha(2-\alpha)]$.

8. a) The log-likelihood is $l(\theta_1, \theta_2) = -n\bar{X}/\theta_1 - n\bar{Y}/\theta_2 - n\log\theta_1 - n\log\theta_2$. Setting derivatives to zero, the maximum likelihood estimators are $\hat{\theta}_1 = \bar{X}$ and $\hat{\theta}_2 = \bar{Y}$. The maximum likelihood estimator $\hat{\theta}_1$ for θ_1 under H_0 maximizes $l(\theta_1, \theta_1/c_0)$. Setting the derivative to zero, $\hat{\theta}_1 = (\bar{X} + c_0\bar{Y})/2$. Then

$$\begin{aligned}\log\lambda &= l(\hat{\theta}_1, \hat{\theta}_2) - l(\tilde{\theta}_1, \tilde{\theta}_1/c_0) \\ &= n\log\left(\frac{\bar{X} + c_0\bar{Y}}{2\bar{X}}\right) + n\log\left(\frac{\bar{X} + c_0\bar{Y}}{2c_0\bar{Y}}\right).\end{aligned}$$

b) The confidence intervals would contain all values c_0 for which $2\log\lambda < 3.84$. (Here 3.84 is the 95th percentile of χ_1^2 .) For the data and sample size given, if c_0 is 2.4, $2\log\lambda = 1.66$. This value is less than 3.84, and so 2.4 is in the confidence interval.

c) The Fisher information matrix is

$$I(\theta) = \begin{pmatrix} \theta_1^{-2} & 0 \\ 0 & \theta_2^{-2} \end{pmatrix}.$$

It seems natural to take $\theta_0 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$, and then $\Delta = \begin{pmatrix} 1 \\ -1 \end{pmatrix} \sqrt{n}/10$. Define an orthonormal basis $v_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix} / \sqrt{2}$ and $v_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix} / \sqrt{2}$. Since Ω_0 is linear, the tangent spaces $V_\theta = V$ at different $\theta \in \Omega$ are all the same, each being the linear span of v_1 . So $P_0 = v_1 v_1'$ and $Q_0 = v_2 v_2'$ are the projection matrices onto V and $V^\perp$. Since Δ lies in $V^\perp$, $Q_0 \Delta = \Delta$. Since $I(\theta_0)$ is the identity, $P_0 + Q_0 I(\theta_0)^{-1} Q_0 = P_0 + Q_0 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$. So the noncentrality parameter is $\delta^2 = \Delta' Q_0 [P_0 + Q_0 I(\theta_0)^{-1} Q_0]^{-1} Q_0 \Delta = \Delta' \Delta = n/50$. The asymptotic distribution of $2\log\lambda$ is $\chi_1^2(\delta^2)$. If $Z \sim N(0, 1)$, then $(Z + \delta)^2 \sim \chi_1^2(\delta^2)$ has this distribution, and so the power is approximately $P[(Z + \delta)^2 > 3.84] = P(Z + \delta > 1.96) + P(Z + \delta < -1.96) = \Phi(\delta - 1.96) + \Phi(-\delta - 1.96)$. The second term here will be negligible, and since $\Phi(1.28) = 0.9$ we will need $\delta - 1.96 = 1.28$, which gives $n = 525$ as the necessary sample size.

9. a) The likelihood L is

$$\frac{\exp\left[-\frac{1}{2\sigma_w^2} \sum_1^n (W_i - \mu_w)^2 - \frac{1}{2\sigma_x^2} \sum_1^n (X_i - \mu_x)^2 - \frac{1}{2\sigma_y^2} \sum_1^n (Y_i - \mu_y)^2\right]}{\sqrt{2\pi}^{3n} \sigma_w^n \sigma_x^n \sigma_y^n},$$

and the Fisher information matrix is

$$I(\mu_w, \mu_x, \mu_y, \sigma_w, \sigma_x, \sigma_y) = \text{diag}\left(\frac{1}{\sigma_w^2}, \frac{1}{\sigma_x^2}, \frac{1}{\sigma_y^2}, \frac{2}{\sigma_w^2}, \frac{2}{\sigma_x^2}, \frac{2}{\sigma_y^2}\right).$$

Maximum likelihood estimators under the full model and under H_0 are given by

$$\hat{\mu}_w = \tilde{\mu}_w = \overline{W}, \quad \hat{\mu}_x = \tilde{\mu}_x = \overline{X}, \quad \hat{\mu}_y = \tilde{\mu}_y = \overline{Y},$$

$$\hat{\sigma}_w^2 = \frac{1}{n} \sum_{i=1}^n (W_i - \overline{W})^2, \quad \hat{\sigma}_x^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \overline{X})^2, \quad \hat{\sigma}_y^2 = \frac{1}{n} \sum_{i=1}^n (Y_i - \overline{Y})^2,$$

and

$$\tilde{\sigma}^2 = \frac{\hat{\sigma}_w^2 + \hat{\sigma}_x^2 + \hat{\sigma}_y^2}{3}.$$

Plugging these in,

$$\lambda = \frac{L(\hat{\theta})}{L(\tilde{\theta})} = \left[\frac{\tilde{\sigma}^3}{\hat{\sigma}_w \hat{\sigma}_x \hat{\sigma}_y} \right]^n.$$

b) The natural choice for θ_0 is $(\mu_w, \mu_x, \mu_y, 2, 2, 2)'$, and then

$$I(\theta_0) = \text{diag}\left(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right).$$

Introduce the orthonormal basis

$$\begin{aligned} v_1 &= \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, & v_2 &= \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, & v_3 &= \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \\ v_4 &= \frac{1}{\sqrt{3}} \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 1 \\ 1 \end{bmatrix}, & v_5 &= \frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \\ -1 \\ 0 \end{bmatrix}, & v_6 &= \frac{1}{\sqrt{6}} \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 2 \\ -2 \end{bmatrix}. \end{aligned}$$

As in Example 17.3, all of the tangent spaces $V = V_\theta$ are the same. Here V is the linear span of v_1, v_2, v_3 , and v_4 , and $Q_0 = v_5 v_5' + v_6 v_6'$ is the orthogonal projection onto $V^\perp$. Noting that $Q_0 I(\theta_0)^{-1} Q_0 = 2Q_0$, that $\Delta = (\theta - \theta_0)\sqrt{n} = (0, 0, 0, -\sqrt{8}, \sqrt{8}, 0)'$, and that $Q_0 \Delta = \Delta$, the noncentrality parameter is

$$\delta^2 = \Delta' Q_0 [P_0 + Q_0 I(\theta_0)^{-1} Q_0]^{-1} Q_0 \Delta = \Delta' \Delta / 2 = 8.$$

The likelihood ratio test with $\alpha \approx 5\%$ rejects H_0 if $2 \log \lambda \geq 5.99$ (the 95th percentile of χ_2^2). The power of this test is thus $P(2 \log \lambda \geq 5.99)$. The distribution of $2 \log \lambda$ should be approximately $\chi_2^2(\delta^2 = 8)$. If F is the cumulative distribution function for this distribution, the power is approximately $1 - F(5.99) = 71.77\%$.

10. a) The log-likelihood is

$$\begin{aligned} l(\mu) &= -\frac{1}{2} \sum_{i=1}^n \|X_i - \mu\|^2 - np \log \sqrt{2\pi} \\ &= -\frac{n}{2} \|\bar{X} - \mu\|^2 - \frac{1}{2} \sum_{i=1}^n \|X_i - \bar{X}\|^2 - np \log \sqrt{2\pi}. \end{aligned}$$

By inspection, $\hat{\mu} = \bar{X}$ is the maximum likelihood estimator under the full model, and the maximum likelihood estimator under H_0 minimizes $\|\bar{X} - \mu\|$ over μ with $\|\mu\| = r$. The natural guess is $\tilde{\mu} = r\bar{X}/\|\bar{X}\|$. That this is indeed correct can be seen using the triangle inequality. Suppose $\|\mu\| = r$. Then if $\|\bar{X}\| > r$, $\|\bar{X} - \mu\| + \|\mu\| \geq \|\bar{X}\| = \|\tilde{\mu}\| + \|\bar{X} - \tilde{\mu}\|$, and so $\|\bar{X} - \mu\| \geq \|\bar{X} - \tilde{\mu}\|$; and if $\|\bar{X}\| < r$, $\|\bar{X}\| + \|\mu - \bar{X}\| \geq \|\mu\| = 1 = \|\bar{X}\| + \|\bar{X} - \tilde{\mu}\|$, and again $\|\bar{X} - \mu\| \geq \|\bar{X} - \tilde{\mu}\|$. So $2 \log \lambda = 2l(\hat{\mu}) - 2l(\tilde{\mu}) = n\|\tilde{\mu} - \bar{X}\|^2 = n(\|\bar{X}\| - r)^2$.

b) By Lemma 14.8, because $\sqrt{n}\bar{X} \sim N(\sqrt{n}\mu, I)$,

$$n\|\bar{X}\|^2 \sim \chi_p^2(n\|\mu\|^2).$$

If F is the cumulative distribution function for this distribution, then the power of the test, assuming $nr^2 > c$, is

$$\begin{aligned} P[n(\|\bar{X}\| - r)^2 > c] &= P(\sqrt{n}\|\bar{X}\| < \sqrt{nr} - \sqrt{c}) \\ &\quad + P(\sqrt{n}\|\bar{X}\| > \sqrt{nr} + \sqrt{c}) \\ &= F((\sqrt{nr} - \sqrt{c})^2) + 1 - F((\sqrt{nr} + \sqrt{c})^2). \end{aligned}$$

c) Since the Fisher information is the identity, the formula for the noncentrality parameter δ^2 is just $\|Q_0\Delta\|^2$, where Q_0 is the projection onto the orthogonal complement of the tangent space at μ_0 and $\Delta = \sqrt{n}(\mu - \mu_0)$. Now μ_0 should be near μ and in the null parameter space, and the most natural choice is just $\mu_0 = \mu/\|\mu\|$. Then $\mu - \mu_0$ is in the orthogonal complement of the tangent space at μ_0 , and the noncentrality parameter is $\delta^2 = \|\Delta\|^2 = n\|\mu - \mu_0\|^2 = n/100$. Arguing as in the Example 17.4, $\delta^2 = 10.51$ will give power 90%, and 1051 observations will be necessary.

12. a) Because Y_i and W_i are linear functions of X_i , ϵ_i , and η_i , they should have a bivariate normal distribution with mean zero and covariances given by

$$\text{Var}(Y_i) = 1 + \beta^2, \quad \text{Var}(W_i) = 1 + \sigma^2,$$

and

$$\text{Cov}(Y_i, W_i) = \text{Cov}(\beta X_i + \epsilon_i, X_i + \eta_i) = \beta.$$

b) The log-likelihood is

$$l(\beta, \sigma^2) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log(1 + \sigma^2 + \sigma^2 \beta^2) - n \frac{(1 + \sigma^2)T_1 - 2\beta T_2 + (1 + \beta^2)T_3}{2(1 + \sigma^2 + \sigma^2 \beta^2)},$$

with

$$T_1 = \frac{1}{n} \sum_{i=1}^n Y_i^2, \quad T_2 = \frac{1}{n} \sum_{i=1}^n W_i Y_i, \quad \text{and} \quad T_3 = \frac{1}{n} \sum_{i=1}^n W_i^2.$$

Maximum likelihood estimators for $\hat{\beta}$ and $\hat{\sigma}^2$ can be obtained solving the following equations, obtained setting partial derivatives of l to zero:

$$1 + \beta^2 + T_1 = \frac{[(1 + \sigma^2)T_1 - 2\beta T_2 + (1 + \beta^2)T_3](1 + \beta^2)}{1 + \sigma^2 + \sigma^2 \beta^2},$$

$$2\beta\sigma^2 - 2T_2 + 2\beta T_3 = \frac{[(1 + \sigma^2)T_1 - 2\beta T_2 + (1 + \beta^2)T_3](2\beta\sigma^2)}{1 + \sigma^2 + \sigma^2 \beta^2}.$$

Explicit formulas do not seem possible. Under $H_0 : \beta = 0$, the maximum likelihood estimator for σ^2 is $\hat{\sigma}^2 = T_3/n - 1$, obtained setting $\partial l(0, \sigma^2)/\partial \sigma^2$ to zero.

c) When $\sigma = 0$, then the least squares estimator for β is T_2/T_1 . By the law of large numbers, $T_2 \xrightarrow{P} EY_1 W_1 = \beta$ and $T_1 \xrightarrow{P} EW_1^2 = 1 + \sigma^2$. So the least squares estimator converges in probability to $\beta/(1 + \sigma^2)$ and is inconsistent unless $\sigma^2 = 0$ or $\beta = 0$.

d) Let $\theta = (\beta, \sigma^2)'$. The Fisher information is

$$I(\theta) = \frac{1}{D^2} \begin{pmatrix} 2 + 2\sigma^2 + 6\beta^2\sigma^4 & \beta + 2\beta^2 + \beta\sigma^2 + \beta^3\sigma^2 \\ \beta + 2\beta^2 + \beta\sigma^2 + \beta^3\sigma^2 & (1 + \beta^2)^2/2 \end{pmatrix},$$

where $D = 1 + \sigma^2 + \beta^2\sigma^2$. Under H_0 , $\beta = 0$ and $I(\theta)$ is diagonal with inverse

$$I(\theta)^{-1} = \begin{pmatrix} \frac{1}{2}(1 + \sigma^2) & 0 \\ 0 & 2(1 + \sigma^2)^2 \end{pmatrix}.$$

If $e_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $e_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$, the standard basis vectors, then the projection matrices in the formula for the noncentrality parameter δ^2 are $P_0 = e_2 e_2'$ and $Q_0 = e_1 e_1'$. Taking $\theta_0 = \begin{pmatrix} 0 \\ \sigma^2 \end{pmatrix}$, $\Delta = \sqrt{n} \begin{pmatrix} \beta \\ \sigma^2 \end{pmatrix}$ and

$$\begin{aligned} \delta^2 &= n\beta^2 e_1' [P_0 + Q_0 I(\theta_0)^{-1} Q_0]^{-1} e_1 \\ &= n\beta^2 e_1' \left(P_0 + \frac{Q_0}{e_1' I(\theta_0)^{-1} e_1} \right) e_1 \\ &= \frac{2n\beta^2}{1 + \sigma^2}. \end{aligned}$$

The power of the test is approximately $P((Z + \delta)^2 > 1.96^2)$ with $Z \sim N(0, 1)$. It decreases as σ^2 increases.

13. a) Viewing (p_1, p_2) as the parameter with p_3 determined as $1 - p_1 - p_2$, the parameter space Ω is the triangular-shaped region where $p_1 > 0$, $p_2 > 0$, and $p_1 + p_2 < 1$, an open set in $\mathbb{R}^2$, and the likelihood function is

$$L(p) = \binom{n}{Y_1, Y_2, Y_3} p_1^{Y_1} p_2^{Y_2} p_3^{Y_3}.$$

Maximum likelihood estimators under the full model are $\hat{p}_i = Y_i/n$, $i = 1, 2, 3$. The parameter space under H_0 is

$$\Omega_0 = \{(1 - e^{-\theta}, e^{-\theta} - e^{-2\theta}) : \theta > 0\}.$$

The maximum likelihood estimator $\tilde{\theta}$ for θ under H_0 maximizes

$$\log L(1 - e^{-\theta}, e^{-\theta} - e^{-2\theta}).$$

Setting the θ -derivative to zero, $\tilde{\theta} = -\log[(Y_2 + 2Y_3)/(2n - Y_1)]$. Plugging in the estimators, $2 \log \lambda$ is

$$2 \left[\sum_{i=1}^3 Y_i \log \frac{Y_i}{n} - (Y_1 + Y_2) \log \frac{Y_1 + Y_2}{Y_1 + 2Y_2 + 2Y_3} - (Y_2 + 2Y_3) \log \frac{Y_2 + 2Y_3}{Y_1 + 2Y_2 + 2Y_3} \right].$$

b) With $Y = (36, 24, 40)$, $2 \log \lambda = 0.0378$ and we accept H_0 . The p -value is $P(\chi^2(1) > 0.0378) = 84.6\%$.

c) Take $\theta_0 = (0.4, 0.24)$, a convenient point in Ω_0 close to $\theta = (0.36, 0.24)$ (any other reasonable choice should give a similar answer). Inverse Fisher information at θ is

$$I(\theta_0)^{-1} = \begin{pmatrix} 0.240 & -0.0960 \\ -0.096 & 0.1824 \end{pmatrix},$$

and points θ in Ω_0 satisfy the constraint $g(\theta) = \theta_1^2 + \theta_2 - \theta_1 = 0$. If V_0 is the tangent space at θ_0 , then $V_0^\perp$ is the linear span of

$$v = \nabla g(\theta_0) = \begin{pmatrix} 2\theta_1 - 1 \\ 1 \end{pmatrix}_{\theta=\theta_0} = \begin{pmatrix} -0.2 \\ 1 \end{pmatrix},$$

and $Q_0 = vv'/\|v\|^2$. Also, $\Delta = \sqrt{n}(\theta - \theta_0) = \sqrt{n} \begin{pmatrix} -0.04 \\ 0 \end{pmatrix}$. So

$$\begin{aligned} \delta^2 &= \Delta' Q_0 (P_0 + Q_0 I(\theta_0)^{-1} Q_0)^{-1} Q_0 \Delta \\ &= \frac{(\Delta \cdot v)^2}{v' I(\theta_0)^{-1} v} = \frac{n(0.008)^2}{0.2304} = \frac{n}{3600}. \end{aligned}$$

For 90% power, δ^2 needs to be 10.51 which leads to $n = 37,836$ as the requisite sample size.

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